SECJ2203: Software Engineering

System Documentation (SD)

Software Test Documentation

Version 6.0

12 FEB

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Faculty of Computing

Prepared by: <SYNAPSE>

Revision Page

a. Overview

The System Documentation (SD) of the current version of Koperasi KADA's Automated System has the system introduction comprising the scope, purpose, objective and feature to be included in the system. SD provides user personas, user stories and system features. Product backlogs, sprint deliverables, user story with support of sequence and activity diagram. also covered are non-functional requirements

b. Target Audience

The target audience are applicants, admin and Ahli Lembaga Koperasi of KADA.

c. Project Team Members

Member Name	Role	Task	Status
Farra Nurzahin	Project Manager	Project planning	Complete
		Project management	Complete
		Create system architecture	Complete
Dayang Farah Farzana	Technical Writer	Gather system and user requirements	Complete
		Create SRS document	Complete
		System documentation	Complete
Nawarrah Auni	System Developer	Develop system	Complete
		System Testing	Complete
		System Implementation	Complete
Safiya Nursyahadah	Designer	System Design	Complete
		Design system functional diagram	Complete
		Design draft system	Complete

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Farra Nurzahin binti Zaharil Anuar	Completed Topic 1(Introduction), Scope, Section 2.1.2 Section 2.2.1, Section 2.3, Section 2.4.2, Section 2.5	30/11/2024

	Dayang Farah Farzana binti Abang Idham	Completed Overview, Section 1.3, Section 2.1.3, Section 2.2 (System Features), Section 2.3, Section 2.4.3	30/11/2024
	Nawwarah Auni binti Nazrudin	Completed Purpose, Specific Requirements, Section 2.1, Section 2.2.2, Section 2.3, Section 2.4, Section 2.4.1, Section 2.6.1, Section 2.6.4	29/11/2024
	Safiya Nursyahadah binti Masnoor	Completed Target Audience, Section 1.5 (overview), Section 2.1.1, Section 2.2.3 (state diagram), Section 2.3, Section 2.6,	30/11/2024
2.0	Farra Nurzahin binti Zaharil Anuar	Completed Section 4.1, Section 4.2, Section 4.3, Section 4.4 Section 5.1, Section 5.1.1, Section 5.1.7, Section 5.1.8, Section 5.1.9, Section 6.1	19/12/2024
	Dayang Farah Farzana binti Abang Idham	Completed Section 4, Section 4.2, Section 4.3, Section 4.4 Section 5, Section 5.1.1, Section 5.1.2, Section 5.1.4, Section 5.1.6, Section 5.1.12, Section 5.1.13, Section 6.1, Section 7	19/12/2024
	Nawwarah Auni binti Nazrudin	Completed Section 3.1, Section 4.2, Section 4.3, Section 4.4 Section 4.5, Section 5.1.2, Section 5.1.10, Section 5.1. 11, Section 5.1.14, Section 6.1, Section 7	19/12/2024
	Safiya Nursyahadah binti Masnoor	Completed Section 3, Section 4.2, Section 4.3, Section 4.4, Section 5.1.3, Section 5.1.5, Section 5.1.15, Section 6, Section 6.1	19/12/2024
3.0	Dayang Farah Farzana binti Abang Idham	Correction of architectural views	31/12/2024
4.0	Farra Nurzahin binti Zaharil Anuar	Correction of ERD and Data Dictionary	14/1/2025
5.0	Farra Nurzahin binti Zaharil Anuar	Completed TC003, TC004, Section 7 sprint 4 and 5, Section 9, sprint 4 and 5	14/1/2025

	Dayang Farah Farzana binti Abang Idham	Completed TC005, TC006, Section 7 sprint 3 and 6, Section 9, sprint 3 and 6	14/1/2025
	Nawwarah Auni binti Nazrudin	Correction of architectural rationale, Completed TC007, TC008, Section 7 (sprint 7 and 8), Section 9 (sprint 7 and 8)	14/1/2025
	Safiya Nursyahadah binti Masnoor	Completed TC001, TC002, Section 7 sprint 1 and 2, Section 9, sprint 1 and 2	14/1/2025
6.0	Farra Nurzahin binti Zaharil Anuar	Correction of 2.1.1, 2.2.1, 2.3, 2.4.1, 4.1, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7, 9 Completed UC010, 8.10	12/2/2025
	Dayang Farah Farzana binti Abang Idham	Correction of 2.1.3, 2.2.1, 2.3, Section 4, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7, 9 Completed UC011, 8.7	12/2/2025
	Nawwarah Auni binti Nazrudin	Correction of 2.1.2, 2.2.1, 2.2.2, 2.3, 2.4.2, 3.1, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7, 9 Completed UC009, 8.11	12/2/2025
	Safiya Nursyahadah binti Masnoor	Correction of 2.1.2, 2.2.3, 2.4.3, 4.2, 4.3, 4.4, Section 5, 5.1, 6.2, 7, 8.2, 9	12/2/2025

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Introduction

This part of System Documentation (SD) provides detailed information about the system's development and implementation processes. This document consists of Software Requirements Specification (SRS) to outline the functional and nonfunctional requirements of the system. This to make sure what kind of software in the Koperasi KADA's system must be achieved. The comprehension of Software Design Description (SDD) is also in this document to provide essential detailed design information and architecture for the system's development. Lastly, Software Test Documentation (STD) is included in this document to describe the test plans to validate and verify the software's functionality and performance.

1.1 **Purpose**

This SD describes the requirements for Koperasi KADA's Automated System. It explains how the system will handle membership and loan management. It also describes the system's features, how users will interact with it, and any rules or limits the system must follow. To help understand the system better, this document uses user stories that describe scenarios to show different users, like staff and members, will interact with the system to complete their tasks.

This document is meant for developers, system administrators, users and stakeholder, Koperasi KADA who need to understand how the system works and how to use it effectively.

1.2 **Scope**

The software product is the Koperasi KADA automated website system which is going to be built to make managing memberships and processing loans easier for cooperative members. It contains the following main software products including membership management system module, loan management system module, report generator module and notification system module.

Firstly, the membership management system module makes sure the processing of the submission, review and approval of membership applications are secured and also provides the applicants with a real-time update. Secondly, the loan management module eases the submission and approval of loan applications that includes the loan calculator as well as real-time update. This module only prioritizes loans within the Koperasi KADA and does not support other external loans. Next, the report generator module covers the preparation of detailed reports and statements regarding loans, membership and other activities for the administrators. Lastly, the notification system module is built for sending an automated notification via applicant's email to keep members updated about their application statuses. It only covers the notification from email and not SMS.

This software is built to make managing memberships and processing loans easier for Koperasi KADA. This product ensures the improvement of operational efficiency by automating the manual processes as well as reduces errors and saves time for administrators and ensures data security for Koperasi KADA by having a built-in encryption and access control. The objective of this document is to:

I. To design a web-based system for KADA's koperasi system.
II. To develop and test the system to ensure the requirements are met.

1.3 Definitions, Acronyms and Abbreviation

This section provides the definitions, acronyms and abbreviations used in the System Documentation (SD)

- ALK : Ahli Lembaga Koperasi - Board members that responsible for decision-making
- Admin : Staff that responsible for managing and reviewing applications
- Applicant : A user that apply membership or loan through the system
- SD : System Documentation - Detailed document

Term	Definition
ALK	Ahli Lembaga Koperasi - Board members that responsible for decision-making
Admin	Staff that responsible for managing and reviewing applications
Applicant	A user that apply membership or loan through the system
SD	System Documentation - Detailed document

1.4 References

[1] A. Dennis, B. H. Wixom & D. Tegarden (2015). *System Analysis and Design, An Object-Oriented Approach with UML,* 5th Edition. [E-book]

1.5 Overview

The System Documentation (SD) for Koperasi KADA's Automated System provides a detailed description of the system's purpose, structure, and features. It includes a system introduction that defines its scope, purpose, objectives, and features. This document uses diagrams like use case, class diagram, state machine diagram to visualize the system's functionality and interactions. Additionally, it covers non-functional requirements and design constraints to ensure the system's quality, performance, and security standard.

The SD is organized into sections for better understanding. It begins with an introduction to the system's goals and requirements, followed by detailed descriptions of personas, system features, and workflows. Specific requirements are explained using user stories supported by diagrams for each use case. The document concludes with sections on performance expectations, design constraints and administrative details like version control and the project team's distributions. This structured approach ensures that the SD is user-friendly and informative for developers, stakeholders and end users.

2. Specific Requirements

This section explains the detailed software requirements necessary for building and testing the system. It provides enough information to guide designers in creating a system that meets these needs and helps testers ensure the system works as intended. Every requirement is described in a way that can be observed or experienced by users, operators, or connected systems. It includes details about all inputs, the action it performs and the outputs.

2.1 Persona

The personas for this system includes applicants, administrators, and members of the Koperasi KADA board (ALK). Each user has specific needs and responsibilities, which are described through their individual stories to provide clear understanding of the user requirements.

2.1.1 Persona 1

User Need

- Applicants need a way to apply for a membership, login into the system, apply for a loan and calculate loan.

User Stories

- As an applicant, I want to apply for a membership and login into the Koperasi Kada Automated System so that I can apply for a loan.
- As a member, I want to fill out a loan application form so that I can provide my details for loan approval.
- As a member, I want to calculate the monthly loan repayment and any applicable dividends so that I can make an informed decision.
- As a member, I want to terminate my membership application so that I can discontinue my association with Koperasi KADA.

When Applicant opens the Koperasi KADA Automated System's website, the interface shows the user's login details. After the Applicant login successfully, the applicant will be taken directly to the main page menu which includes activities such as membership application, loan application, and calculation of loan features.

If the applicant clicks on ‘Apply for Membership’, the applicants are taken to the membership form page from the menu option. The applicant has to complete details in the first part of the form and then press on the ‘next’ button to go to the next page. On the following page, applicants fill in other parts of the form and click the ‘submit’ button at the end of the page. Submission shows their membership status where applicants are informed if their application has been reviewed or not.

However, to apply for a loan, an applicant has to login as a member first. On the home page after login applicants will be directed to the loan application page where they can input the amount and type of loan, and the loan duration to get estimated monthly repayments using the loan calculator.

Additionally, in order to proceed further, the applicant has to accept the agreement by clicking on the agreement option. After that, the applicant has to give the details of a guarantor and make sure that guarantor agrees to the terms and conditions. Once an applicant completes the form and submits the application, the system will show the loan application status.

If a member wishes to terminate their membership, they can do so by selecting ‘Borang Berhenti’ under the ‘Permohonan Berhenti’ option from the menu. Once the member submitted their application, the system will show the application status.

2.1.2 Persona 2

User Need

Admin need a way to review membership and loan applications, generate an online statement and report and send automatic notifications to applicants

User Stories

- As an admin, I want to review submitted membership applications so that I can verify the applicant’s details.
- As an admin, I want to review and validate loan applications so that I can ensure it aligns with the Koperasi KADA’s loan criteria.
- As an admin, I want to review submitted membership termination so that I can verify the main reason for termination.
- As an admin, I want to manage transactions of the members of Koperasi KADA’s so that financial records remain accurate and up-to-date for reporting.

- As an admin, I want to generate online statements and reports so that I can provide correct reports' records to the stakeholders.
- As an admin, I want to send automated email notifications to applicants so that they are updated about the status of their applications.
- As an admin, I want to send financial statements to applicants so that they are updated about their financial balance.

Admin has a task on managing the system effectively including reviewing submitted membership, membership termination and loan applications, generating statements and reports as well as sending notifications via email. Admin logs in the system as admin and redirects to the admin page.

If admin chooses 'Membership List' or 'Loan Application List' or 'Membership Termination', they will be redirected to the list of the applicant's name page. Admin click on the not reviewed button to review applicant's details and make sure all details provided are correct and complete.

In both cases, they can click on the 'Complete' if the application is complete and correct and 'Incomplete' if the application is incomplete and not accurate.

If the admin approves a member through 'Approve Member by ALK,' they will be redirected to the 'Status Submission' page to send the application status to the applicant. Approved applicants can then access the 'Financial Statement' page to generate their statement.

The admin can also generate the statement by clicking on 'Generate Statement'. After that, a pop-up of an generated statement will be displayed on the admin screen to confirm the statement. Automated notifications are then sent to the approved applicant through their registered email by clicking on 'Send' button.

2.1.3 Persona 3

User Need

- ALK needs a way to approve membership and approve loan applications

User Stories

- As an ALK, I want to approve applicants' membership so that I can ensure that only eligible individuals can become membership and earn membership benefits
- As an ALK, I want to approve loan application so that I can let only the eligible applicants to receive loan in line with company's policies

- As an ALK, Iwant to approve membership termination application so that I can deactivate members' membership

ALK is responsible for membership and loan approval and the decision made is fair inline with company policies. ALK logs in to the system and then is forwarded to a personalized page for ALK.

If ALK decides to review membership applications, then the system will display a pending list for membership applications. The same applies to when ALK chose to review loan applications.

ALK then selects any applications that appear on the pending list of membership and loan approval. ALK evaluates the applications and decides for both membership and loan approval or rejection based on the company's policies. For membership, if the status of the reviewed application is 'Incomplete', then ALK will reject immediately.

In the case of loan approval, ALK is allowed to use the 'Compare' button to compare loan applications with loan terms. The system compares and displays whether or not applicants pass the loan terms. ALK also can directly approve or reject the application without using the button.

ALK is also responsible for terminating membership application approval. Similar to membership and loan application, the termination request is reviewed by Admin before forwarding it to ALK for final decision. ALK the decide to approve or reject the application.

After making decisions for membership, loan and membership termination application, the system will update the applicant's status, notify the applicants and the status is passed to admin for further report.

2.2 System Features

The Koperasi KADA Automated System is a software solution designed to enhance operation efficiency. Access to the system is possible on standard desktop and mobile devices, operating on widely used browsers and operating systems. It puts at the disposal of the staff to manage the membership, loan and termination application more effectively. In changing from manual procedures to an automated system, the new system guarantees in increasing the efficiency of operation, security of data and more user-friendly in usage for all relevant parties. The system features a real-time database, integrated for easy tracking and management applications, with ease of access both for applicants and KADA's staff. The system features are as follows ;

1. Membership Management

   Description :
   - The system supports membership registration, review membership, membership approval and record membership data.

   Requirements :
   - Allow secure submission and review of membership applications
   - Allow ALK to approve or reject membership applications
   - Allow applicants to monitor the status of their application in real-time
   - Record the data

2. Loan Management

   Description :
   - The system supports loan application, loan review and loan approval as well as record loan data.

   Requirements :
   - Allow members to submit loan application, admin to review and ALK to approve loan application
   - Allow applicants to monitor the status of their application in real-time
   - Record the data

3. Terminate Membership Management

   Description :
   - The system supports a termination membership application, review and approval as well as record data.

   Requirements :
   - Allow members to submit terminate membership applications, admin to review and ALK to approve the application.
   - Allow applicants to monitor the status of their application in real-time
   - Record the data

4. Report Generator

   Description :
   - Generate detailed report for members

Requirements :

- Allow administrators to generate detailed financial statements and reports for Koperasi KADA members focusing on savings balance and loan balance

5. Notification

   Description :

   - Automated email notification related to loan and membership status

   Requirements :

   - Sends automated email notification to applicants regarding membership and loan status of their applications
   - Sends generated financial statements through email notification

6. Financial Transaction

   Description :

   - A system for recording, updating, and reconciling financial transactions

   Requirements :

   - Allow administrators to records all saving transactions in the koperasi
   - Allow administrators to keep track of all loan repayments

7. Status Modification

   Description

   - A function to change membership and loan status

   Requirements :

   - Allow ALK to change membership status from Diterima to Ditolak and vice versa
   - Allow ALK to change loan status from Diterima to Ditolak and vice versa

2.2.1 Use Case

Figure 2.3 below is a use case that illustrates the structure of the Koperasi KADA Automated System. This use case highlights key modules such as membership application module, loan application module, review module, approval module and statement report production module. Each module covers a particular use case such as LogIn to Apply Membership (UC01) in the membership application module, Apply for Loan (UC04) and Calculate Loan and Dividend (UC05) in loan application module, Review Applicant’s Form (UC02) in review module, Approve Membership (UC03) and Approve Loan Application (UC06) in approval module, Prepare Statement and report (UC07) and Notify via Email (UC08) in statement report generation module as well as Manage Transactions (UC09) in financial tracking module. The detailed description of each module and functions is tabulated in Table 2.1.

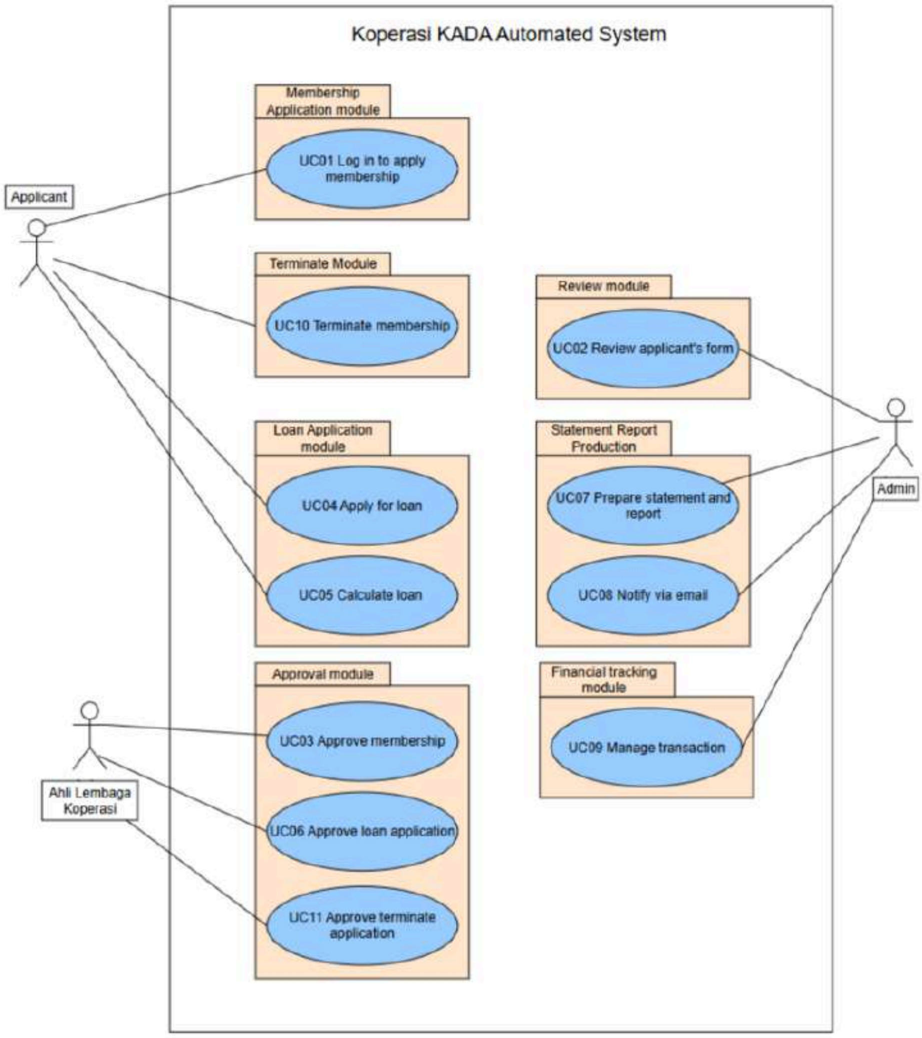


Figure 2.1: Use Case Diagram for Koperasi KADA's Automated System

Table 2.1: Description of Module and Functions for Koperasi KADA's Automated System

Use Case	Function	Description
UC01	Login to apply membership	This use case enables applicants to login, complete and submit a membership form within the system.
UC02	Review applicant's form	This use case ensures the admin to thoroughly check and validate each applicant's form before final

		submission.
UC03	Approve Membership	This use case enables Ahli Lembaga Koperasi (ALK) to approve or reject membership applications submitted by applicants.
UC04	Apply Loan	This use case enables members to submit loan applications online.
UC05	Calculate Loan and Dividend	This use case automates the calculation of monthly repayment and applicable dividends based on applicant's input.
UC06	Approve Loan Application	This use case enables the Ahli Lembaga Koperasi (ALK) staff to review and make a decision on loan applications submitted by Koperasi members.
UC07	Prepare Statement and Report	This use case allows the system to generate a detailed statement report for loan applicants based on their approved loan applications.
UC08	Notify via Email	This use case enables the system to send notifications to applicants regarding their loan statement and other important updates.
UC09	Manage Transaction	This use case enables the system to handle and record all savings and loan repayment transactions whether on a monthly or irregular basis efficiently.
UC10	Terminate Membership	This use case enables the applicant to request membership termination.
UC11	Approve Terminate Application	This use case enables the ALK to approve or reject membership termination applications for members that wish to end their membership.

2.2.2 Domain Model: Class Diagram

Figure 2.2 illustrates a domain model representing the key entities and relationships in a membership and loan management system. The model includes users, administrators, and ALK staff, who manage membership applications, approvals, and modifications. Applicants provide personal, family, and bank information when applying for membership, loans and membership terminations. The system tracks loan details, repayments, and guarantees while maintaining historical records of modifications. It also incorporates savings accounts categorized by type. Feedback mechanisms are included for application reviews. The diagram showcases the interactions and dependencies among these entities to facilitate an organized and efficient management process.

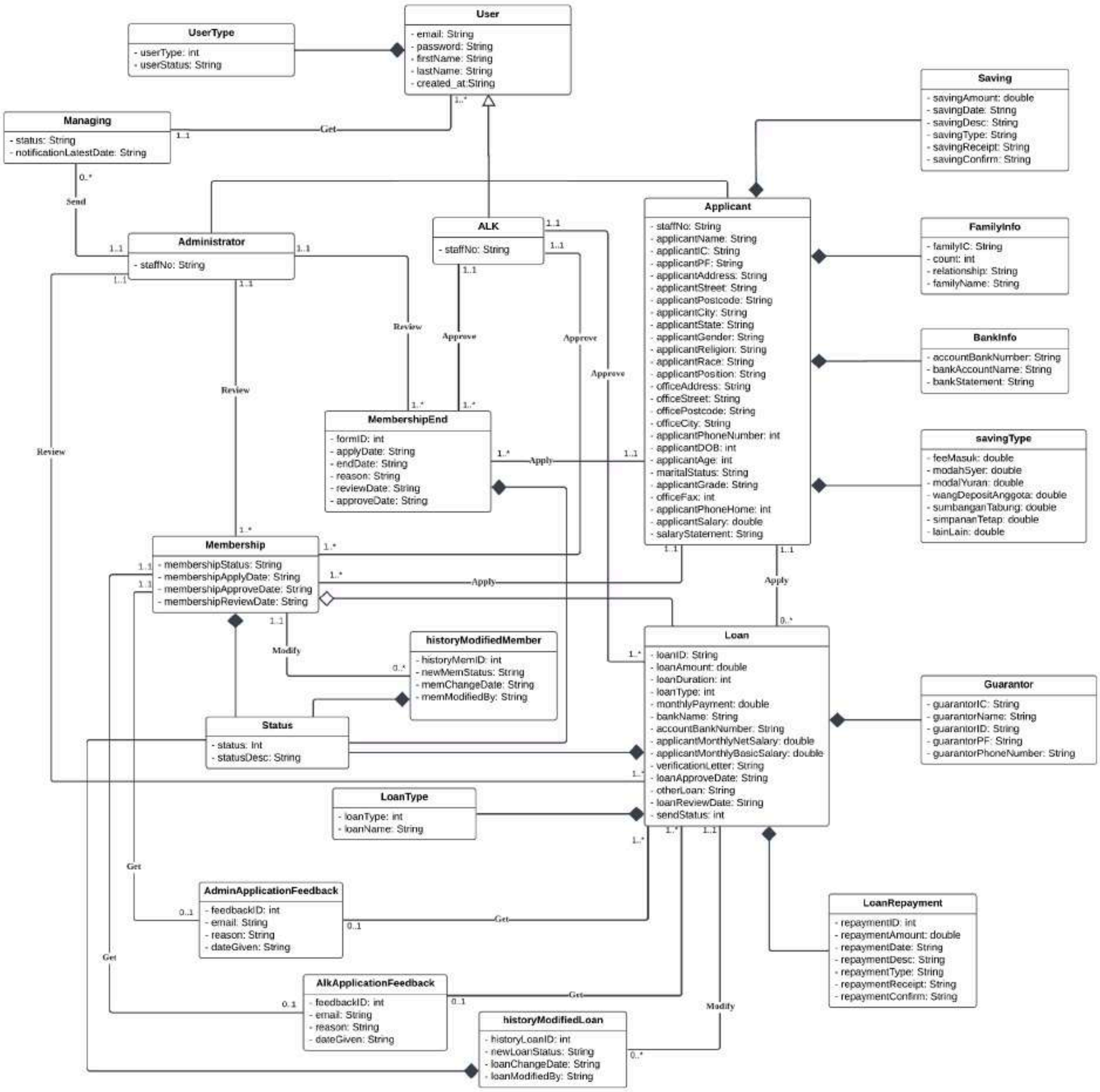


Figure 2.2: Domain Model for Koperasi KADA's Automated System

2.2.3 State Diagram

Figure 2.3 shows the Loan State Machine Diagram. It is the flow of loan application from the time an application is made to the time the final decision is made. It starts when an applicant completes the loan application and becomes the 'Loan Under Review' state where the ALK needs to approve the application. If the loan is rejected the applicant can reapply the application. If the loan has been approved, the loan status will change to ‘Loan Active’. When the loan is fully paid, the process goes to ‘Loan Completed’, which closes the loan.

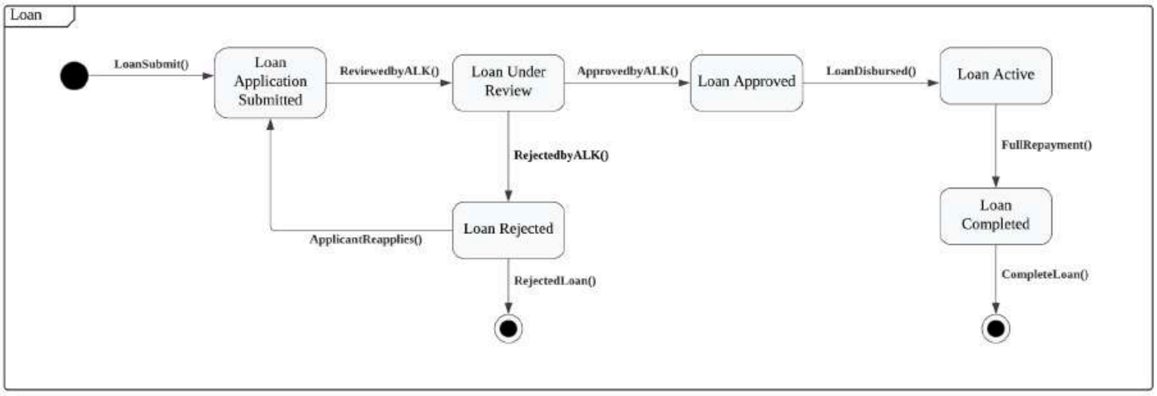


Figure 2.3: State Machine Diagram of Loan for Koperasi KADA’s Automated System

Figure 2.4 shows the Membership State Machine Diagram’s sequence of an application for membership. The process begins when a user submits the application, and it goes to ‘Membership Under Review’, the ALK either approves it or denies it. If approved, the membership becomes ‘Membership Active’. From here, applicants can allow the membership to lapse and be ‘Membership Expired’. The rejected applications enable applicants to reapply and go through the process all over again.

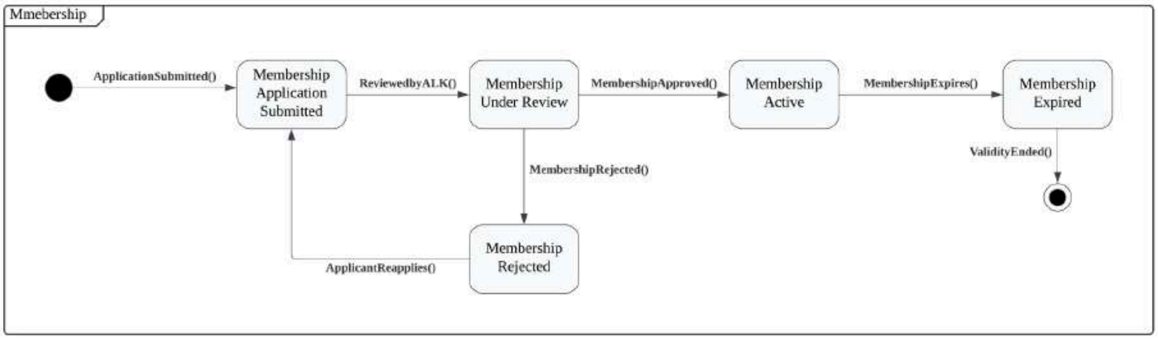


Figure 2.4: State Machine Diagram of Membership for Koperasi KADA’s Automated System

Figure 2.5 shows the Report Generation State Machine Diagram. This is the process of creating and distributing the report. It starts with ‘Report Initiated’ each time the Admin initiates the creation of a report. After being generated, the report can go to the recipient and be in the state of ‘Report Sent' or be stored in the system at the state of ‘Report Stored’. Notifications are sent with the report, and status changed to ‘Notification Created’ and when the recipient reads the notification, it ends at ‘Notification Read’

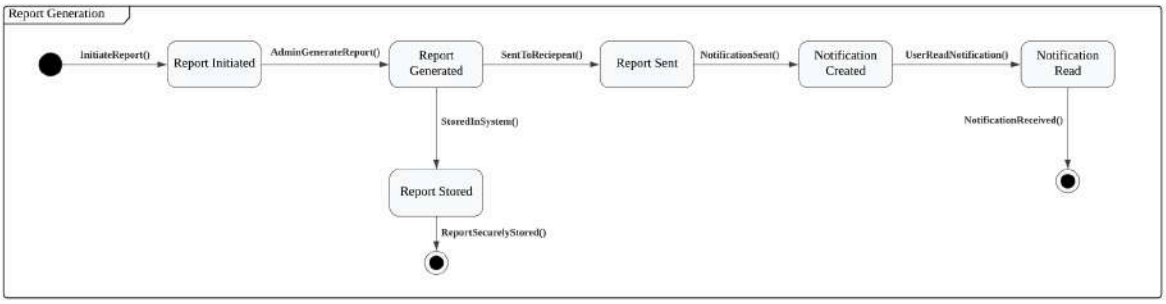


Figure 2.5: State Machine Diagram of Report Generation for Koperasi KADA's Automated System

Figure 2.6 shows the Membership Termination State Machine Diagram. The process begins when a member submits a termination request, entering the 'Membership Termination Application Submitted' state. The request is then reviewed, leading to either 'Membership Termination Approved' or 'Membership Termination Rejected'. If rejected, the applicant can reapply, restart the application process. Once approved, the membership ends and moves to 'Membership Termination Completed', concluding when validity expires.

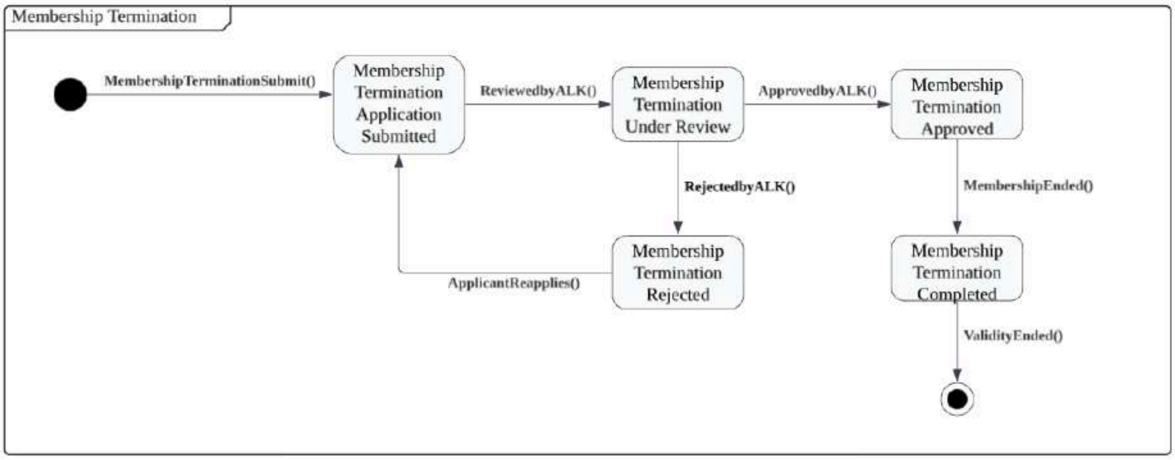


Figure 2.6: State Machine Diagram of Membership Termination for Koperasi KADA's Automated System

UC001: Use Case Login to Apply Membership

Table 2.2 user story description allows applicants to access the membership application system securely. After entering valid login credentials, users are redirected to a dashboard where they can apply for membership by filling out a detailed application form. The system verifies the submitted information and ensures all required fields are completed. If the information is valid, the application is successfully submitted. If any issues are detected, the system provides feedback to guide users in correcting their errors.

Table 2.2: Use Case Description for Login to Apply Membership

Use Case: Login to Apply Membership
ID: UC001
User Story Description **As an** applicant, **I want** to log into the membership application system, **So that** I can apply for membership.
Flow of events: 1. The Applicant navigates to the login page of the system. 2. The system prompts the Applicant to enter their username and password. 3. The Applicant enters their login details and submits the form. 4. The system verifies the login information. 4.1 If the login details are correct, the system grants access to the Applicant. 4.2 If the login details are incorrect, the system displays an error message. 5. Once logged in, the Applicant can access the membership application form.
Alternative flow *n*: n.1 : If the applicant forgets their login details, they can click on the “Forgot Password” option. n.2 : The system guides the Applicant through a password recovery process by sending a recovery link to their registered email.
Acceptance Criteria Postcondition : 1. The Applicant is successfully logged into the system and gains access to the membership application form. Precondition : 1. The Applicant must be registered in the system with a valid username and password 2. The system must be accessible for login attempts. Other conditions : 1. The system must ensure that all login details are encrypted for security purposes. 2. The login page must support both desktop and mobile devices.
Exception flow: 1. If the Applicant enters invalid login details three times, the system locks the account temporarily and prompts Applicant to use the “Forgot Password” option. 2. If the system is unavailable due to maintenance, the Applicant can see a maintenance notice and cannot login until the system is recovered.

Figure 2.7 illustrates the applicant’s login process. The applicant accesses the login page, enters their details and the system will validate the details. If the details are valid, access to the

membership application form is granted. Otherwise, an error message is displayed. The membership form is then accessed and shown to the applicant.

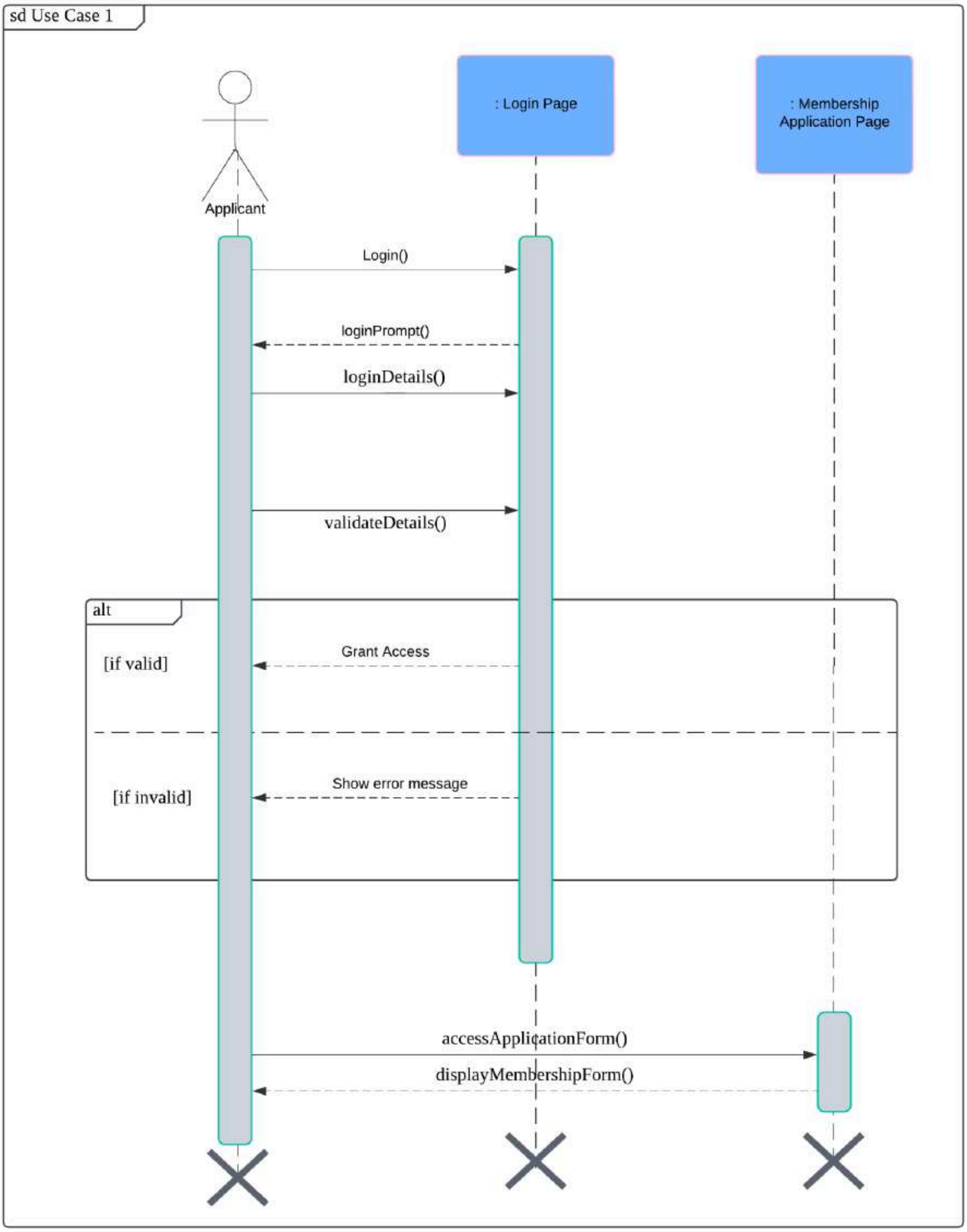


Figure 2.7: Sequence Diagram for Login to Apply Membership

UC002: User Story Review Applicant’s Form

Table 2.3 user story description allows administrators to review multiple membership, loan, or membership termination applications in batches. The process begins with the administrator selecting a batch of applications for review. Based on the review, the admin can approve or reject the application based on the provided information. For rejected applications, the system allows the admin to provide specific reasons for rejection. Approved applications are sent to the next stage of processing.

Table 2.3: Use Case Description for Review Applicant’s Form

Use Case: Review Applicant’s Form
ID: UC002
User Story Description **As an** admin, **I want** to review the membership, loan and membership termination application forms submitted by applicants, **So that** I can verify applicant details and determine whether the forms are complete or incomplete for further processing.
Flow of events: 1. The Admin logs into the system. 2. The Admin navigates to the section containing submitted membership, loan or membership termination 3. The system displays a list of all pending applications. 4. The Admin selects an application form to review. 5. The system displays the details of the selected application. 6. The Admin reviews the information provided by the Applicant.
Alternative flow *n*: n.1 : If the Admin encounters an incomplete or invalid application form : 1. The Admin flags the application for follow-up and adds a note specifying the missing or incorrect information. n.2 : If the Admin needs to pause the review and continue later : 1. The Admin saves the progress of the review without finalizing the status. n.3 : If the Admin accidentally selects the wrong application : 1. The Admin uses the “Back” button to return to the list of applications.
Acceptance Criteria Postcondition : 1. The admin successfully reviews the application form and determines the validity of applicant’s details. Precondition : 1. The Admin must have valid login details to access the system. 2. Applicants must have submitted their membership applications forms for review. Other conditions :

1. The system must allow the Admin to view all details of the selected application form. 2. The system must ensure that only pending applications are displayed for review.
Exception flow: 1. If the system is down, the Admin cannot access the application forms and is shown a maintenance message. 2. If an application form is incomplete or contains errors, the Admin flags it for further action.

Figure 2.8 shows the process of an admin reviewing membership, loan and termination membership applications. The admin logs in, navigates to the application list, and views batch application details. The system loops through each application in the batch, marking it as either “reviewed” or “incomplete”.

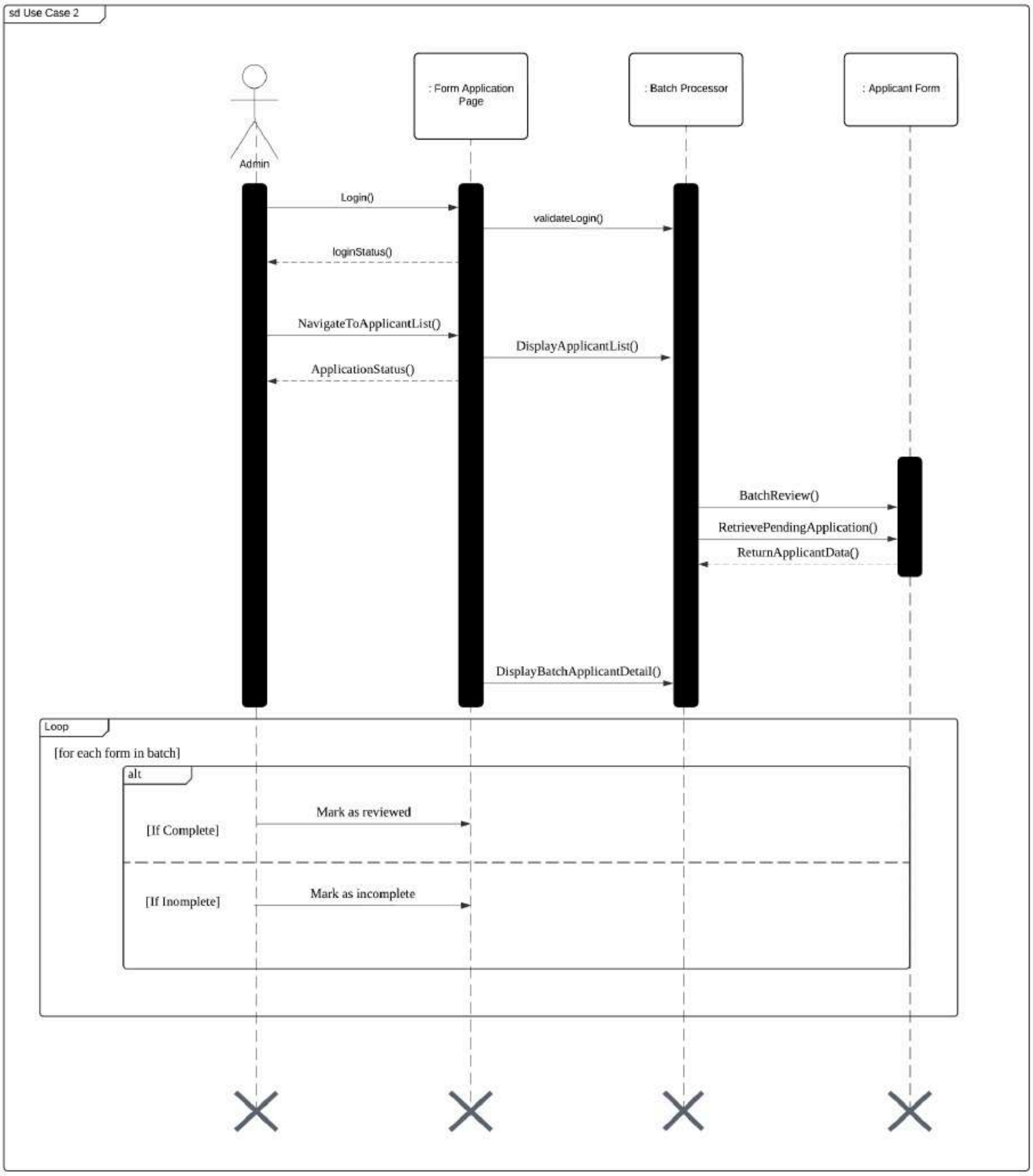


Figure 2.8: Sequence Diagram for Review Applicant's Form

UC003: Use Case Approve Membership

Table 2.4 describes the process of ALK in approving membership applications. ALK accesses the pending applications, assesses them and makes a decision on their approval or rejection, basing their judgement on the applicants' eligibility. If applications are incomplete, they are rejected. If further discussion is required, they are flagged for clarification. When a decision is reached, the system will update the applicant's status regarding membership and record this. The preconditions are that the admin must view the applications beforehand and ALK would have logged on. System shall ensure applicant documents access is secure.

Table 2.4: Use Case Description for Approve Membership

Use Case: Approve Membership
ID: UC003
User Story Description **As an** Ahli Lembaga Koperasi (ALK), **I want** to approve applicants' membership , **So that** I can ensure that only eligible individuals can become membership and earn membership benefits
Flow of events: 1. ALK accesses the pending list of membership applications 2. ALK chooses a membership applications to evaluate 3. The ALK can choose batch processing for approval simultaneous 4. ALK makes decision to approve or reject the membership application 5. ALK chooses next membership application
Alternative flow *n*: n.1 : If the application status forwarded by admin is "Incomplete", system will reject immediately n.2 : If ALK requires additional discussion, the application is flagged for further clarification in meeting
Acceptance Criteria Postcondition : 1. The applicant's membership status is updated. 2. The system records the applicant's membership status. Precondition : 1. Admin has reviewed the membership applications. 2. The ALK must be logged into the system. Other conditions : 1. The system must allow ALK to view all documents uploaded by applicants
Exception flow: 1. If the system is unavailable, both admin and ALK cannot process applications and error message will be displayed

Figure 2.9 shows a sequence diagram for use case approved membership. ALK logs into the system, which either grants access or reuters an error. ALK then selects any

membership application from the pending list, evaluates and decides whether to approve or not. ALK also can use batch processing to select more than one application for approval simultaneously. Finally, the system updates the loan status

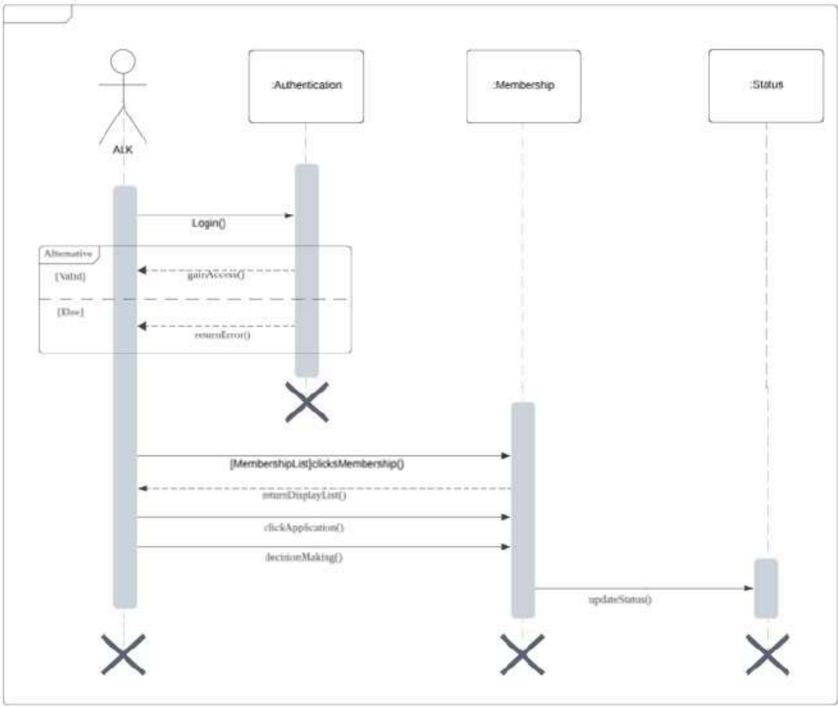


Figure 2.9: Sequence Diagram for Approve Membership

UC004: Use Case Apply Loan

Table 2.5 user story description emphasises on enabling members of Koperasi KADA to submit loan applications seamlessly through an online platform. The process starts with the applicant filling out the compulsory details in the loan application form. The system validates the application to ensure all required information is correct and complete once it is submitted. If the validation is successful, the application is stored in the loan database for further processing. Applicants will receive a confirmation message after the submission and can track their application status in real-time. This user story description plans for the data accuracy while aims to provide an outlined and user-friendly exposure for loan applications.

Table 2.5: Use Case Description for Apply Loan

Use Case: Apply Loan
ID: UC004

User Story Description
As an member of Koperasi KADA, **I want** to apply for loan through the Koperasi KADA Automated System, **So that** I can request financial support for my needs.
Flow of events: 1. The member logs into the system by accessing the login page and entering valid login credentials. 2. The member selects the “Apply Loan” option after logged in. 3. The system displays the loan application form. 4. The member completes the loan application form. 1. The member submits the loan application form to the system.
Alternative flow *n*: n.1 : If any required fields are empty or invalid, the system will ask the member to re-enter the correct or valid information before submit.
Acceptance Criteria Postcondition : 1. The loan application is successfully submitted and recorded in the system. 2. The member receives a confirmation notification. Precondition : 1. The member must be logged into the system. 2. The member must have an approved membership status. Other conditions : 1. The member must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the system detects a problem with the member’s account, an error message is displayed and the members cannot proceed with the application.

Figure 2.10 is the sequence diagram for the application loan process, representing the interaction between the applicant and the system for applying for the loan. It starts when the user logs in and accesses the loan application page. The user fills out the loan application with compulsory details, then the system validates it. If the details are correct, the application is submitted and saved into the system’s database. The system notifies the user after accepting the successful submission. This diagram illustrates the important steps involved, including user interactions, form validation and database operations.

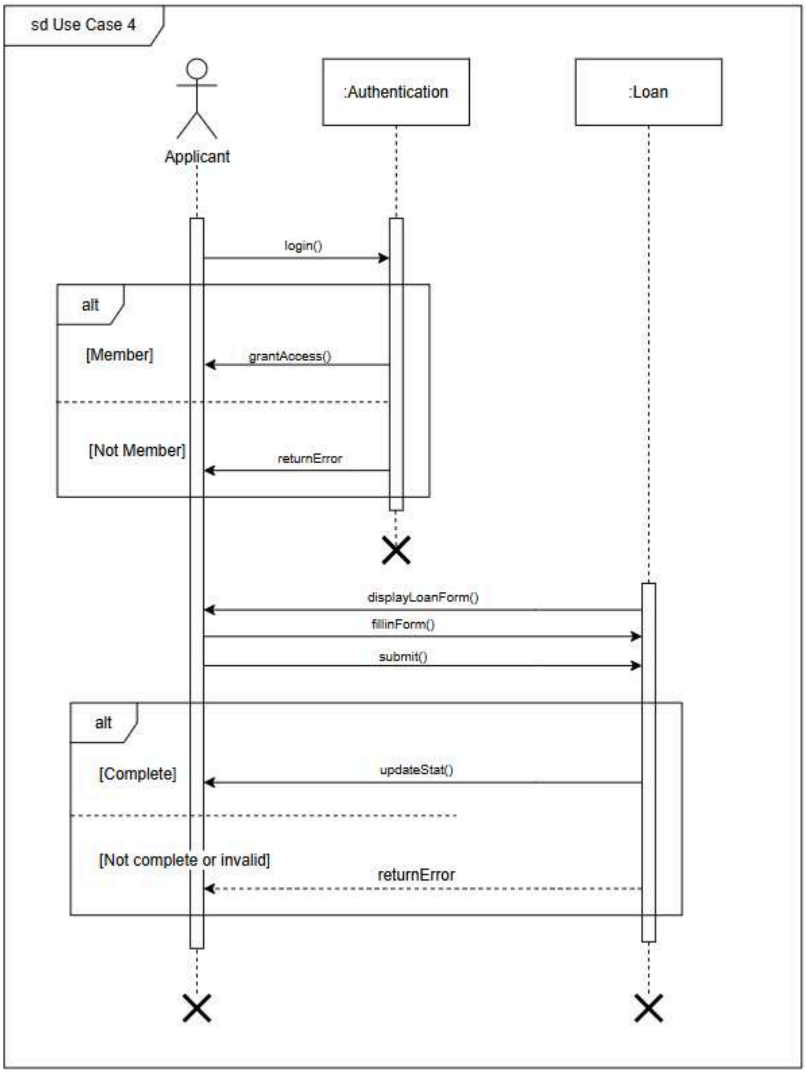


Figure 2.10: Sequence Diagram for Apply Loan

UC005: Use Case Calculate Loan

Table 2.6 focuses on the members of Koperasi KADA who want to calculate their loan repayment to know about their financial commitments and benefits. The table outlines the process that includes the member logging into the system, selects the “Loan Calculator” button features, enters the loan details like amount, duration and interest rate. Then, click the “Calculate” button and the system validates the inputs. If valid, the system displays the repayment results. If invalid, it shows an error message.

Table 2.6: Use Case Description for Calculate Loan and Dividend

Use Case: Calculate Loan
ID: UC005
User Story Description **As an** member of Koperasi KADA, **I want** to calculate my loan repayment and dividend returns, **So that** I know about my financial commitments and also benefits.
Flow of events: 1. The member logs into the system by accessing the login page and entering valid login credentials. 2. The member selects the “Loan Calculator” option after logged in. 3. The system displays the loan calculator page. 4. The member enters the loan desired amount, duration. 5. The member chooses the interest rate from the list of chosen loan types. 6. The member clicks on the “Calculate” button.
Alternative flow *n*: n.1 : If the member enters invalid amount, an error message will be displayed.
Acceptance Criteria Postcondition : 1. The system calculates and displays the amount of loan to the member. Precondition : 1. The member must be logged into the system. Other conditions : 1. The member must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the system is unable to process calculations, an error message will be displayed.

Figure 2.11 is a sequence diagram for calculating loan and dividend highlights the interaction between members and the system. The process begins with the member logging into the system by entering their details. The system then verifies the details. If valid, the system grants access to the members. If invalid, the system returns an error message. The member selects the ”Loan Calculator” button, and the system rate then submits the details. The Loan Calculator calculates and generates the loan repayment by processing these inputs. If the inputs are valid, the system displays the results. If invalid, it returns an error message that

requires the member to reenter the inputs. The interaction makes sure the member has correct financial calculations based on their provided details.

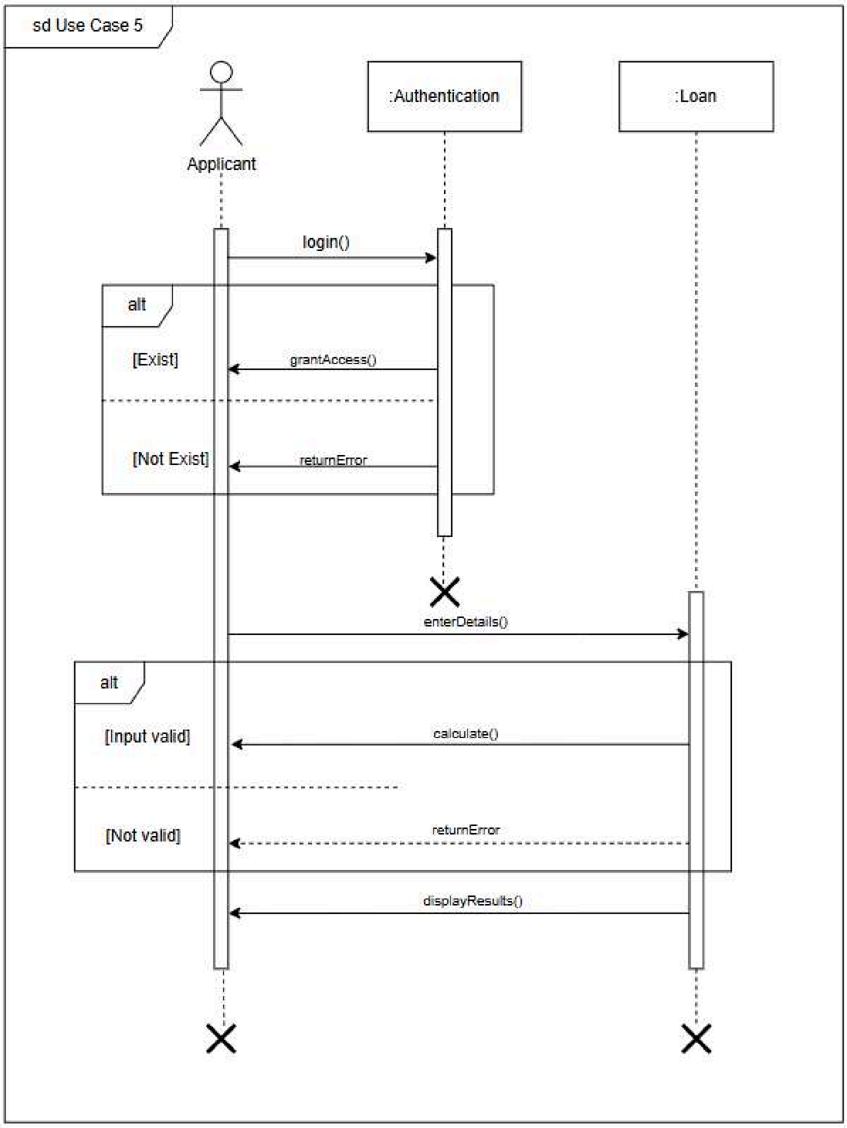


Figure 2.11: Sequence Diagram for Calculate Loan and Dividend

UC006: Use Case Approve Loan Application

Table 2.7 shows a use case description of the procedure of loan application approval by ALK. The system is viewed for pending applications by ALK, evaluates applications based on the company's policies, after which a decision to either approve, reject or use the compare feature before finalizing. Then, ALK can proceed to the next application or log out. Alternative flows include rejecting incomplete applications reviewed by admin or postponing decisions.

Postconditions ensure the loan status is updated, recorded and applicants notified. Preconditions required admin's prior review and ALK's login. Exception flows address system downtime or missing data, returning applications to admin for clarification.

Table 2.7: Use Case Description for Approve Loan Application

<table>
<tr><td>Use Case ID: UC006</td></tr>
<tr><td>User Story Description:

As an Ahli Lembaga Koperasi
I want to approve loan application
So that i can let only the eligible applicants to receive loan in line with company's policies</td></tr>
<tr><td>Flow of Events:
1. The ALK staff member accesses the list of pending loan applications.
2. The ALK selects a specific loan application to review the applicant's details
3. The ALK can choose batch processing for approval simultaneous
4. The ALK staff member evaluates the application based on established criteria.
5. The ALK makes a decision to either directly approve or reject or use the compare button to compare the applicant's form with loan terms before approving or rejecting the loan form.
6. The ALK chooses the next application or log out of the system.</td></tr>
<tr><td>Alternative Flow n:
n.1 : If the application is marked "Incomplete" by Admin, ALK will reject immediately
n.2 : If ALK need more time to review particular application, ALK will press press back button and do decision later</td></tr>
<tr><td>Acceptance Criteria:
Postcondition:
1. Loan application status is updated
2. The system record the loan application status
3. Applicants are notified of the status
Precondition:
1. The Admin has reviewed the loan applications
2. ALK must have logged into the system
Other condition:
1. All applicants' data must be encrypted and only accessible to authorized personnel</td></tr>
<tr><td>Exception Flow:
1. If the system is down, ALK cannot evaluate the applications, causing delays
2. If ALK cannot make a decision due to missing data, the application is sent back to the Admin for further discussion</td></tr>
</table>

Figure 2.12 shows a sequence diagram for use case approved loan applications. ALK logs into the system, which either grants access or reuters an error. ALK then selects any loan application from the pending list or utilizes batch processing, evaluates and decides whether to approve or not. If necessary, ALK uses the compare button to compare against loan terms before finalizing the decision. Finally, the system updates the loan status.

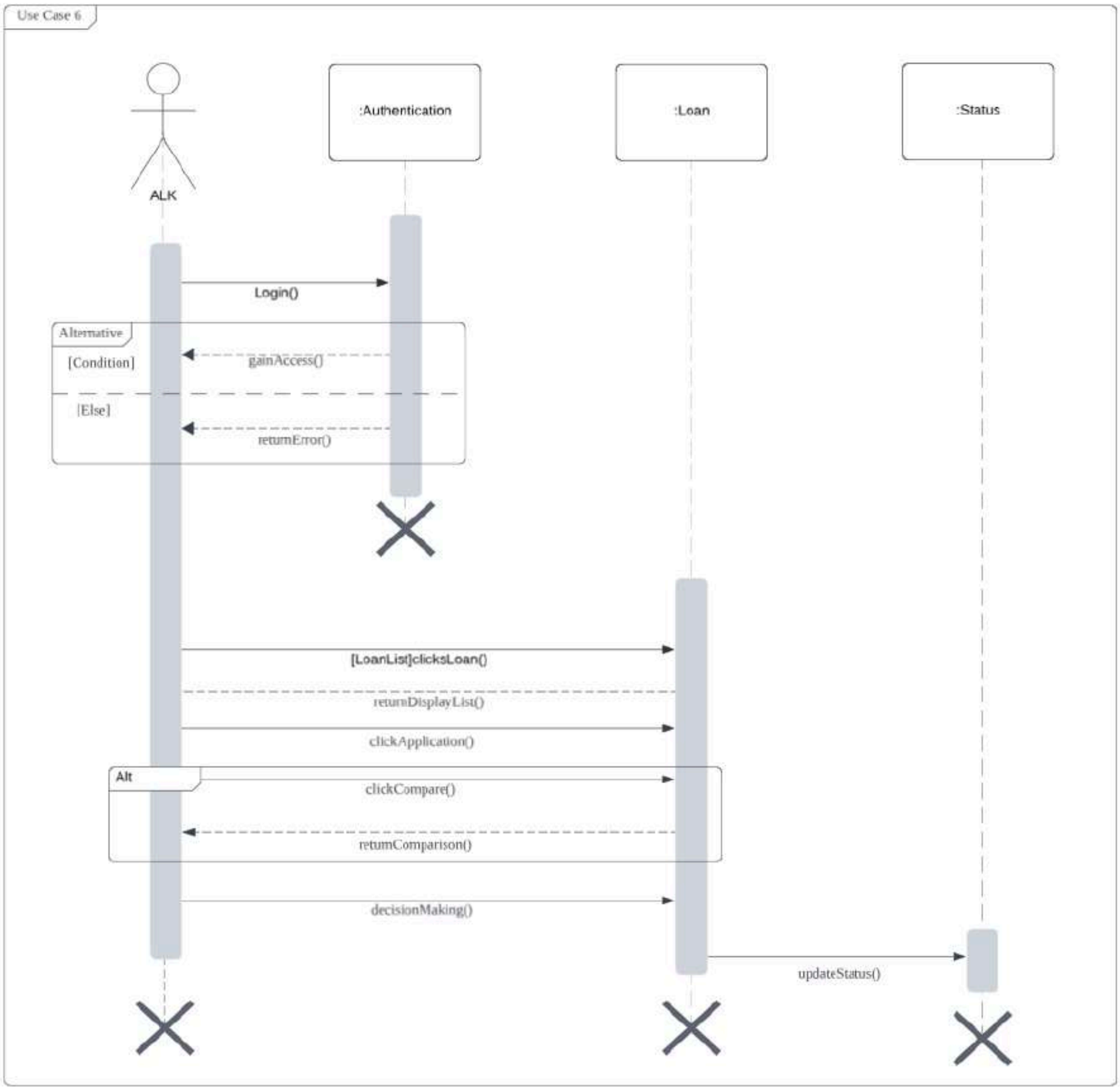


Figure 2.12: Sequence Diagram for Approve Loan Application

UC007: Use Case Prepare Statement and Report

Table 2.8 provides a detailed description of the use case for the Prepare Statement and Report feature, which describes how an admin generates statements and reports to provide applicants with details about their application status, loan history, and membership information. The admin can choose between a financial statement or a corporation report. The system retrieves relevant data from the database and processes the report based on the selected member. The generated report is then saved for future reference. If the database is unavailable or a system error occurs, the admin will be notified and given options to retry or troubleshoot the issue.

Table 2.8: Use Case Description for Prepare Statement and Report

Use Case: Prepare Statement and Report
ID: UC007
User Story Description **As an** admin, **I want** prepare statements and reports for applicants, **So that I** can provide them with detailed information about their application status, loan history, and membership details.
Flow of events: 1. Select whether the report is for financial statement or corporation report 2. If the report is for generated statement: 2.1. Admin retrieves data for approved member and loan. 2.2. The system fetches the relevant applicant data from the database. 2.3. Admin clicks the "Generate Statement" button to initiate the statement preparation process. 2.4. The system prepares the statement report based on the retrieved data. 2.5. The system displays successful report generation. 3. If the report is for corporation: 3.1. The system fetches the relevant information about membership applications, loan applications and terminate membership applications. 3.2. The system displays the information based on monthly and yearly views.
Acceptance Criteria Postcondition : 1. The statement or report is successfully generated and saved in the system for future reference. Precondition : 1. Admin must be logged into the system with the appropriate permissions. 2. Required data for the statement and report must be accessible within the system. Other conditions : 1. The system must have sufficient access control to ensure only authorized admins can generate reports. 2. The admin must be able to view and select the specific applicant for whom the statement or report is to be generated.

Exception flow: 1. If the database is temporarily unavailable, the system will display a message indicating the issue and allow the admin to retry the process later. 2. If the system fails to generate the statement or report due to system errors, the admin will be notified with a detailed error message and instructed on the next steps.

Figure 2.13 shows the sequence diagram for the process of preparing financial statements and reports by an admin, interacting with the Authentication and Report systems. The admin begins by logging in, where successful authentication grants access to the admin page, while failure triggers an error message. Once authenticated, the admin retrieves member data and generates financial statements for selected members. If statement generation is successful, a confirmation message is displayed; otherwise, an error message appears. Additionally, corporate reports containing membership, loan, and termination statistics are generated, providing monthly and yearly financial summaries.

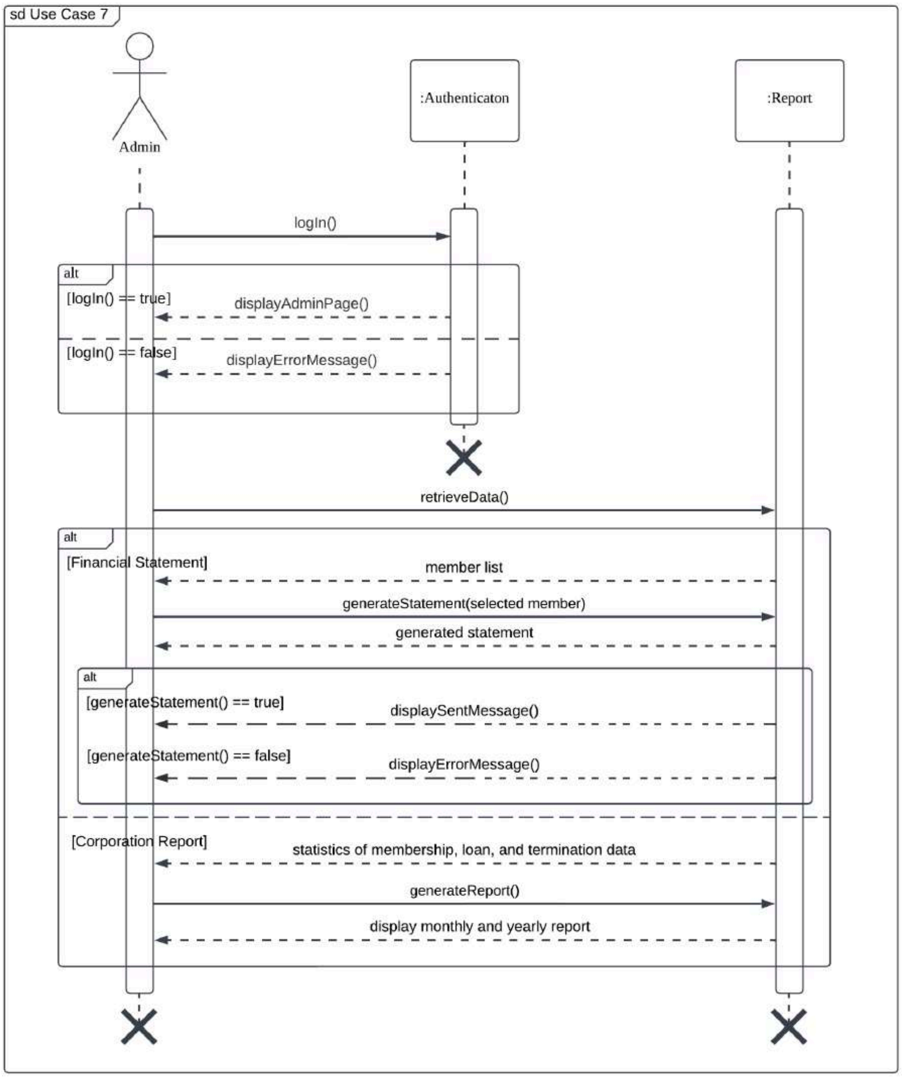


Figure 2.13: Sequence Diagram for Prepare Statement and Report

UC008: Use Case Notify via Email

Table 2.9 describes the Notify via Email use case, where the admin reviews prepared statements and sends notification emails to applicants. The system allows the admin to select whether the email notification is for a generated statement or an updated application status. Once selected, the system generates and sends the appropriate email, logging the status for tracking purposes. The system also supports batch processing for multiple applicants and handles exceptions such as failed email generation or invalid email addresses by notifying the admin for corrective actions.

Table 2.9: Use Case Description for Notify via Email

Use Case: Notify via Email
ID: UC008
User Story Description **As an** admin, **I want** to review prepared statements and send notification emails to applicant, **So that** I can ensure the applicants are informed about their loan details.
Flow of events: 1. Select whether the email notification is for a generated statement or state notification. 2. If the notification is for generated statement: 2.1. The system detects that a loan statement and report have been prepared. 2.2. The Admin reviews the prepared statement and clicks the “Send” button. 2.3. The system generates a notification email with statement details. 2.4. The system sends the email to the selected members. 2.5. The system logs the email status in the admin interface. 3. If the transaction is irregular: 3.1. The Admin tap on “Send” at desired applicants. 3.2. The system generates an email notification based on the updated status. 3.3. The system sends the email to the selected applicants.
Acceptance Criteria Postcondition : 1. Recipients are notified via email, with access to the statement or report as needed. Precondition : 1. The system must have a list of recipients with valid email addresses. 2. The statement or report has been prepared in the system. Other conditions : 1. The system must allow the Admin to view all details of the selected application form before sending the notification email.
Alternative Flow 1: 1. If the Admin wants to batch process for application status or generated statements. 2. Admin selects and processes multiple applicants at once. 3. The system should then send email notifications to each applicant.
Exception flow: 1. If the system fails to generate a notification email, it will notify the admin of the failure and allow the admin to retry sending the email. 2. If the email address is invalid, the system will prompt the admin to correct the address before retrying the process.

Figure 2.14 illustrates the sequence diagram for the use case Notify via Email, regarding prepared statements and status updates. The Admin initiates the notification process by selecting a notification type, triggering interactions between the Notification and Report components. If a Generated Statement Notification is selected, the system retrieves the

prepared statement for the selected member and sends it via email, followed by updating the notification log. Similarly, for Status Notification, an email is sent to the selected applicant, and the send status is updated. The process also supports batch processing, where multiple members or applicants are notified simultaneously. The final step verifies whether the email was successfully sent, displaying either a success or error message accordingly.

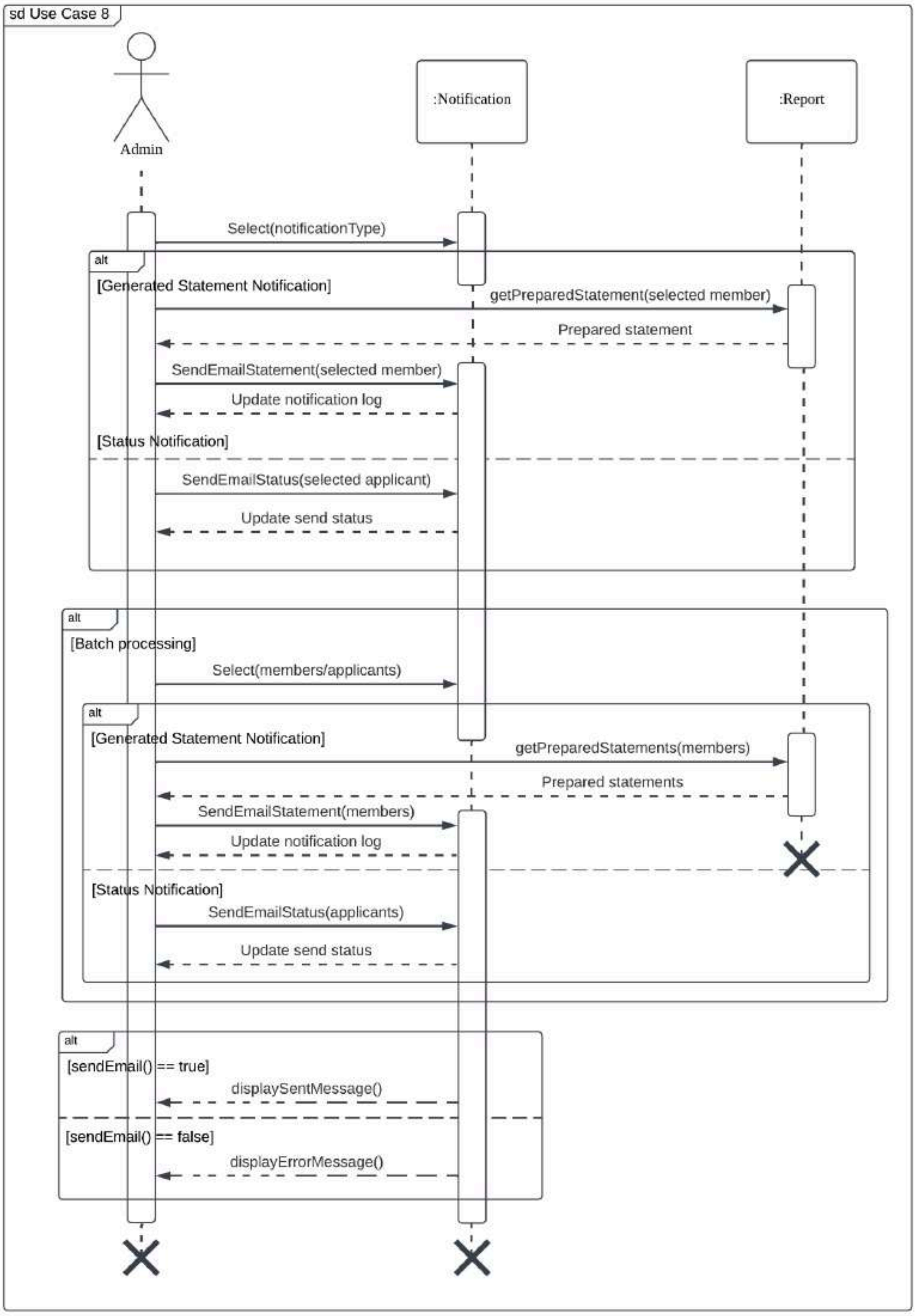


Figure 2.14: Sequence Diagram for Notify via Email

UC009: Use Case Manage Transactions

Table 2.9 describes the Manage Transaction use case. This use case explains how an admin handles and records savings and loan repayment transactions to keep financial records correct and updated. The admin can choose between monthly or irregular transactions. For monthly transactions, the admin selects a member and processes the update. For irregular transactions, the admin fills out a form and submits it. The system also allows batch processing for multiple monthly transactions. If there are missing details, incorrect member information, or system errors, the system will show a warning and prevent the transaction until fixed.

Table 2.9: Use Case Description for Manage Transactions

Use Case: Manage Transactions
ID: UC009
User Story Description **As an** admin, **I want** to manage and record all savings and loan repayment transactions, whether on a monthly or irregular basis, **So that** I can ensure the financial records are accurate and up to date.
Flow of events: 1. Select whether the transaction is monthly or irregular. 2. If the transaction is monthly: 2.1. Choose the member involved in the transaction and click "Keluarkan". 2.2. The system updates financial transactions. 3. If the transaction is irregular: 3.1. Select Savings or Loan Repayment. 3.2. Fill in the transaction form. 3.3. Submit the form for processing.
Acceptance Criteria Postcondition : 1. The transaction is successfully recorded in the system. Precondition : 1. The admin is logged into the system. 2. The membership application status is approved in the system. Other conditions : 1. Only authorized administrators can process transactions.
Alternative Flow 1: 1. If the Admin wants to batch processing for monthly transactions 2. Admin selected and processes multiple monthly transactions at once. 3. The transactions are recorded and updated.

Exception flow:

1. If required fields are missing in the transaction form, the system prompts an error.
2. If the selected member does not exist, the system prevents transaction processing.
3. If there is a system error, the transaction is logged for review.

Figure 2.15 illustrates the process of managing financial transactions by an admin, interacting with the Transaction and Financial Record systems. The admin selects a transaction type, which can be a Monthly Transaction, an Irregular Transaction, or Batch Processing Monthly. For monthly transactions, the system deducts the salary of a selected member, while irregular transactions involve selecting a member and specifying savings or loan repayment. In batch processing, multiple members are selected for financial updates. Each transaction update is confirmed, and the system displays either a success or error message based on the update status.

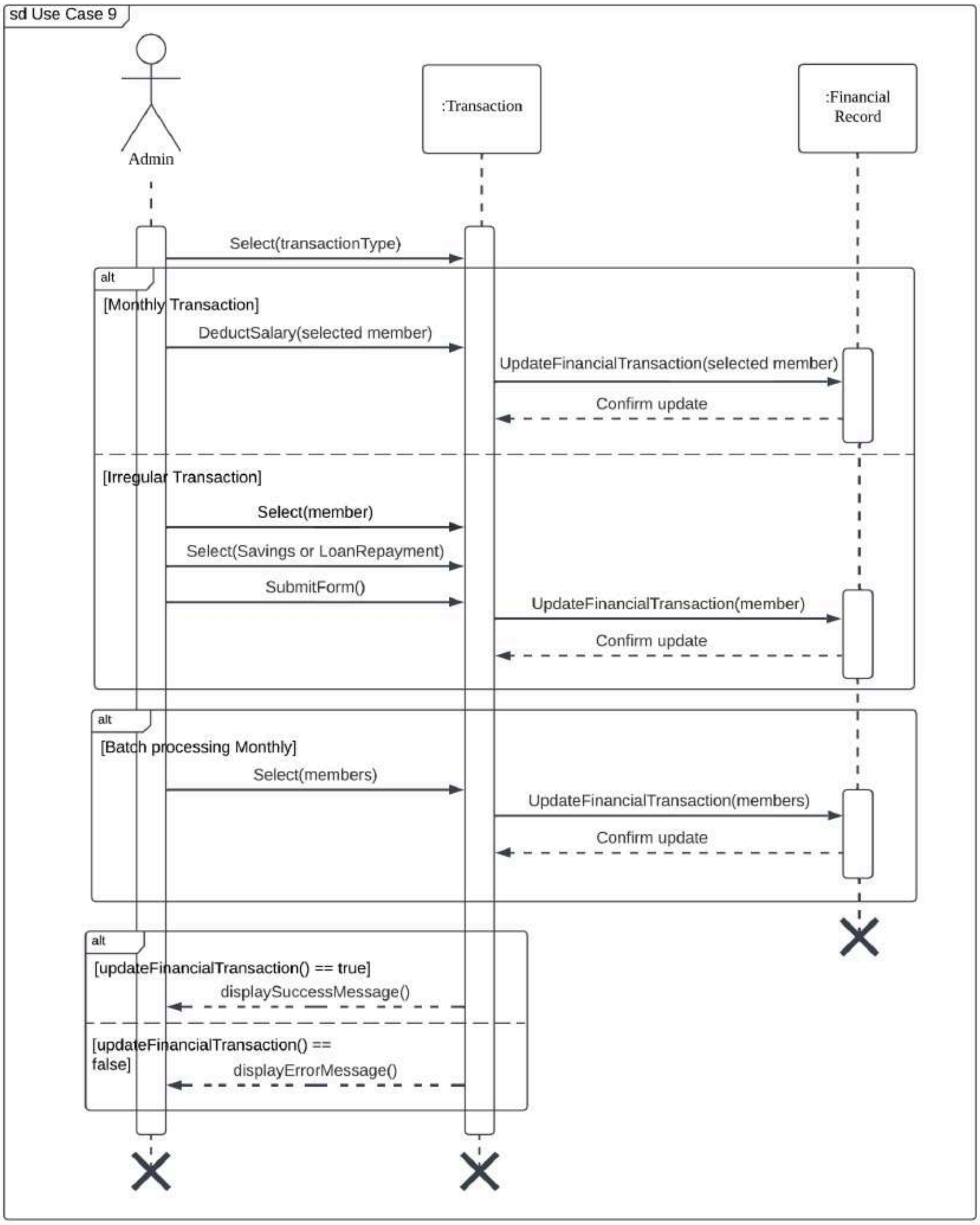


Figure 2.15: Sequence Diagram for Manage Transactions

UC010: Use Case Terminate Membership

Table 2.1 is a use case description that enables an Applicant to request termination of the Koperasi KADA membership. The Applicant submits the required details through the system, which validates the request and then forwards it to the Admin for review. Once they submit the form, the status of the application can be displayed on the status page.

Table 2.11: Use Case Description for Terminate Membership

Use Case ID: UC010
User Story Description: **As an** Applicant **I want** to terminate my membership **So that** i can discontinue my association with the cooperative
Flow of Events: 1. The Applicant logs into the system. 2. The Applicant navigates to the membership termination session. 3. The Applicant submits a termination request by providing the required details. 4. The system validates the termination request for completeness. 5. The system notifies the Applicant of the decision via email or system notification.
Alternative Flow n: n.1 : If the application is marked “Incomplete” by Admin, the system notifies the Applicant to resubmit with all required details.
Acceptance Criteria: Postcondition: 1. The membership termination request is processed. 2. The Applicant is notified of the outcome. Precondition: 1. The Applicant has an active membership. 2. The Applicant is logged into the system. Other condition: 1. All applicants’ data must be encrypted and only accessible to authorized personnel.
Exception Flow: 1. If the system is down, the Applicant cannot submit the termination request, causing delays.

Figure 2.16 is a sequence diagram for Terminate Membership use case that illustrates the interaction between the Applicant, Authentication system and Terminate Membership module. The Applicant logs in and then the system verifies their membership status. If access is granted, the termination form is displayed. The Applicant fills in and submits the form. The system will validate the request, if complete, it updates the status of application. Otherwise, it returns an error to the Applicant.

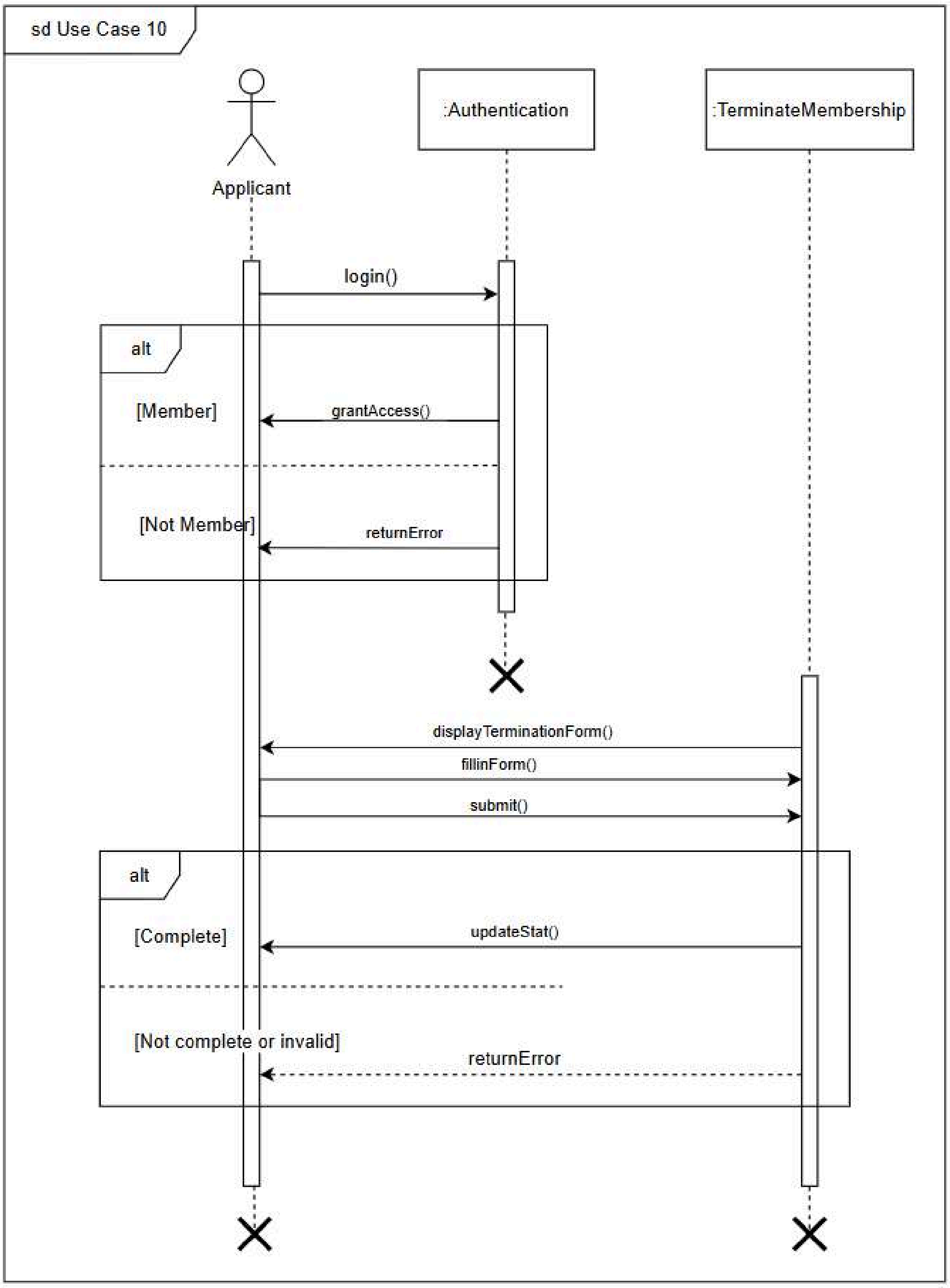


Figure 2.16: Sequence Diagram for Terminate Membership

UC011: Use Case Approve Terminate Application

Table 2.1 is a use case description that enables an ALK to approve termination membership application. The application must be reviewed by Admin beforehand. Then, ALK will make decisions based on circumstances to approve or reject applications. If the application is rejected, then the membership continues as usual otherwise the membership ends immediately.

Table 2.12: Use Case Description for Approve Terminate Application

Use Case ID: UC011
User Story Description: **As an** Alk **I want** to approve terminate application **So that** i can discontinue member's membership
Flow of Events: 1. The ALK logs into the system. 2. The ALK navigates to the terminate application page. 3. The ALK performs decision-making to approve or reject application. 4. The system notifies the Applicant of the decision via email or system notification.
Alternative Flow n: n.1 : If the application is marked "Incomplete" by Admin, the application will not be forwarded to ALK
Acceptance Criteria: Postcondition: 1. Status is updated 2. The Applicant is notified of the outcome. Precondition: 1. Application reviewed by Admin Other condition: 1. All applicants' data must be encrypted and only accessible to authorized personnel.

Figure 2.17 is a sequence diagram for the Approve Terminate Application use case that illustrates the interaction between the ALK and Terminate Membership module. The ALK will log in and then the system will display the termination application reviewed by the admin. The ALK then performs decision-making whether to approve or reject the application. Then the latest status will be updated to keep the applicant informed of real-time status.

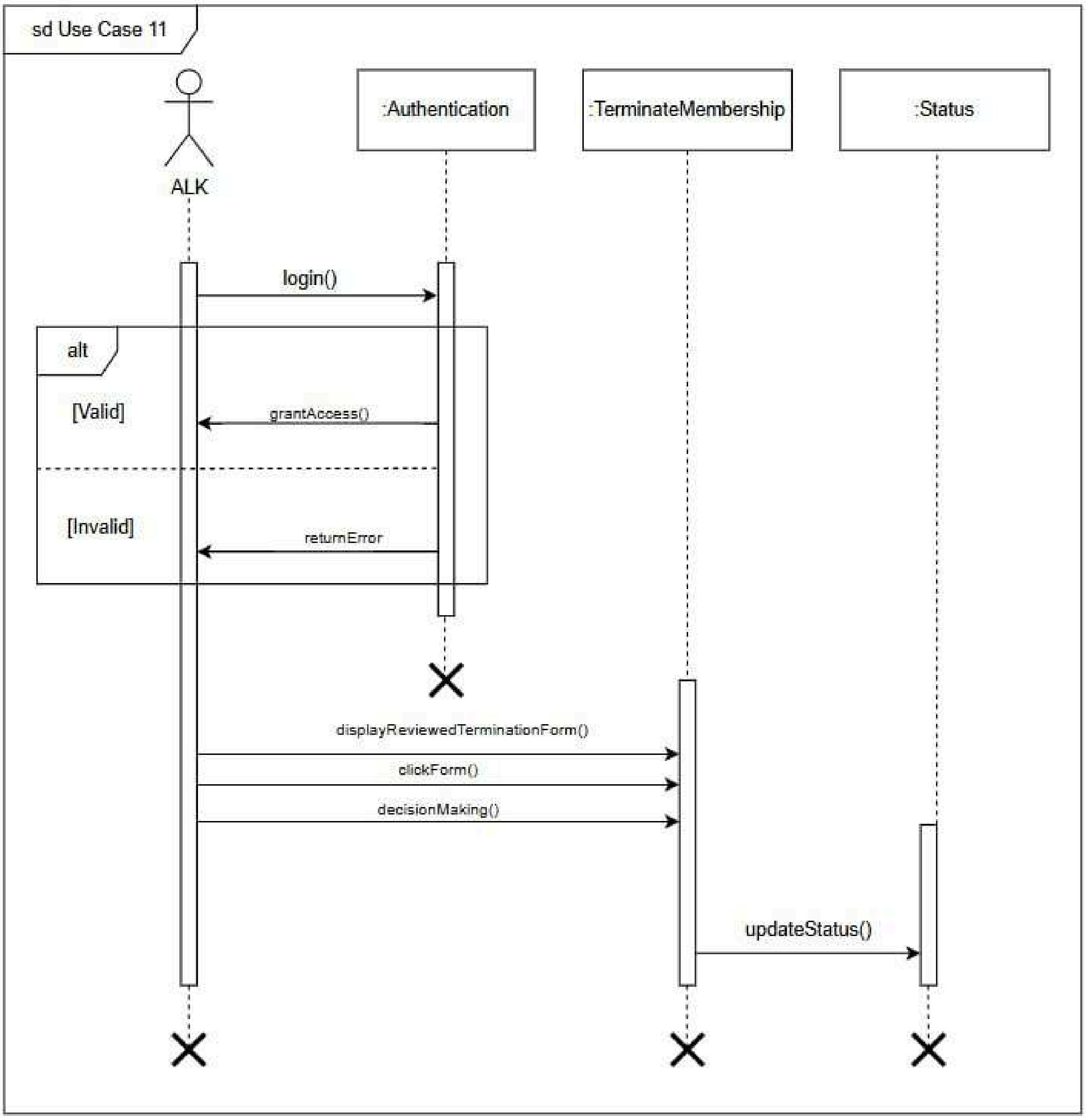


Figure 2.17: SD011 Sequence Diagram for < Approve Terminate Application> Subsystem

2.3 Launch Phase

Product backlog is a list of tasks and features that need to be done to create a product. It's arranged by priority, with the most important tasks at the top. Each task is written as a user story, explaining what a user wants from the system. These tasks are picked and worked on during sprints, which are short time periods, usually lasting one to four weeks. During sprints, team members finish tasks, update the system, and make incremental improvements. Each team member is given specific tasks or features to work on, which helps keep things organized and ensures progress is made. Table 2.13 shows the sprint sequence, user story, and which member is responsible for it for Koperasi KADA's Automated System.

Table 2.13: Product Backlog

Sprint	**Team members assigned**
Sprint #1 Applicant - Login to Apply Membership	Safiya Nursyahadah
Sprint #2 Admin - Review Applicant's Form	Safiya Nursyahadah
Sprint #3 ALK - Approve Membership	Dayang Farah Farzana
Sprint #4 Applicant - Apply Loan	Farra Nurzahin
Sprint #5 Applicant - Calculate Loan	Farra Nurzahin
Sprint #6 ALK - Approve Loan Application	Dayang Farah Farzana
Sprint #7 Admin - Prepare Statement and Report	Nawwarah Auni
Sprint #8 Admin - Notify via Email	Nawwarah Auni
Sprint #9 Admin - Manage Transaction	Nawwarah Auni
Sprint #10 Applicant - Terminate Membership	Farra Nurzahin
Sprint #11 ALK - Approve Terminate Application	Dayang Farah Farzana

2.4 User Story Details

This section represents how an applicant, an admin and an ALK would interact with the system performing their respective tasks. It consists of user story description including flow of events, alternative flows, acceptance criteria and exception flow, assuring that every expectation of a user is well handled. It includes uniquely identified user stories with supporting sequence and activity diagrams to elaborate on the process. The structured approach ensures the system is user-centered and in line with the operational objectives of the Koperasi.

2.4.1 US001: User Story Applicant

The user story in Table 2.14 describes the role and actions of the applicant in the koperasi KADA's automated system. The applicant starts by logging into the system with their username and password. If the login is successful, the applicant can access the membership application form and apply for it. Then if membership is approved, applicants can fill out and submit the loan form. If anything is missing while the applicant is filling out the membership and loan form, the system will prompt the applicant to enter the correct information. Applicants also can calculate and display the loan amount along with financial commitments and benefits. Once a member, the applicant also has the option to submit a termination application if they wish to discontinue their membership. The system will guide the applicant through the termination form to ensure all necessary information is met before finalizing the request.

Table 2.14: User Story Description for Applicant

User story: Applicant
ID: US001
User Story Description **As an** applicant, **I want** to log into the membership application system, apply for loan and calculate my loan repayment as well as submit a termination application, **So that I** can apply for membership, request financial support for my needs and to know about my financial commitments and can terminate my membership when necessary.
Flow of events: 1. The Applicant navigates to the login page of the system. 2. The system prompts the Applicant to enter their username and password. 3. The Applicant enters their login details and submits the form. 4. The system verifies the login information. 4.1 If the login details are correct, the system grants access to the Applicant. 4.2 If the login details are incorrect, the system displays an error message. 5. Once logged in, the Applicant can access the membership application form. 6. After approval, applicants can select the "Borang Pembiayaan" option. 7. The system displays the loan application form.

8. Applicant completes the loan application form. 9. Applicant submits the loan application form to the system. 10. Applicants click on “Kalkulator Anggaran Bayaran Balik” and enter input details. 11. The system displays calculated loans based on applicant input. 12. If the Applicant wishes to terminate their membership, they navigate to the “Borang Berhenti” section. 13. The system displays the termination application form. 14. The Applicant completes and submits the termination application form.
Alternative flow 1: 1. If any required fields are empty or invalid, the system will ask the applicant to re-enter the correct or valid information.
Acceptance Criteria Postcondition : 1. The Applicant is successfully logged into the system and gains access to the membership application form. 2. The loan application is successfully submitted and recorded in the system. 3. The system calculates and displays the amount of loan to the member. 4. The termination application is successfully submitted and recorded in the systems. Precondition : 1. The Applicant must be registered in the system with a valid username and password 2. The system must be accessible for login attempts. 3. For loan application, applicants must have an approved membership status. Other conditions : 1. The system must ensure that all login details are encrypted for security purposes. 2. Applicants must meet loan requirements as defined by Koperasi KADA.
Exception flow: 1. If the Applicant enters invalid login details three times, the system locks the account temporarily and prompts Applicant to use the “Forgot Password” option. 2. If the system is unavailable due to maintenance or connection, the applicant can see an error message. 3. If the system is unable to process calculations, an error message will be displayed.

The sequence diagram in Figure 2.26 illustrates the interactions between the Applicant, Authentication, Membership, Loan and Terminate Membership components in handling use cases 1, 4, 5 and 10. It begins with the Applicant attempting to log in through the login process. Depending on the outcome, either the applicant page is displayed for successful login or an error message for failure. Upon accessing the membership section, the admin can interact with the Membership component by accessing and submitting the membership form, followed by receiving membership status. For loan management, if the user is a member, the system facilitates actions like displaying the loan form, submitting it, calculating loan details, and presenting the results. If the user is not a member, an error message is shown. For termination, if a member wishes to terminate their membership, they can submit a termination request.

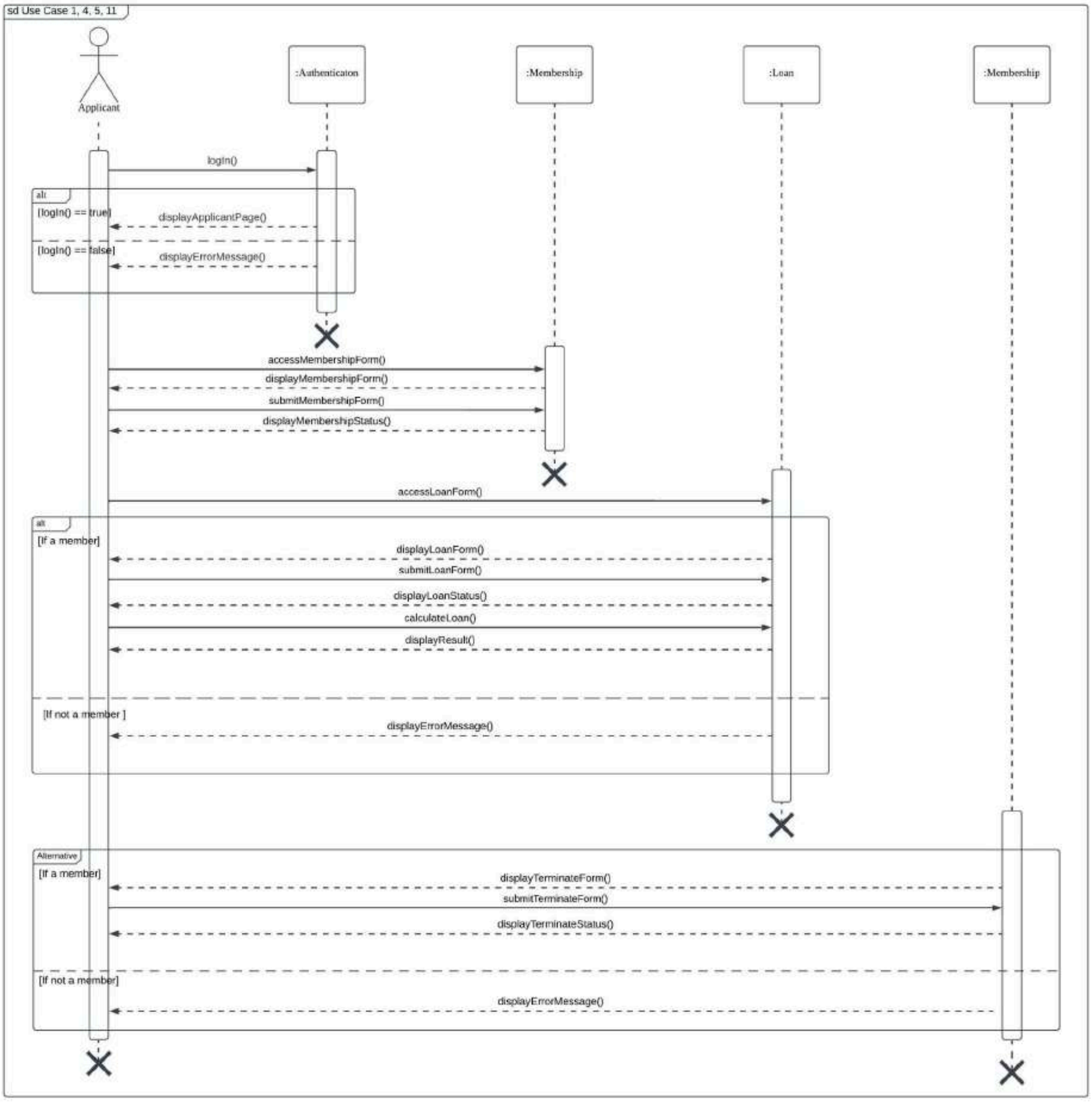


Figure 2.18: Sequence Diagram for Applicant

The activity diagram in Figure 2.19 explains the steps an applicant follows when using the system for membership and loan applications. It starts with the applicant logging in. If the applicant is not authorized, an error message is displayed, otherwise, they are redirected to the applicant page. For membership, the applicant selects "Apply Member," fills in the membership form, and submits it. If the form is incomplete, they need to re-enter the missing details. Once complete, the system displays the membership status. For loan applications, the applicant selects "Apply Loan," fills in the loan form, and submits it. If the form is incomplete, they re-enter the missing details. Afterward, the applicant can use the loan calculator to input loan details. If the input is valid, the system calculates and displays the calculated loan and dividend. For termination, the applicant selects “Borang Berhenti”, fills in the termination form and submits it.

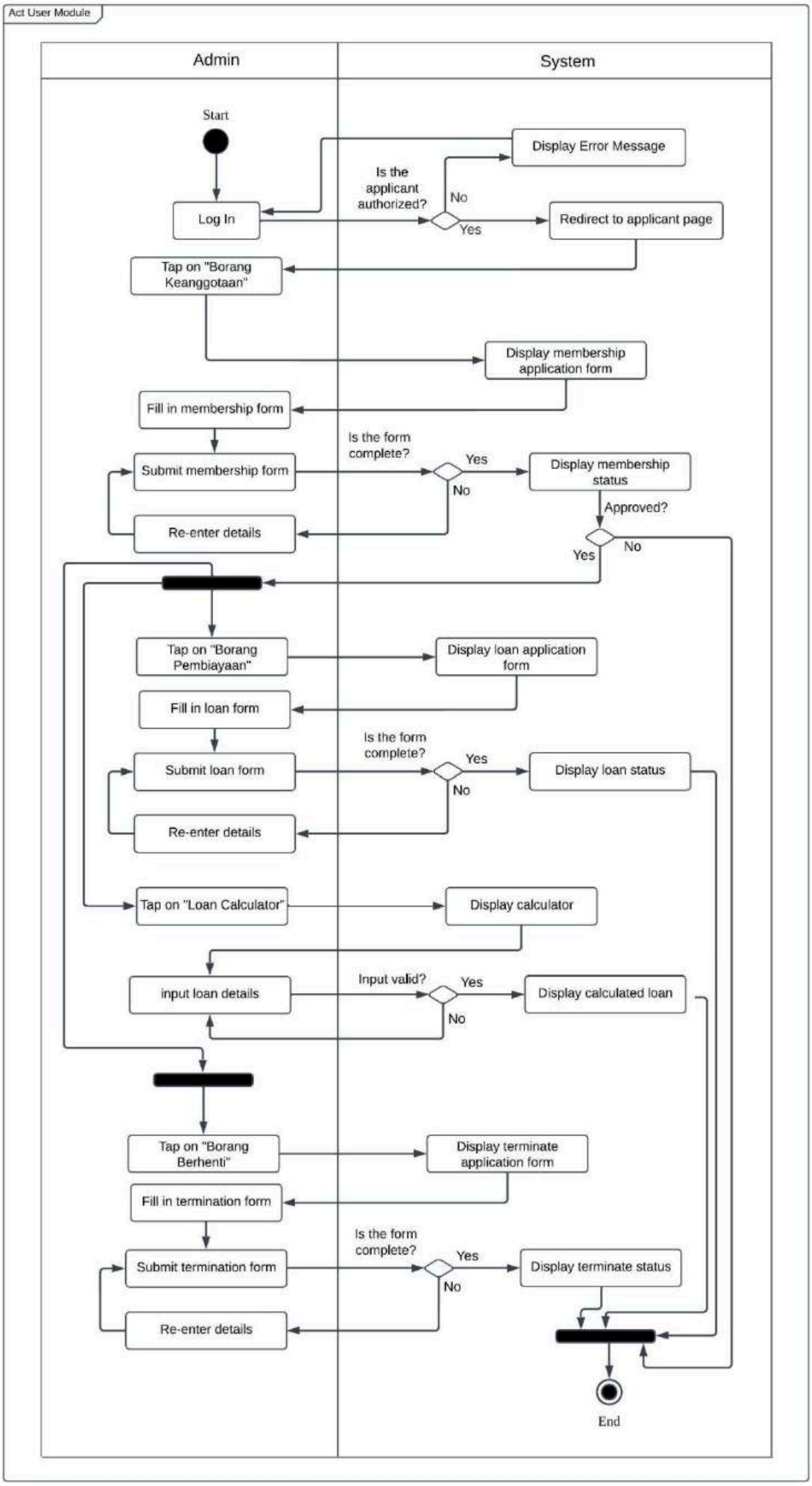


Figure 2.19: Activity Diagram for Applicant

2.4.2 US002: User Story Admin

Table 2.15 is a user story description for an admin managing the system to handle tasks such as reviewing membership and loan application, generating statements and sending automated notifications. The admin starts by logging into the system where their login details are verified. If the login details are correct, the admin is granted access to the admin page. If incorrect, an error message is displayed. The admin can select from various options at the admin page once logged in. If the admin chooses to review applications, they can click to either the 'Membership List' or 'Loan Application List'. If the admin selects the 'Approve Member by ALK', they can generate an online statement. This process includes creating the statement by clicking 'Generate Statement' and then sending an automated notification to the applicant by clicking 'Push email to applicant'.

Table 2.15: User Story Description for Administrator

User story: Administrator
ID: US002
User Story Description **As an** Admin, **I want** to review and process membership applications, loan applications, and membership termination requests, as well as manage transactions, prepare statements, and send notifications, **So that** I can ensure the accuracy, completeness, and proper handling of financial, membership and loan related processes.
Flow of events: 1. The Admin logs into the system. 2. The Admin navigates to the section for reviewing membership applications, loan applications, or termination requests. 3. The system displays a list of pending applications. 4. The Admin selects an application to review. 5. The system displays the application details. 6. The Admin verifies the provided information. 7. If the application is complete and valid, 7.1. it is marked as approved. 8. If the application is incomplete or contains errors, 8.1. it is flagged for follow-up. 9. The Admin prepares statements and reports based on applicant data. 10. The Admin generates and sends notification emails regarding application status and financial statements. 11. The Admin manages and records financial transactions, whether monthly or irregular.
Acceptance Criteria Postcondition :

1. The application review process is completed, and the applicant's details are verified. 2. Statements and reports are successfully generated and saved. 3. Email notifications are sent to applicants. 4. Financial transactions are recorded accurately. Precondition : 1. The Admin must have valid login details to access the system. 2. Applicants must have submitted their membership and loan applications forms for review. 3. Required data for the statement and report must be accessible within the system. 4. The system must have a list of recipients with valid email addresses. 5. The statement or report has been prepared in the system. Other conditions : 1. The system must allow the Admin to view all details of the selected application form. 2. The system must ensure that only pending applications are displayed for review. 3. The system must have sufficient access control to ensure only authorized admins can generate reports. 4. The admin must be able to view and select the specific applicant for whom the statement or report is to be generated.
Exception flow: 1. If an application form is incomplete or contains errors, the Admin flags it for further action. 2. If the system fails to generate the statement or report due to system errors, the admin will be notified with an error message. 3. If the system fails to generate a notification email, it will notify the admin of the failure and allow the admin to retry sending the email.

Figure 2.20 shows the sequence of steps when an admin uses the system, from logging in to managing applications, transactions, and notifications. First, the admin logs in, and the system checks if the details are correct. If valid, the admin can view pending applications, review them, whether the application is completed or uncompleted. After that, the admin can process financial transactions, update records, and generate reports. If everything is successful, a confirmation message appears. The system also allows sending notifications to applicants. If something goes wrong, like a failed transaction or error in generating a statement, an error message is shown to inform the admin.

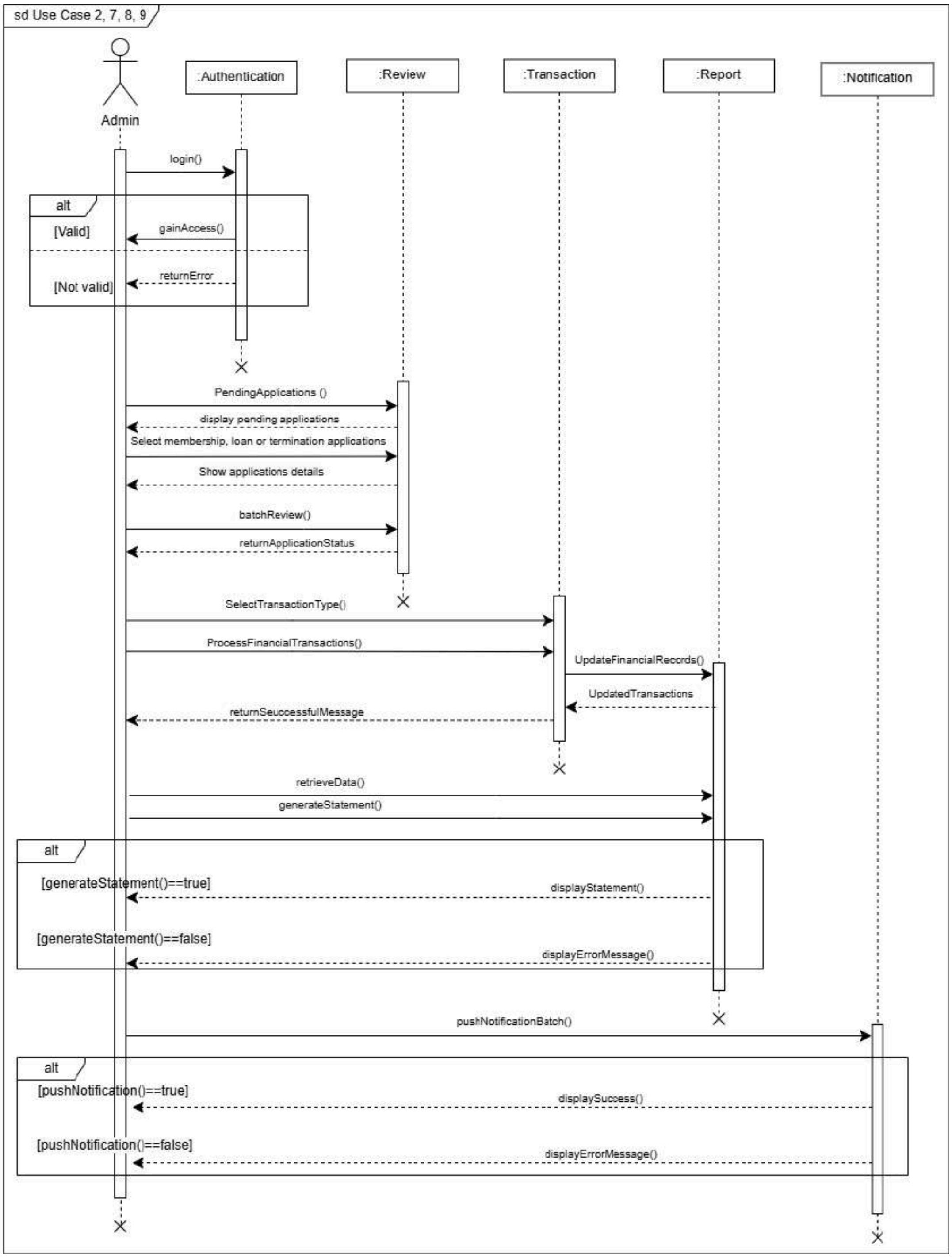


Figure 2.20: Sequence Diagram for Admin

Figure 2.21 is an activity diagram that presents the flow of how an admin manages user applications and financial transactions. The process starts with the admin logging in, where the system verifies the credentials. After logging in, the admin can review membership or loan applications, generate statements, or send notifications to applicants. Applications are either marked as complete or incomplete based on the provided details during the review. If an application is approved, the admin can proceed with financial transactions, and the system updates the records accordingly. Once the transactions are completed, the admin generates statements, and the system prepares a confirmation for review before sending them to approved members. Notifications are then sent to applicants, informing them about their application status or financial updates. If any step fails, the system displays an error message to guide the admin. The diagram ensures a clear and organized process for handling all admin tasks efficiently.

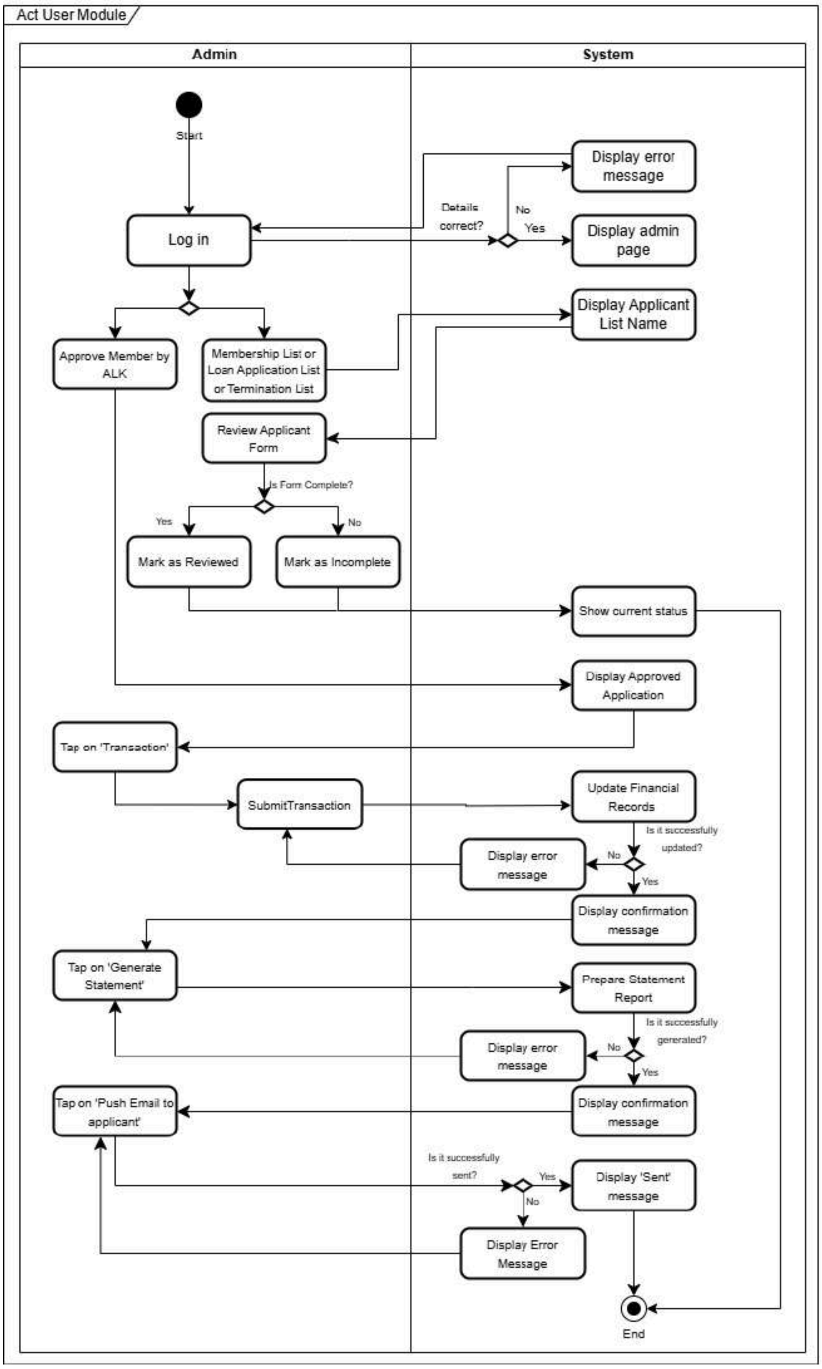


Figure 2.21: Activity Diagram for Admin

2.4.3 US003: User Story Ahli Lembaga Koperasi (ALK)

Table 2.16 shows a process of ALK in the membership,loan and membership termination approval. In this system, ALK will log in, choose a membership application, loan application, or membership termination application to review the pending applications for approval or rejection. An incomplete application marked by Admin is directly rejected by the system, and applications which need further discussion are marked for meeting actions. Once the decisions are made, the system updates and records the statuses. The Table also states the precondition that Admin has to go through the applications beforehand.

Table 2.16: User Story Description for ALK

User story: ALK
ID: US003
User Story Description **As an** Ahli Lembaga Koperasi (ALK), **I want** to approve applicants' membership and loan application, **So that** I can ensure that I can ensure only eligible individuals can become membership and receive loan from the company
Flow of events: 1. ALK logs into the system 2. ALK choose to evaluate membership, loan application or terminate membership application 2.1. If ALK chose membership 2.1.1. ALK accesses the pending list of membership applications 2.1.1.1 If ALK choose batch processing 2.1.1.2 ALK tick checkbox 2.1.1.3 ALK approve or reject selected applications 2.1.2. ALK selects any application from the pending list 2.1.3. ALK evaluate the application 2.1.4. ALK decides to approve or reject the membership application 2.1.5. ALK chooses next membership application 2.1.6. ALK log out of the system 2.2. If ALK chose loan 2.2.1. ALK accesses the pending list of loan applications 2.2.1.1 If ALK choose batch processing

2.2.1.2 ALK tick checkbox 2.2.1.3 ALK approve or reject selected applications 2.2.2. ALK selects any application from the pending list 2.2.3. ALK evaluate the application 2.2.4. ALK decides to approve or reject the loan application 2.2.5. ALK chooses next loan application 2.2.6. ALK log out of the system 2.3. If ALK chose membership terminate application 2.3.1. ALK accesses the pending list of membership terminate applications 2.3.1.1 If ALK choose batch processing 2.3.1.2 ALK tick checkbox 2.3.1.3 ALK approve or reject selected applications 2.3.2. ALK selects any application from the pending list 2.3.3. ALK evaluate the application 2.3.4. ALK decides to approve or reject the membership terminate application 2.3.5. ALK chooses next membership terminate application 2.3.6. ALK log out of the system
Alternative flow *n*: n.1 : If the application status forwarded by admin is “Incomplete”, system will reject immediately n.2 : If ALK requires additional discussion, the application is flagged for further clarification in meeting
Acceptance Criteria Postcondition : 1. The applicant’s membership and loan status is updated. 2. The system records the applicant’s membership and loan status. Precondition : 1. Admin has reviewed the membership and loan applications. Other conditions : 1. The system must allow ALK to view all documents uploaded by applicants
Exception flow: 1. If the system is unavailable, both admin and ALK cannot process applications and error message will be displayed 2. If ALK cannot make a decision due to missing data, the application is sent back to the Admin for further discussion

Figure 2.22 is a sequence diagram that shows the interaction between ALK and the system components. The process starts when ALK logs into the system. The system checks the

validity and grants access or returns an error. After logging in, ALK can either access the Membership, Loan and Membership Termination Module. In the Membership module, ALK starts this process by selecting the Membership List. The system opens a list of pending membership applications. ALK opens an application by selecting, makes a decision to approve or reject and the system updates the membership status. ALK also can utilize batch processing by selecting intended membership applications to be approved or rejected simultaneously. In the loan module, ALK selects Loan List, then reviews a pending loan application and may want to compare the application details with loan terms. After that, the system provides the comparison results. ALK then approves or rejects the loan application, then the system will update the status of the loan application. ALK is also allowed to use batch processing for the approval process. In the Membership Termination Module, ALK starts by choosing the Termination Application List. The system displays a list of pending membership applications. ALK opens an application by selecting, makes a decision to approve or reject and the system updates the membership status. ALK also can use batch processing by selecting intended membership termination applications to be approved or rejected simultaneously.

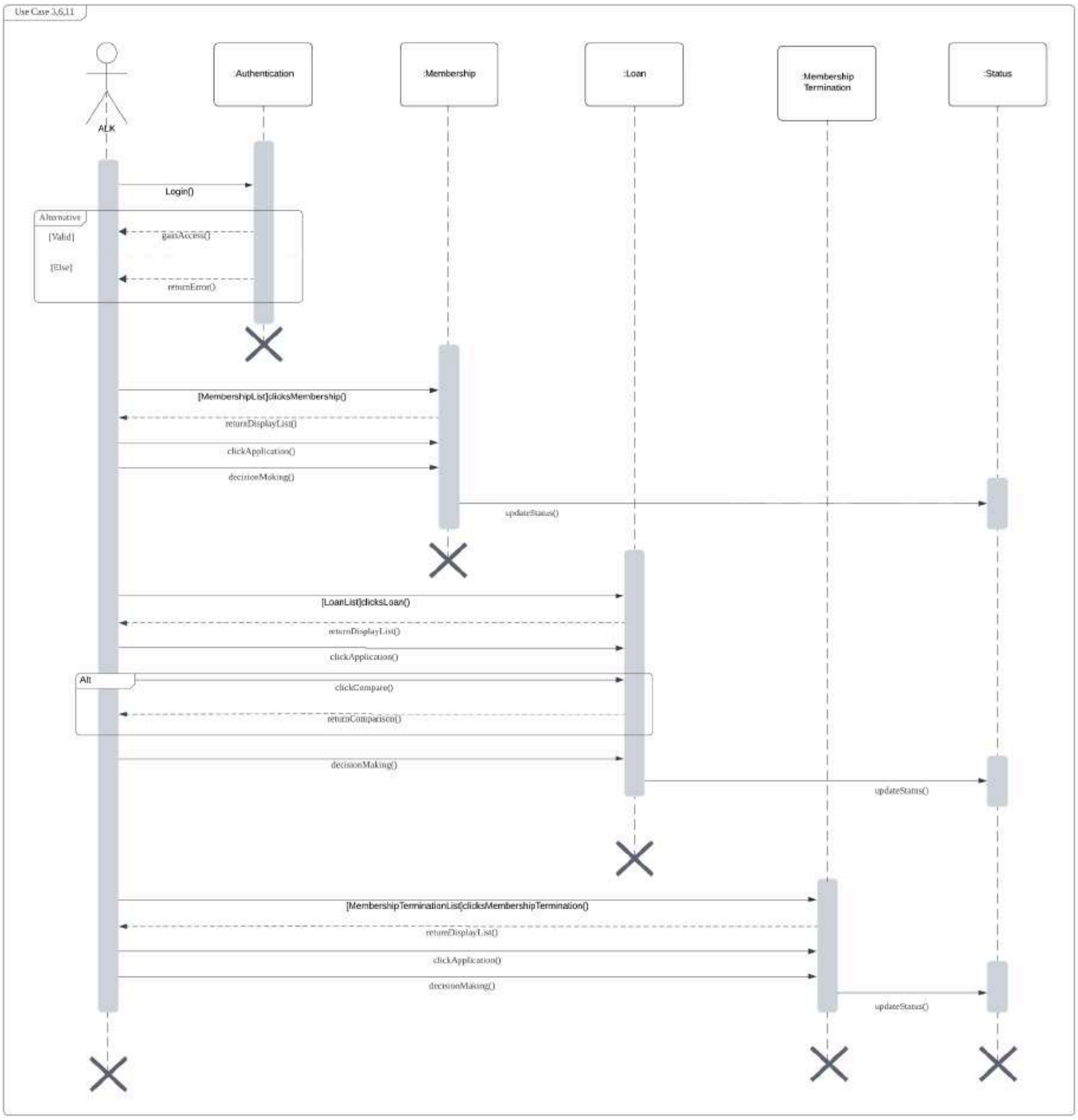


Figure 2.22: Sequence Diagram for ALK

Figure 2.23 shows the activity diagram of the workflow between ALK and the system for managing membership and loan applications. ALK logs in, and upon successful access, selects to review membership, loan or membership termination applications. In the case of membership, ALK will assess pending applications, approve or reject them and the system will update the status. For loans, ALK may directly assess or compare to evaluate. The system will show the result of comparison to assist ALK in decision making. Approved or rejected statuses are recorded, thus ensuring systematic and informed application management. For membership terminations, ALK will assess pending applications, approve or reject the applications and the status will be updated into the system.

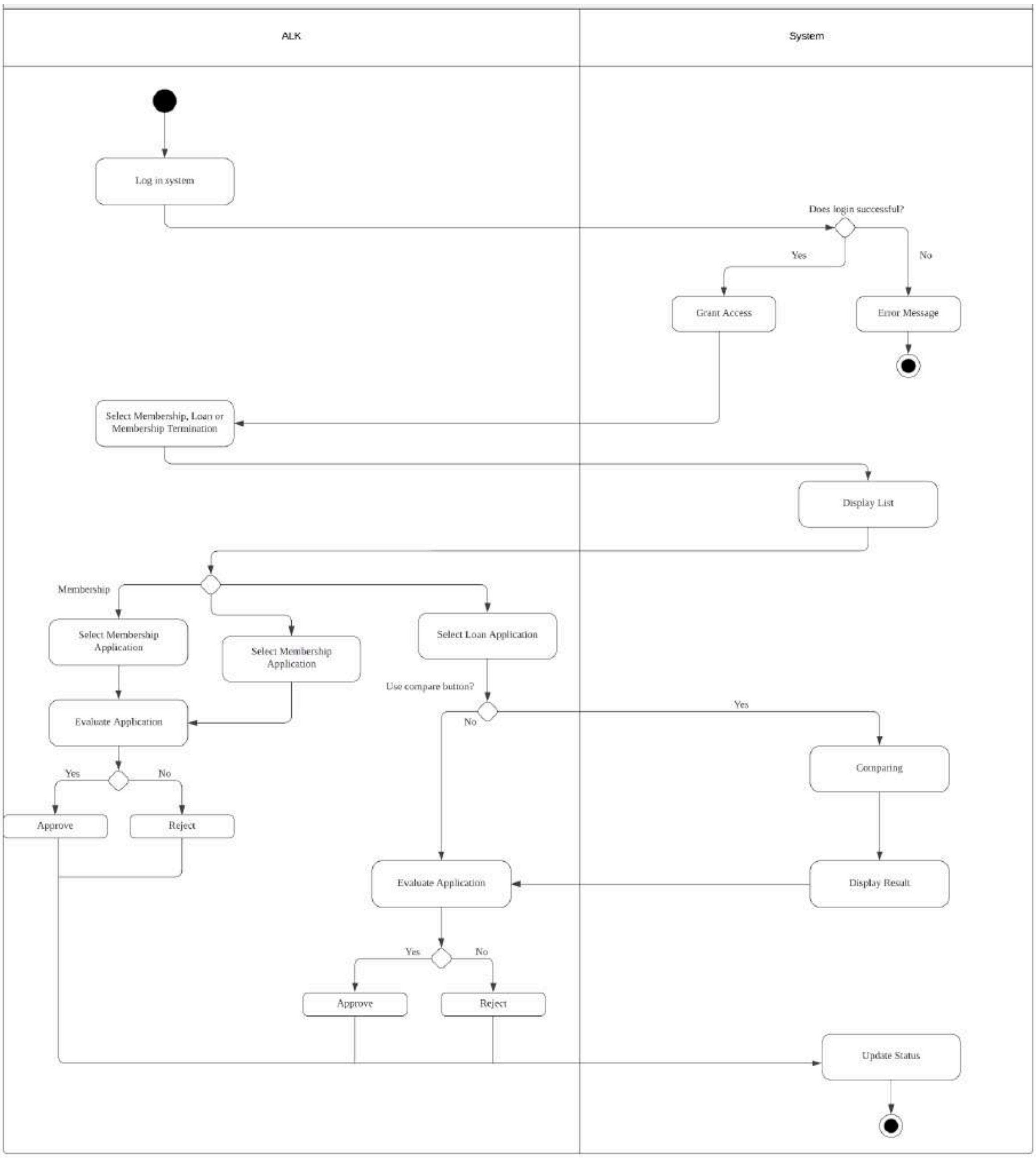


Figure 2.23: Activity Diagram for ALK

2.4 Performance and Other Requirements

Non-functional requirements for this project system define how the system should work and focus on that rather than show its specific functionalities. It is essential to have non-functional requirements to make sure the system fulfills quality standards and can work in various environments. These requirements are classified into Software System Attributes, Performance and Other Requirements according to the ISO/IEC/IEEE 29148 standard.

Performance requirements specify how quickly the system can handle data and react to user requests. These are requirements that belong to the system's performance under different conditions such as:

- Response time: The response time for any user request should not exceed 2 seconds under normal operating conditions.
- Throughput: The system should handle a significant number of concurrent users without performance degradation to ensure timely processing of membership and loan applications.
- Capacity: The system must store a large number of member records and loan applications while maintaining optimal performance.
- Availability: The system should remain accessible and functional even under interruptions or heavy loads.

Software System Attributes define the overall quality and behavior of the software. These attributes include:

- Usability: The system's interface should enable new users to complete tasks without requiring a lot of training.
- Reliability: The system must secure data consistency and correctness, with a monthly downtime tolerance of no more than 0.5%.
- Maintainability: The system should enable updates such as bug fixes by laying objectives to finish major updates in a week and minor updates in 1 to 2 days.
- Portability: The system's software needs to work with a variety of hardware and software operating systems, including different Windows and macOS versions.
- Safety: The system includes specific safety requirements to mitigate potential loss, damage, or harm associated with its use, particularly in handling sensitive financial and personal data.
- Security: The system will require multi-factor authentication (MFA) to help prevent unauthorized access.

2.5 Design Constraints

The design constraints for the system focus on ensuring it works efficiently with standard computers and devices, while maintaining security for user data. It should be compatible with common hardware and across different operating systems and devices. Additionally, the system should support multiple users and provide quick access to information.

- **Hardware Constraints:** The system must be designed to operate on standard desktop computers with a minimum of 8GB of RAM and a dual-core processor to ensure efficient performance. It should also be compatible with existing printers and scanners to make document management easy. The system will need to have the proper USB or network ports to connect with these devices without any issues.
- **Security Constraints:** The system must follow data privacy rules and have security measures to protect user information. This includes secure login, encryption of personal data, and access controls to stop unauthorized users from accessing information. Additionally, regular security updates should be applied to maintain system integrity and prevent potential threats.
- **Compatibility Constraints:** The system should be compatible with the hardware and software that are mostly used by applicants, admin and ALK. It has to be compatible with different operating systems, browsers and devices in order to be accessible to everyone. The system should be optimized for both desktop and mobile devices to cater to different user needs.
- **Performance Constraints:** The system must be able to handle at least 500 simultaneous users without a noticeable decrease in performance. Data retrieval and report generation should be completed in under 5 seconds for typical queries to provide fast access to information. Additionally, the system must maintain 99.9% uptime, ensuring that it is always available for daily operations without interruptions.

3. Architectural Rationale

An architectural style refers to a set of design principles and patterns that define the structure and organization of a software system. It provides a standardized way to design system components, their relationships, and communication methods. For Koperasi KADA's automated system, Layered Architecture is used.

3.1 Architecture Style and Rationale

Choosing the right system design is important for making sure a software works well and is easy to manage. The Koperasi Kada Automated System needs a reliable and organized structure to handle membership, loan applications, and statement generation. To achieve this, the system follows a Layered Architecture, which divides it into three main parts which is View Layer, Domain Layer, and Data Access Layer.

The View Layer is the part that users interact with. It displays information and allows users to enter data, such as applying for loans or checking membership details. The Domain Layer is responsible for processing data and applying the system's rules, like approving a loan based on certain conditions. The Data Access Layer connects to the database, making sure that information is stored and retrieved correctly. Separating these layers makes the system more organized and easier to update, since changes in one layer do not affect the others.

Using a Layered Architecture makes the system easier to maintain, test, and expand. For example, if the design of the View Layer needs improvement, it can be updated without affecting the business logic or database. This also helps with security, since the Data Access Layer ensures proper handling of sensitive information. Additionally, this structure allows developers to work on different parts of the system at the same time without causing problems. By following this approach, the Koperasi Kada Automated System can efficiently replace the manual system with a faster and more reliable solution. Figure 3.1 illustrates the Layered Architecture Diagram for Koperasi KADA Automated System.

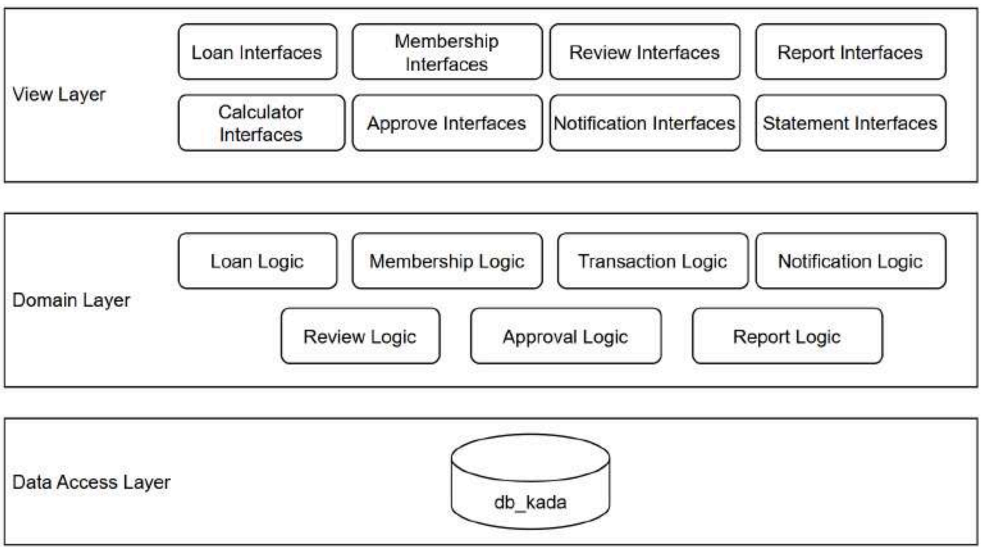


Figure 3.1: Layered Architecture Diagram for Koperasi KADA Automated System

4. Architectural Views

The 4+1 Architectural View Model is an architectural framework that organizes a system into five different views: Logical, Development, Process, Physical and Scenarios (Use Case View). The Logical View represents the system's functionality and object-oriented design, defining key classes and relationships. The Development View focuses on the organization of software modules, including components and overall code structure. The Process View handles dynamic aspects of the system such as the synchronization mechanisms. The Physical View represents how software components are deployed across hardware nodes, addressing performance and scalability. Lastly, the Scenarios serve as a bridge among all the views, describing the system's functionality from the user's perspective. Figure 4.1 illustrates the 4+1 Architectural View Model.

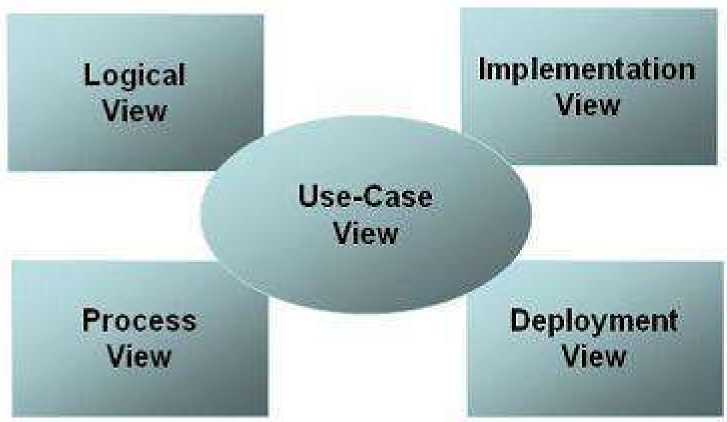


Figure 4.1: 4+1 Architectural View Model

Table 4.1 shows the diagram used for each view. Use Case is used in Scenario, Class Diagram is used in Logical View, Sequence Diagram is used in Process View, Package Diagram is used in Implementation View and Deployment Diagram is used in Deployment View. Further descriptions are described in the table below.

Table 4.1: Type of Diagram used

Type of View	Type of Diagram	Description
Scenario	Use Case	shows the interaction between users and the system
Logical	Class	represent the structure of the system, focusing on business logic and data relationships
Process	Sequence	shows the flow of operations between different components during specific processes
Implementation	Package	represents the organization of system components and how they relate to the storage and management of data
Deployment	Deployment	shows the physical distribution of system components that includes data servers and storage systems

4.1 Scenario (Use Case View) - Use Case

This view acts as a connecting thread across the other three views, maintaining consistency and completeness. It shows the functional requirements and connects the views through real-world scenarios or use cases. These use cases serve as a link between the system's conceptual design and technical execution, connecting various components including data models, processes and user interfaces. FInally, this viewpoint contributes to a Broad view by ensuring that all system components work together effortlessly to achieve the desired results while meeting the specific requirements. Figure 4.2 illustrates a use case diagram with its module for Koperasi KADA's Automated System.

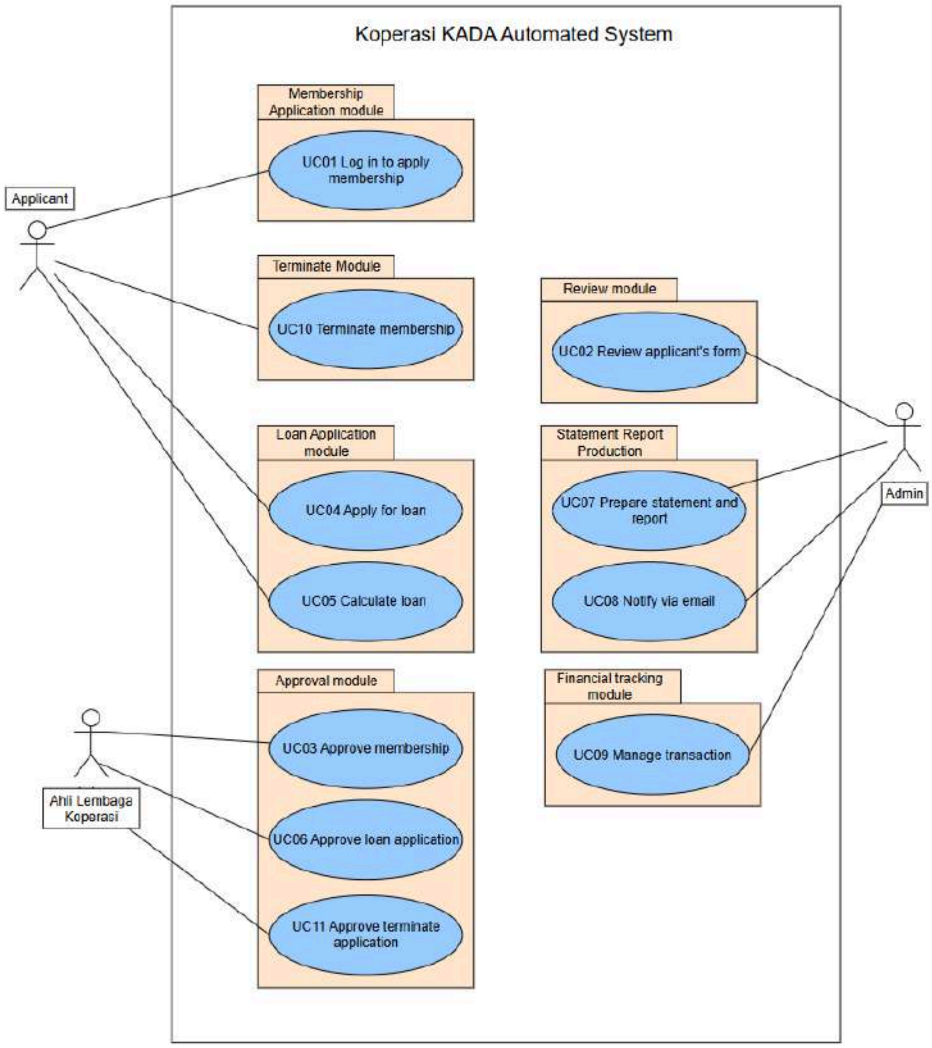


Figure 4.2: Use Case Diagram for Koperasi KADA Automated System

4.2 Implementation View - Package Diagram

Figure 4.3 is the architectural view which highlights the system's software modules that break down into high-level subsystems and their functions which arrange within its development setting. The system includes four main subsystems which are Membership Application Module, Loan Application Module, Review Module, Approval Module and Statement Report Production Module.

The Membership Application Module handles user authentication and processes membership applications. The Loan Application Module allows members to apply for loans, calculates monthly repayments loan amount and stores approved loans in a Loan List.Moreover, the Terminate Module allows users to terminate their membership of Koperasi KADA. The Review Module is in charge of going through applications from the Membership List and Loan

List, making sure they are properly assessed before being accepted or rejected. The Approval Module receives the reviewed applications after which it makes final choices by accepting or rejecting them and updating the corresponding lists. Finally, data from other subsystems is used by the Statement Report Production Module to create reports and send email notifications.

These subsystems work together seamlessly to provide the intended functionality. To guarantee that only authorised members are able to apply for loans, access to the Loan Application Module and Terminate Module requires the Membership List generated by the Membership Application Module. The Review Module processes the applications for the Membership and Loan lists ensuring a thorough validation before forwarding to the Approval Module. By combining data from every subsystem, the Statement Report Production Module creates thorough reports and alerts relevant users.

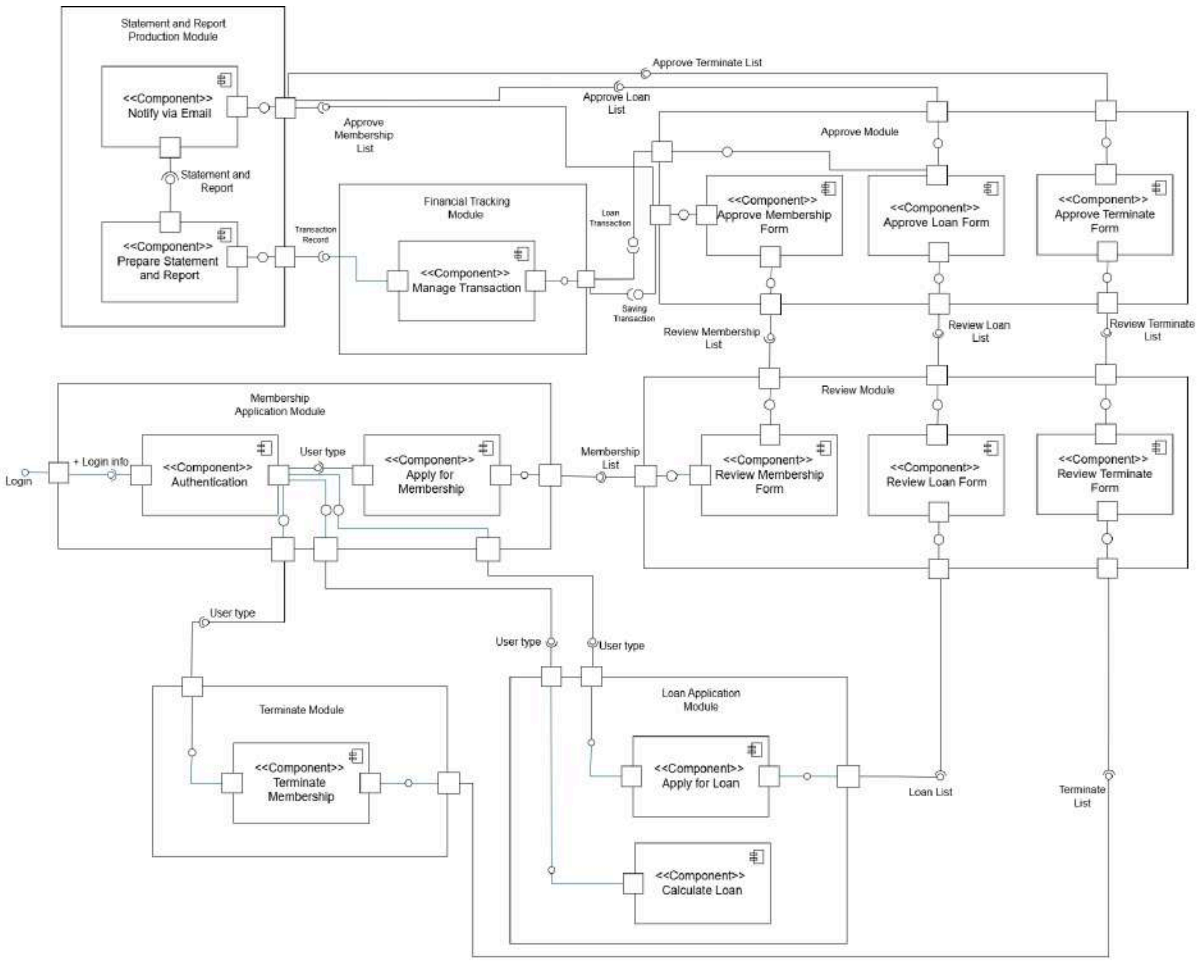


Figure 4.3: Architecture Diagram of <Koperasi KADA Automated System>

Figure 4.4 shows the subsystems in Koperasi KADA Automated System in Package Diagram form. The User Interface Framework handles user interaction and connects to the other subsystems like the Membership Module, Loan Application Module, Terminate Module, Review Module and Approval Module to process and manage data. The Statement Report Production Module generates reports, which are then sent back to the User Interface Framework for users to access.

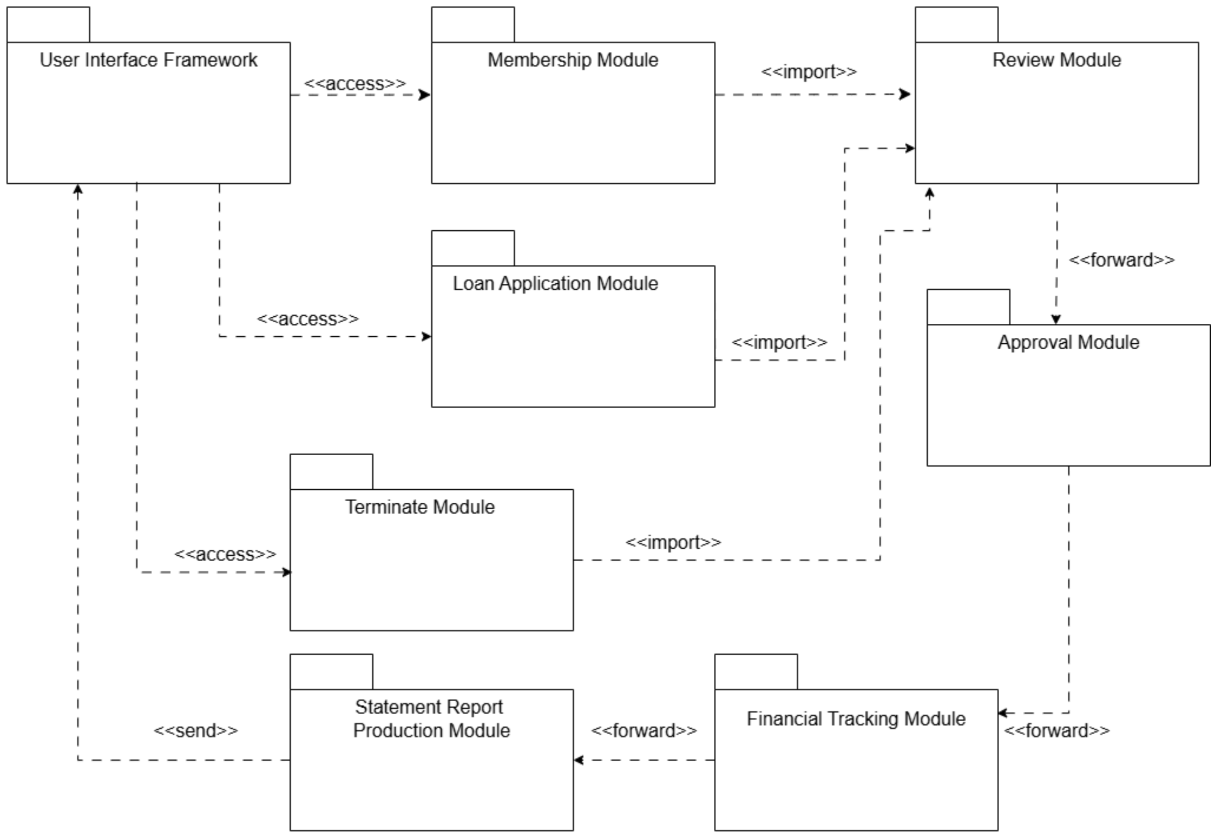


Figure 4.4: Package Diagram for <Koperasi KADA Automated System>

In Figure 4.5 is the Package Diagram for the Membership Application Module. The View Layer consists of user interfaces such as AuthenticationInterfaces and UserInterface and MembershipStatus Interface, which enables users to interact with the system by logging in, applying for membership and viewing the status of membership application. The Domain Layer includes components such as Applicant, Membership and FamilyInfo, which handle core business logic like processing applicant details and managing membership applications. Lastly, the Data Access Layer manages database interactions through components like MembershipData and familyInfoData, ensuring efficient retrieval and management of membership application data.

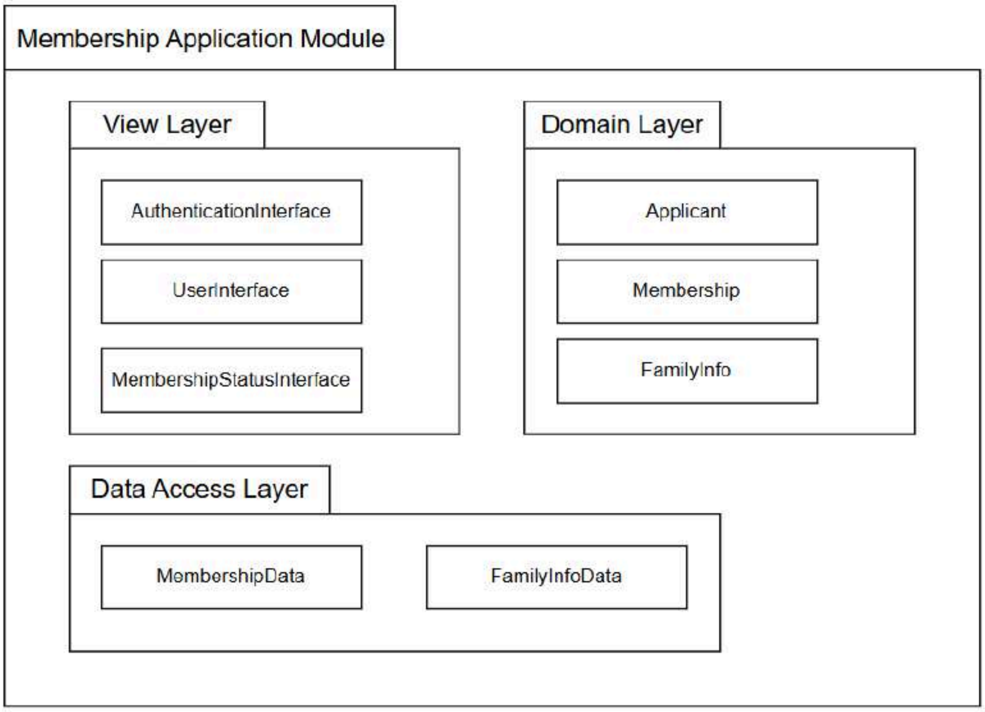


Figure 4.5: P001 Package Diagram for <Membership> Subsystem

In Figure 4.6 is the Package Diagram for the Review Module. The View Layer features interfaces such as AdministratorInterface, MembershipStatusInterface, Loan StatusInterface and TerminateStatusInterface, enabling administrators to interact with the system to review membership application, loan application, terminate membership application and update the status for all the applications. The Domain Layer consists of components such as Loan, GuarantorInfo, Membership, FamilyInfo, and ApplicationManagement which handle core business logic for loan, membership, and membership termination application reviewing process. Lastly, the Data Access Layer manages database interactions through components like Loan Data, GuarantorInfoData, MembershipData, familyInfoData, and TerminateMembershipData ensuring efficient retrieval and management of application data.

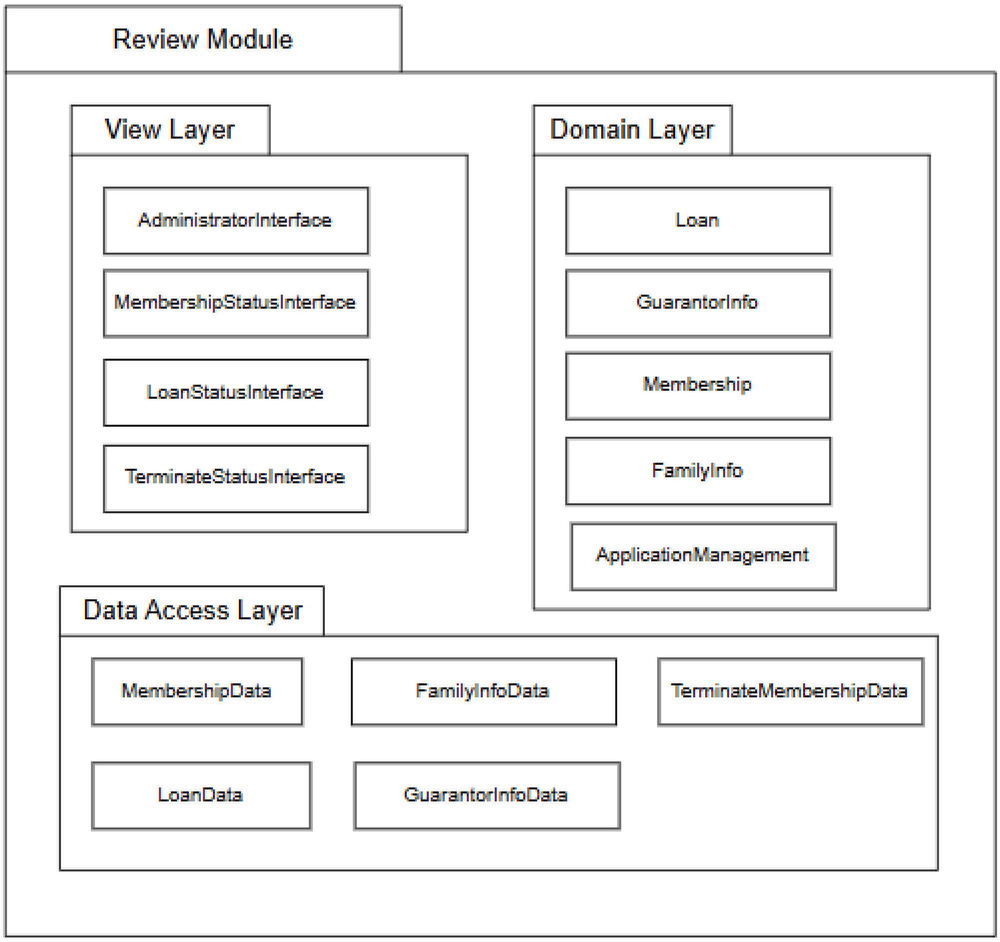


Figure 4.6: P002 Package Diagram for <Review> Subsystem

In Figure 4.7 is the Package Diagram for Loan Application Module. The view layer has user interfaces such as LoanApplicationInterfaces, LoanStatusInterface and LoanCalculatorInterface, which enables users to interact with the system by applying loans, checking loan status and calculating monthly loan repayment amount. Finally, the Data Access Layer coordinates database interactions within components such as LoanData, LoanStatusData and ApplicantData. This layer assures data storage and retrieval for loan and applicant information.

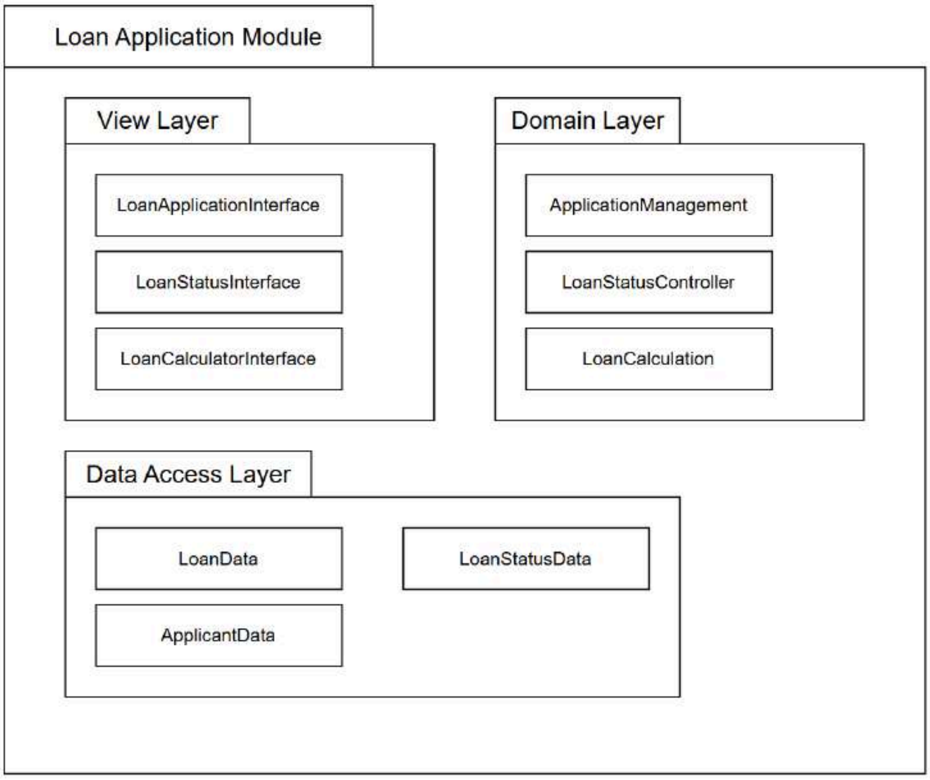


Figure 4.7: P003 Package Diagram for <Loan Application> Subsystem

Figure 4.8 shows the Package Diagram for Approval subsystem. The view layer represents the view component that interacts with the user. The layer includes ListMembershipUI to display the pending list of membership approval, ListLoanUI to display the pending list of loan approval, MembershipApprovalUI to interact in performing membership approval, LoanApprovalUI to interact in performing loan approval, historyMembershipUI to display past membership applications and historyLoanUI to display past loan applications. Then, the Domain Layer contains the business logic for data processing which includes MembershipApprovalController, LoanApprovalController, LoanComparisonProcessor and ModificationController. All of these are involved in approving or rejecting membership and loan applications. Lastly, the Data Access Layer is used to handle data operations. This layer includes MembershipDataRecord, LoanDataRecord and TerminateMembershipData.

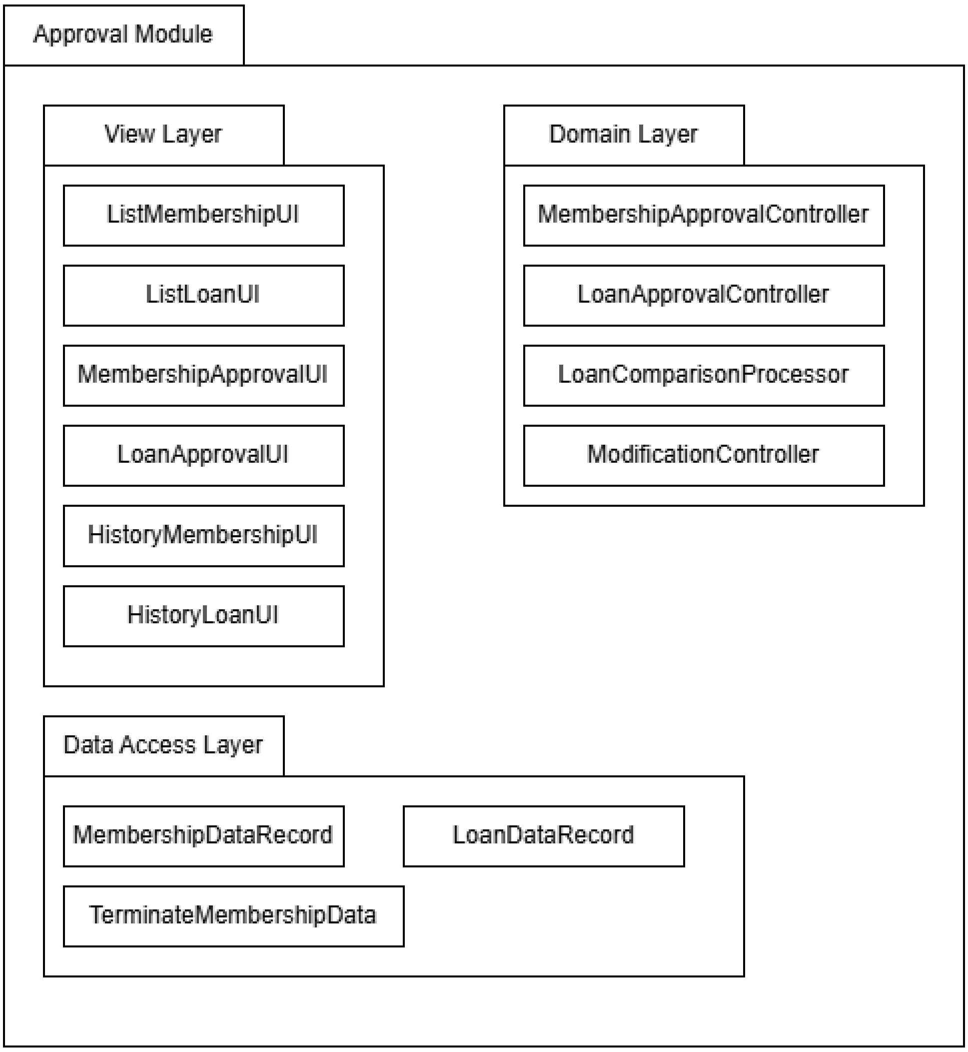


Figure 4.8: P004 Package Diagram for <Approval> Subsystem

Figure 4.9 illustrates the Package Diagram for the Statement Report Production subsystem. The view layer represents the components that interact with the user, including GenerateReportInterface, which provides a button for the admin to generate the statement report, NotificationInterface, which includes a button for sending email notifications, and TransactionListInterface, which displays the list of transactions related to loans and memberships. The Domain Layer contains the business logic responsible for data processing, including GenerateStatementController, which manages the process of generating the report, NotificationController, which handles the creation and sending of email notifications, and ExportController, which manages the exporting of the report into various formats like PDF or Excel. Finally, the Data Access Layer manages data operations and includes LoanTransactionData, LoanType, SavingTransactionData, and MembershipData, which handle the retrieval and storage of financial transaction information.

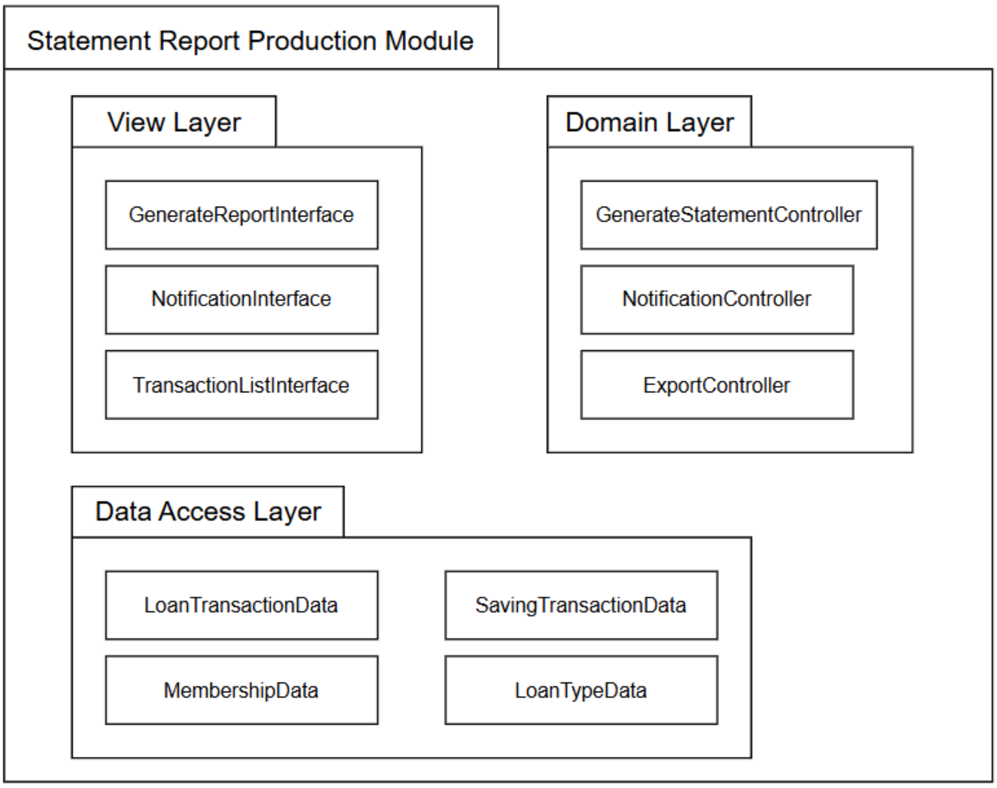


Figure 4.9: P005 Package Diagram for <Statement Report Production> Subsystem

Figure 4.10 is the Terminate Module packet diagram that organizes the use case into three layers. The View Layer contains interfaces for submitting termination requests and viewing their status. The Domain Layer handles the business logic including termination management and status updates. The Data Access Layer manages data related to members, termination requests and their statuses to ensure seamless interaction across the module.

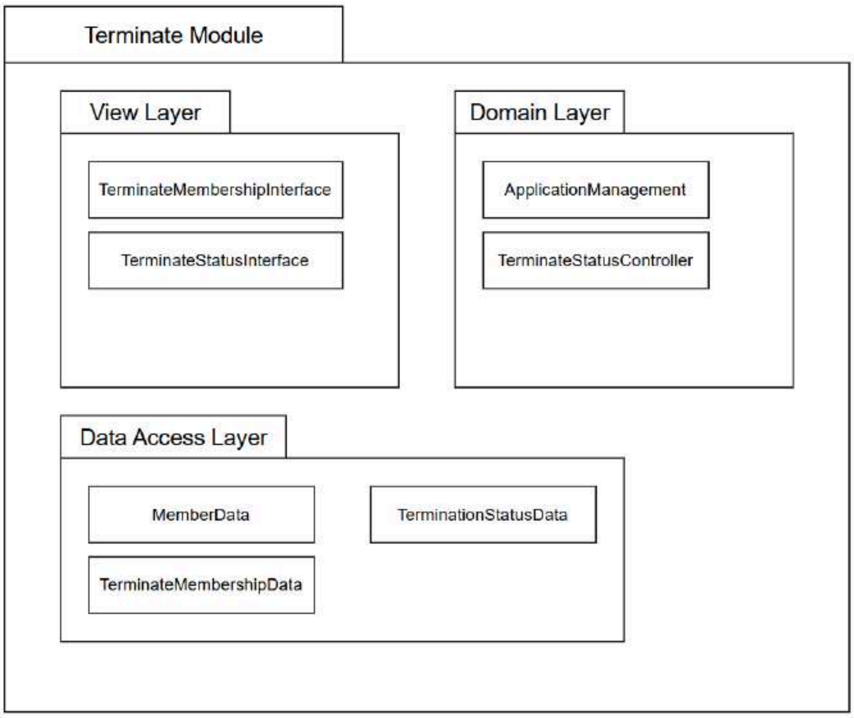


Figure 4.10: P006 Package Diagram for <Terminate Module> Subsystem

4.3 Logical View - Class Diagram

The logical view provides a perspective on the system's structure from the developer or designer's point of view. It highlights the major abstractions in the system, which in an object-oriented design approach would normally be modeled as objects or object classes. This view is represented through class diagrams, which shows the structure of the system by showing the relationships among classes, along with their attributes and methods.

Figure 4.11 class diagram shows the class diagram for Login to Apply Membership . This Authentication class has a one-to-one relationship with the User class, ensuring each user has unique login credentials. The Applicant class has an association relationship with the User class. The Applicant class has an aggregation relationship with the Membership class, meaning membership details can exist independently of the Applicant. Then, the Membership class has a composition relationship with the familyInfo class, meaning the family information is an essential part of the membership and cannot exist separately.

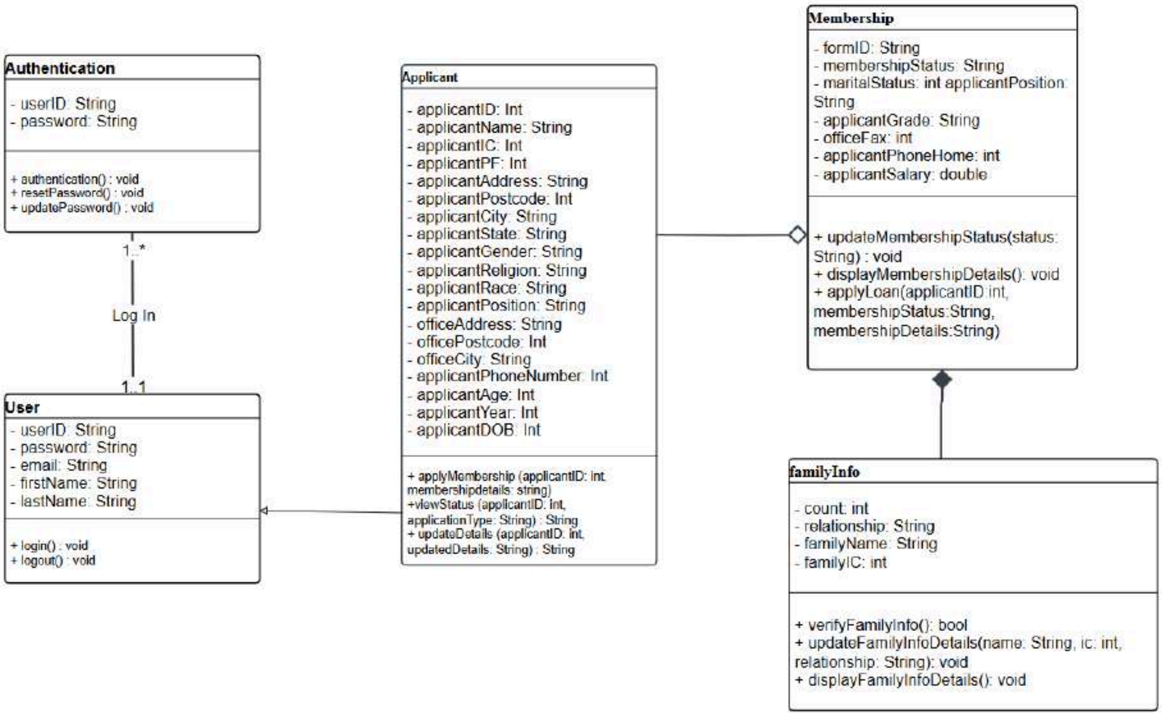


Figure 4.11: Class diagram for <Login to Apply Membership> Subsystem

Figure 4.12 class diagram shows the class diagram for Review Applicant's Form. The Administrator class has a one-to-many relationship with both Loan and Membership classes, meaning one administrator can review multiple loans and membership. The loan class has a composition relationship with the GuarantorInfo class, indicating that a guarantor is tightly bound to the loan and cannot exist independently. Similarly, the Membership class has a composition relationship with the familyInfo and saving class, meaning the family information is an essential part of the membership and cannot exist separately.

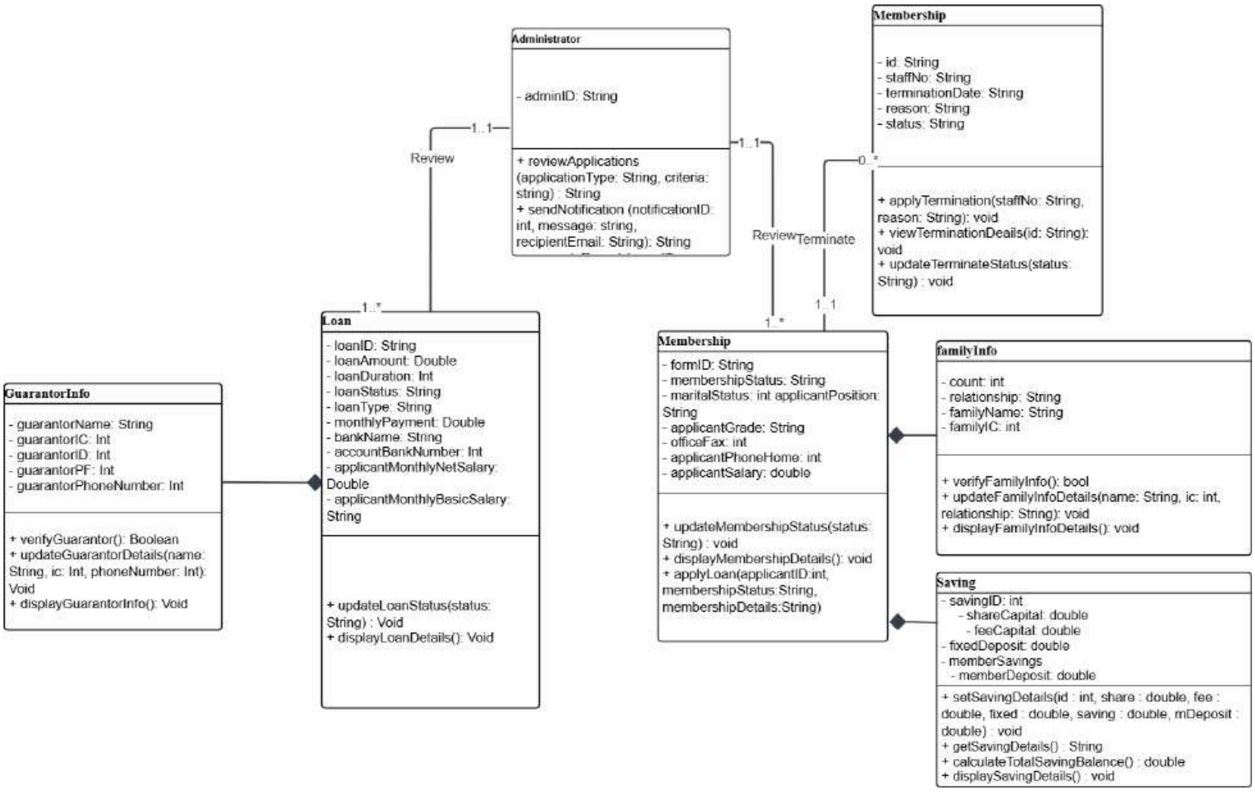


Figure 4.12: Class Diagram for <Review Applicant's Form> Subsystem

Figure 4.13 shows the class diagram for Approve Membership which includes the relationship between ALK class and Membership class. ALK class provides methods like approveMembership() to manage membership approvals and clarifications. The membership class stores the membership details through attributes. The relationship between the two classes is one-to-many, where an ALK can approve multiple membership applications.

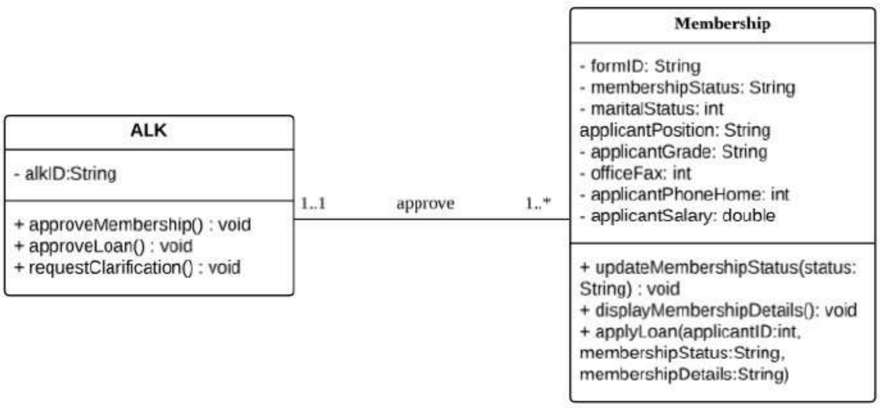


Figure 4.13: Class Diagram for <Approve Membership> Subsystem

Figure 4.14 class diagram shows the class diagram for Apply for Loan that shows two key relationships. First, the Membership and Loan classes have a one-to-many relationship indicating that a single membership can apply for several loans. Second, there is a composition relationship between the Loan and GuarantorInfo classes, which means that each loan application must include guarantor information because the guarantor is a critical component of the loan process.

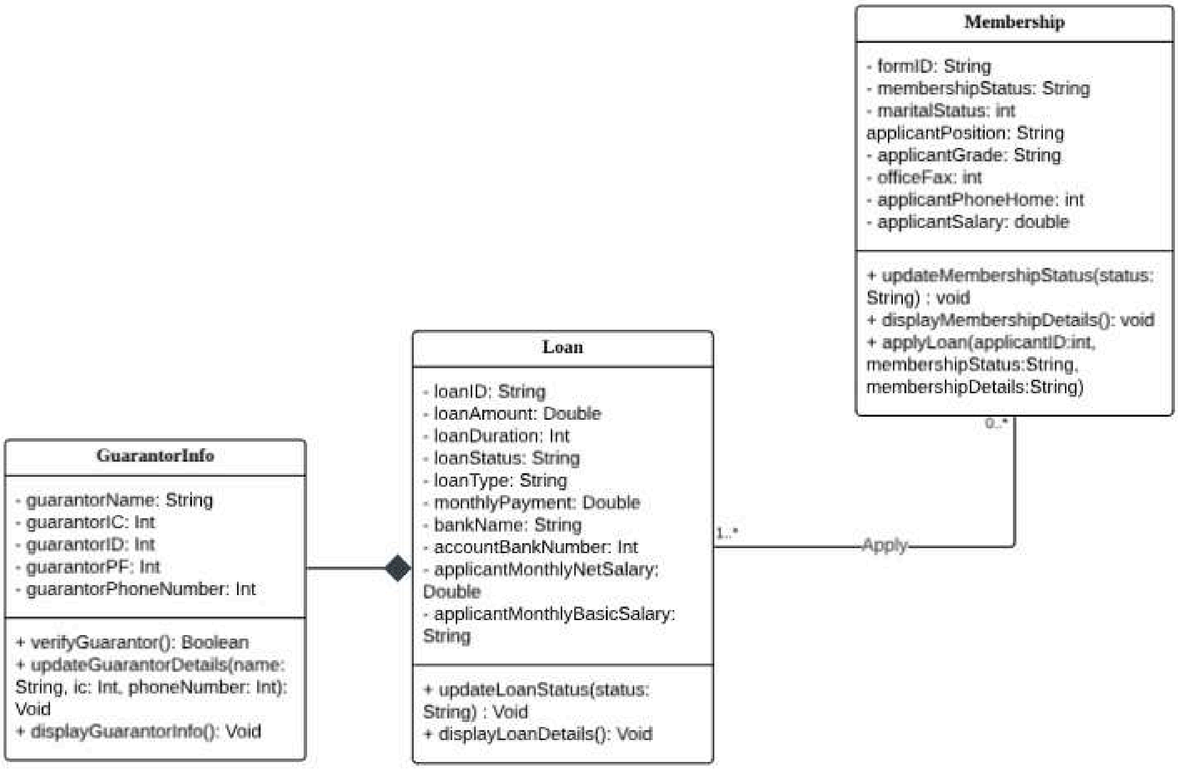


Figure 4.14: Class Diagram for <Apply for Loan> Subsystem

Figure 4.15 class diagram shows the class diagram for Calculate Loan that shows one key relationship. The user class has a one-to-one relationship with the loanCalculator class which allows users to calculate the estimation of monthly repayment.

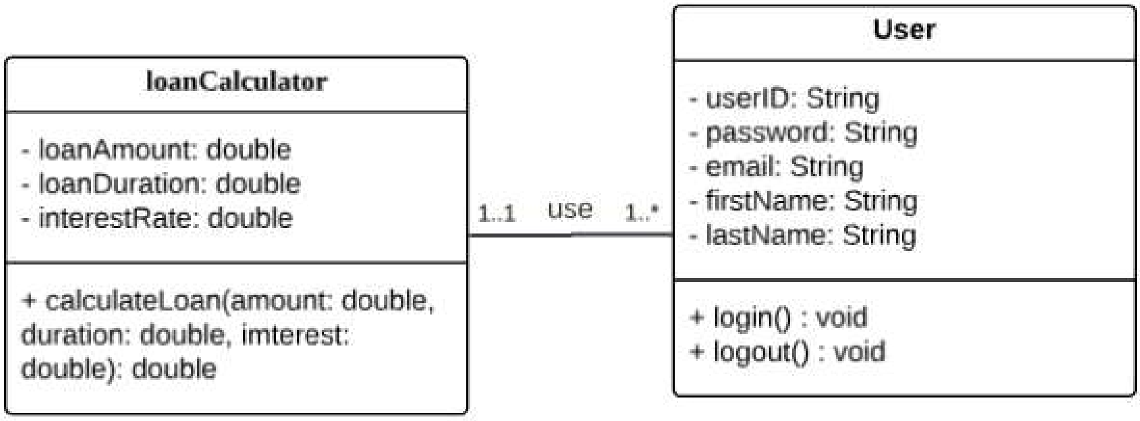


Figure 4.15: Class Diagram for <Calculate Loan> Subsystem

Figure 4.16 shows the class diagram for Loan Approvla which includes the relationship between ALK class and Loan class. ALK class provides methods like approveLoan() to manage loan approvals. The loan class stores the membership details through attributes. The relationship between the two classes is one-to-many, where an ALK can approve multiple loan applications.

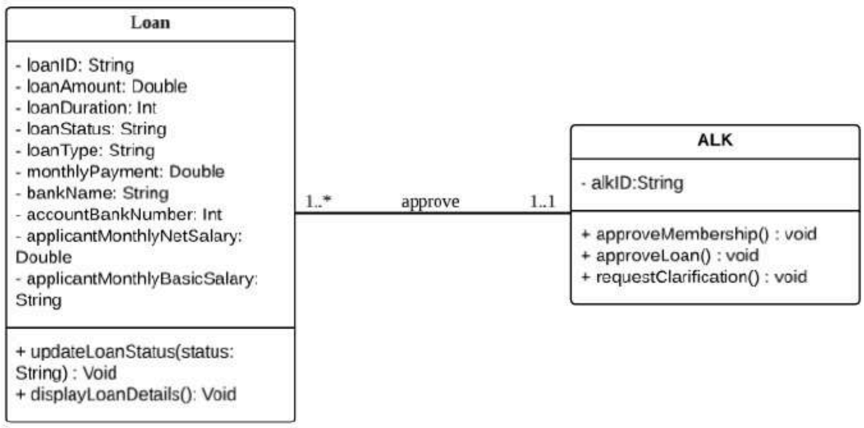


Figure 4.16: Class Diagram for <Loan Approval> Subsystem

Figure 4.17 illustrates the class diagram for preparing statements and reports, detailing the relationships between various entities involved in loan, savings, and membership management. The diagram includes classes such as LoanRepayment, Loan, Applicant, Membership, Saving, and SavingType, each with specific attributes and methods. The methods provided facilitate updates and retrieval of relevant data, supporting efficient statement and report generation.

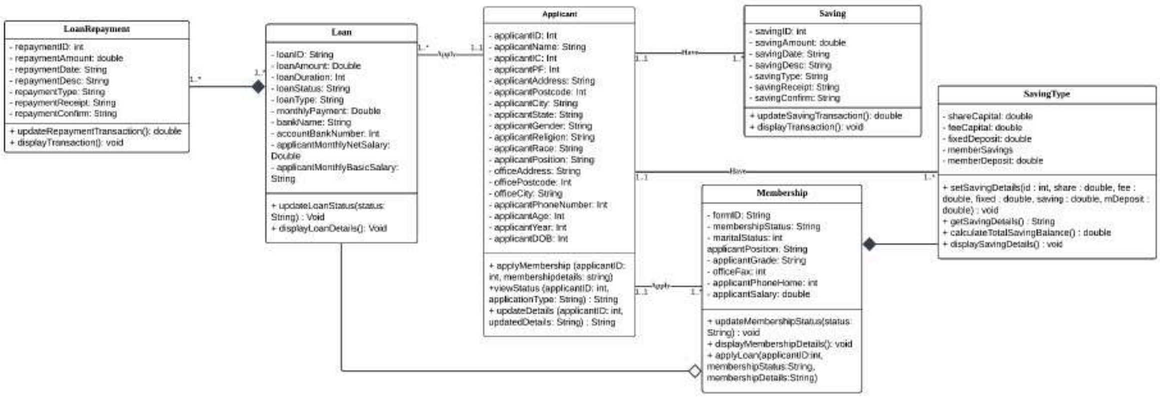


Figure 4.17: Class Diagram for <Prepare Statement and Report> Subsystem

Figure 4.18 presents the class diagram for the Notify via Email. In this diagram, there are two primary classes: Admin and Notification. The Admin class is responsible for sending notifications, establishing a one-to-many relationship with the Notification class. This means that one Admin can send multiple notifications to different recipients. The Notification class holds the details about each notification, such as the content and recipient information.

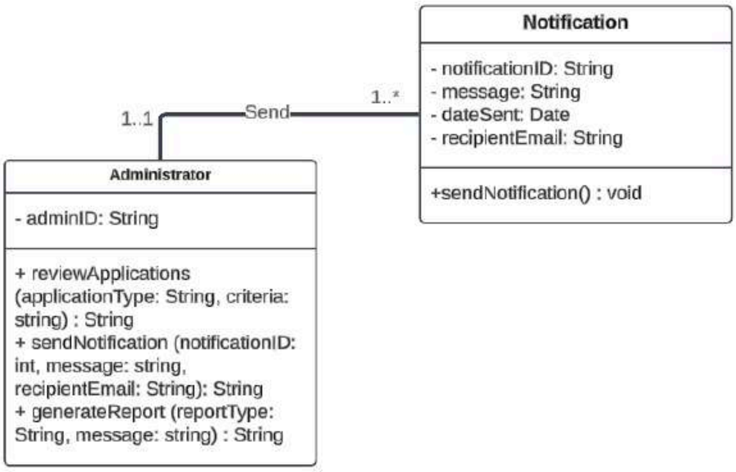


Figure 4.18: Class Diagram for <Notify via Email> Subsystem

Figure 4.19 is the Terminate entity representing the process of membership termination, linking it to the Membership Entity. Each membership can have one termination request while multiple termination requests can exist for different memberships. The entity includes attributes such as id, staffNo, terminationDate, reason and status ensuring a structured termination process.

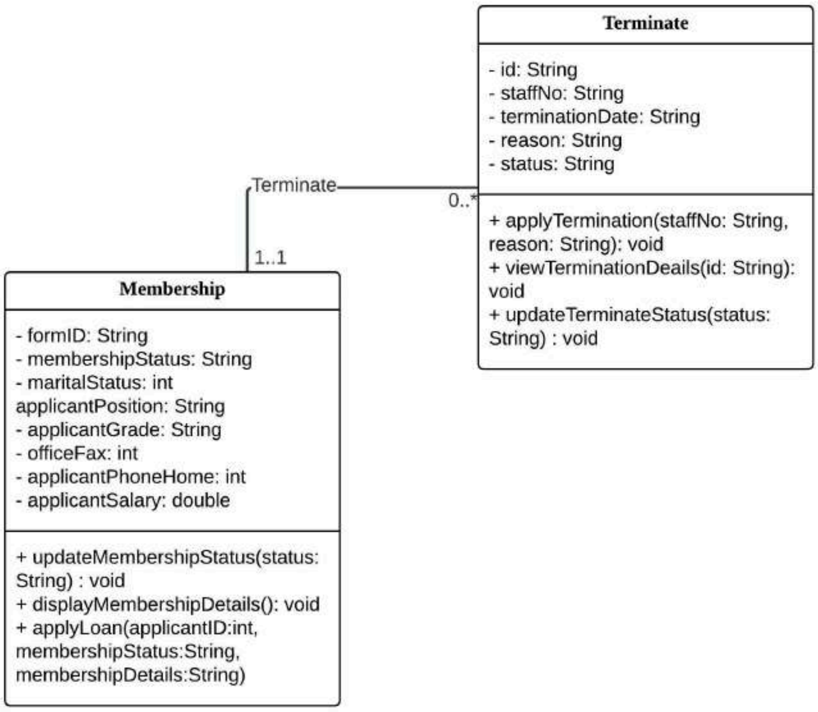


Figure 4.19: Class Diagram for <Terminate Membership> Subsystem

Figure 4.20 shows the class diagram for managing transactions. It includes classes like LoanRepayment, Loan, Membership, and Saving, each handling different financial activities. These classes store important details and have functions to update and display transaction records, making it easier to manage loan repayments, savings, and memberships.

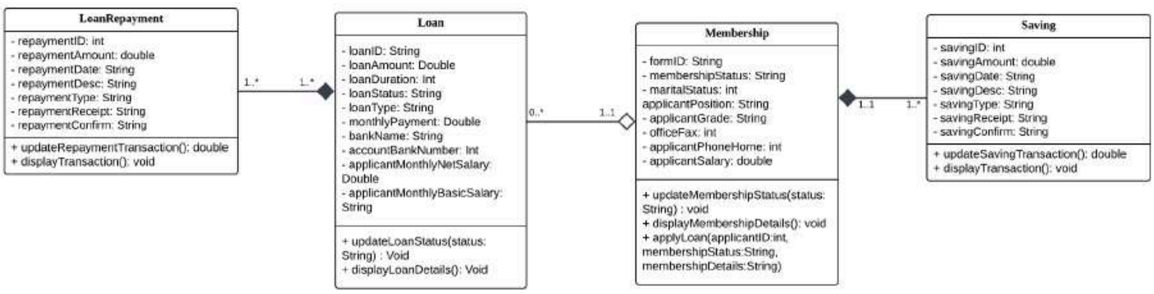


Figure 4.20: Class Diagram for <Manage Transaction> Subsystem

Figure 4.21 shows the class diagram for approval to terminate membership application. It includes classes like ALK, Terminate and Membership. Each application will be undergoing approval process by ALK. Each application will have updated status once ALK approved or rejected application.

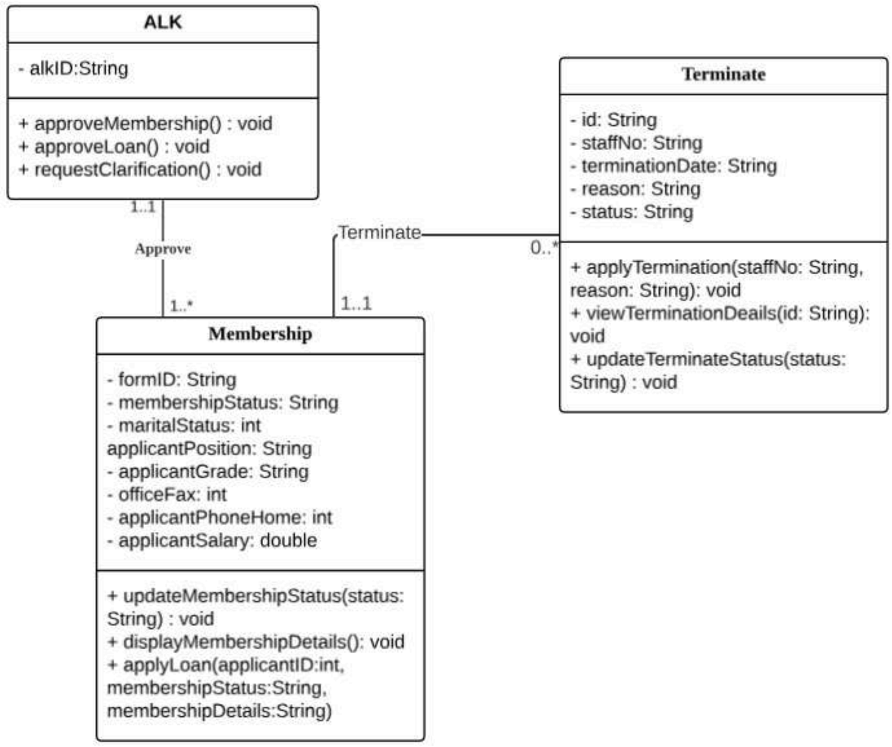


Figure 4.21: Class Diagram for <Approve Terminate Application> Subsystem

4.4 Process View - Sequence Diagram

This view illustrates how the system operates at runtime and focuses on how its components interact through communication and collaboration. It shows how different processes work together to achieve the system's goals, with each process responsible for specific tasks. To represent these interactions, sequence diagrams will be used for relevant use cases, showing the step-by-step flow of messages between components. Branching will capture alternate flows within the same diagram, and if the diagram becomes too cluttered, separate diagrams will be created for each scenario or alternate flow. This view provides a clear understanding of the system's dynamic behavior and the relationships between its components.

Figure 4.22 illustrates the applicant's login process. The applicant accesses the login page, enters their details and the system will validate the details. If the details are valid, access to the membership application form is granted. Otherwise, an error message is displayed. The membership form is then accessed and shown to the applicant.

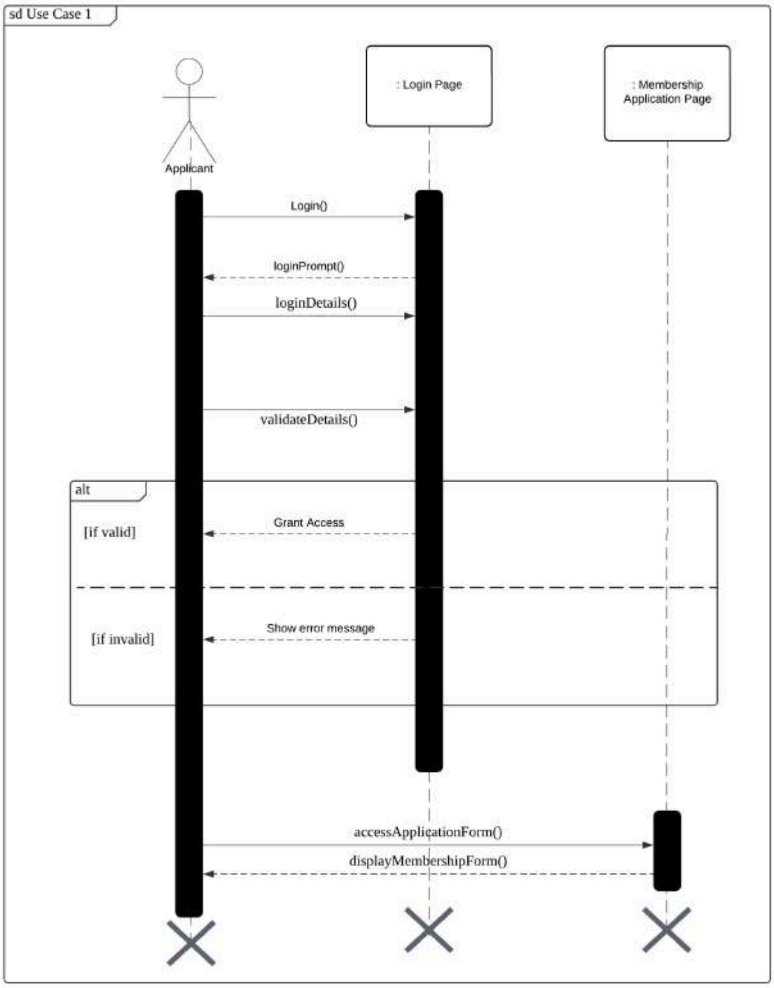


Figure 4.22: SD001 Sequence Diagram for <Login for Apply Membership>

Figure 4.23 shows the process of an admin reviewing membership, loan, or membership termination applications. The admin logs in, navigates to the membership list, and views batch application details. The system loops through each application in the batch, marking it as either "reviewed" or "incomplete".

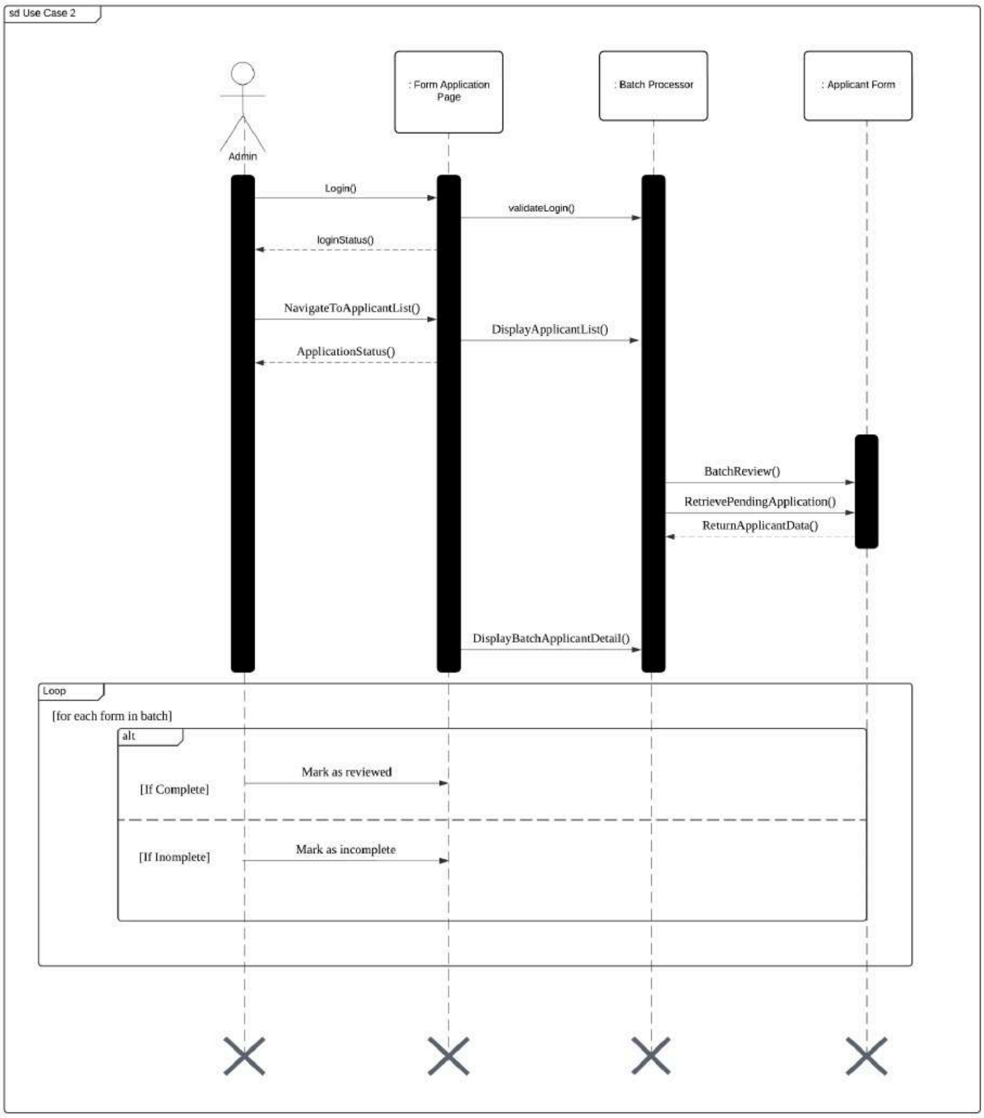


Figure 4.23: SD002 Sequence Diagram for <Review Applicant's Form> Subsystem

Figure 4.24 shows a sequence diagram for use case approved membership. ALK logs into the system, which either grants access or reuters an error. ALK then selects any membership application from the pending list, evaluates and decides whether to approve or not. Finally, the system updates the membership status

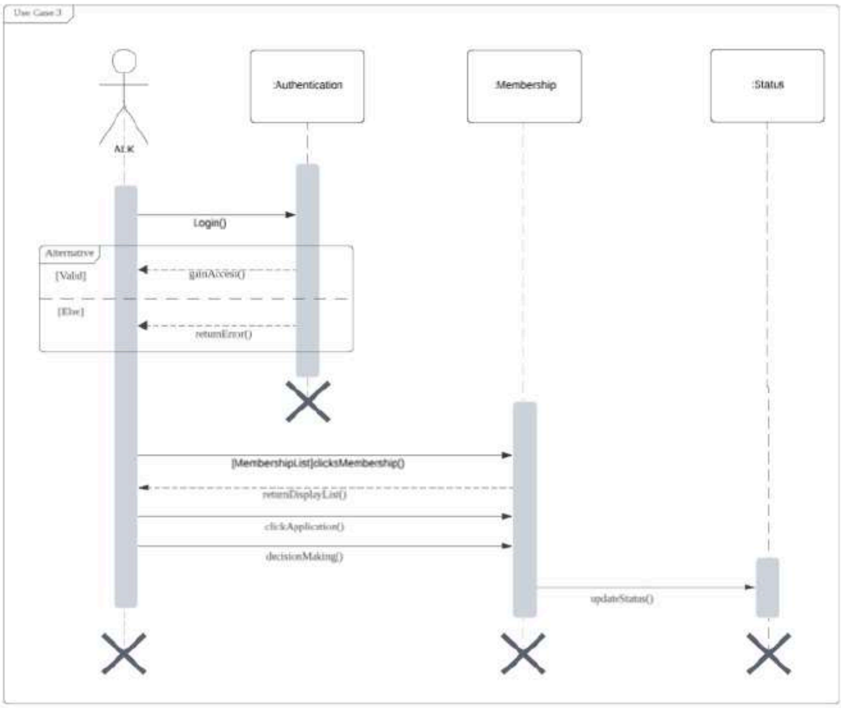


Figure 4.24: SD003 Sequence Diagram for <Approve Membership> Subsystem

Figure 4.25 is the sequence diagram for the application loan process, representing the interaction between the applicant and the system for applying for the loan. It starts when the user logs in and accesses the loan application page. The user fills out the loan application with compulsory details, then the system validates it. If the details are correct, the application is submitted and saved into the system's database. The system notifies the user after accepting the successful submission. This diagram illustrates the important steps involved, including user interactions, form validation and database operations.

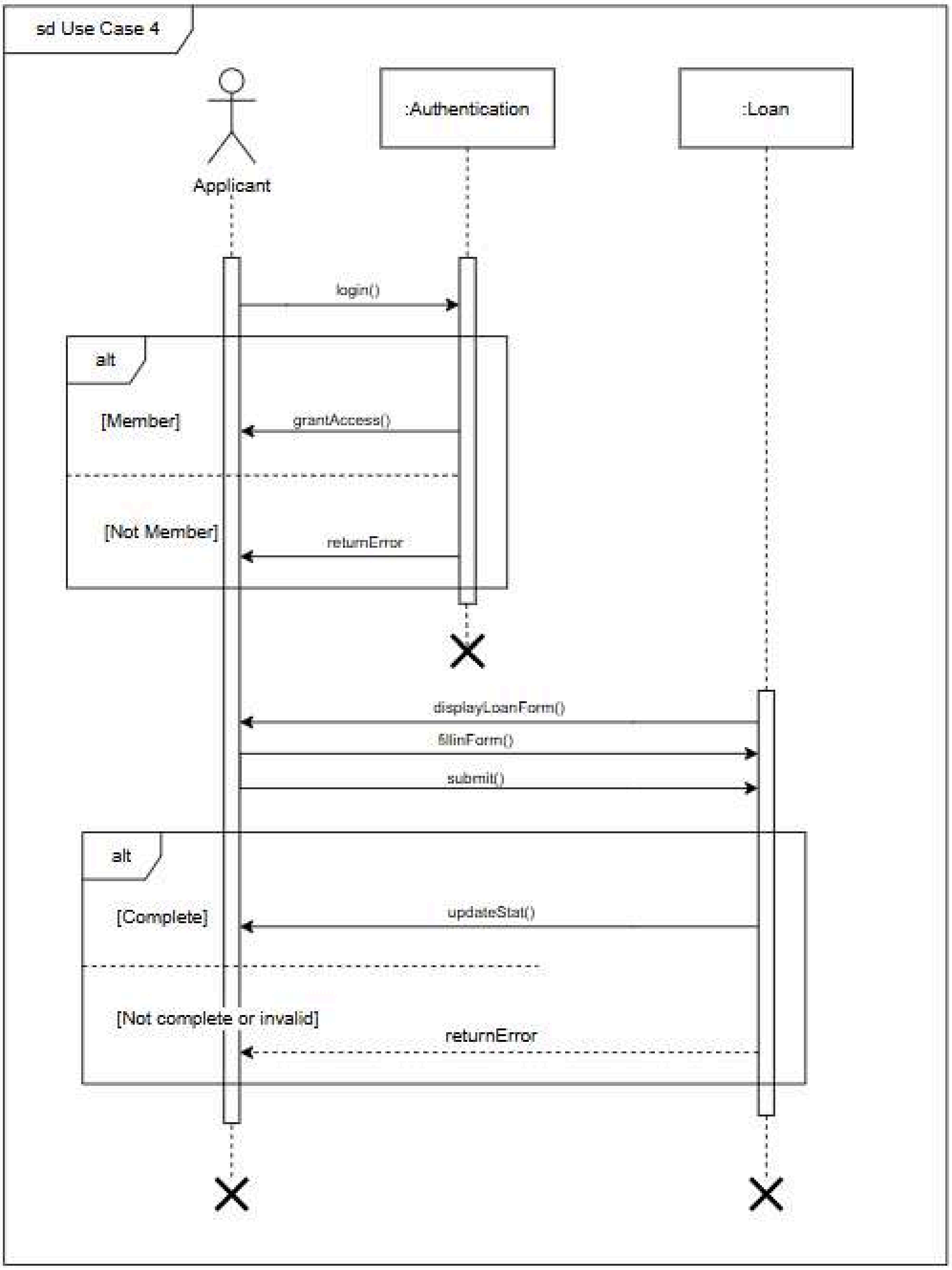


Figure 4.25: SD004 Sequence Diagram for <Apply for Loan> Subsystem

Figure 4.26 is a sequence diagram for calculation loan highlighting the interaction between members and the system. The process begins with the member logging into the system by entering their details. The system then verifies the details. If valid, the system grants access to the members. If invalid, the system returns an error message. The member selects the "Loan Calculator" button, and the system rate then submits the details. The Loan Calculator calculates and generates the loan repayment by processing these inputs. If the inputs are valid, the system displays the results. If invalid, it returns an error message that requires the member to reenter the inputs. The interaction makes sure the member has correct financial calculations based on their provided details.

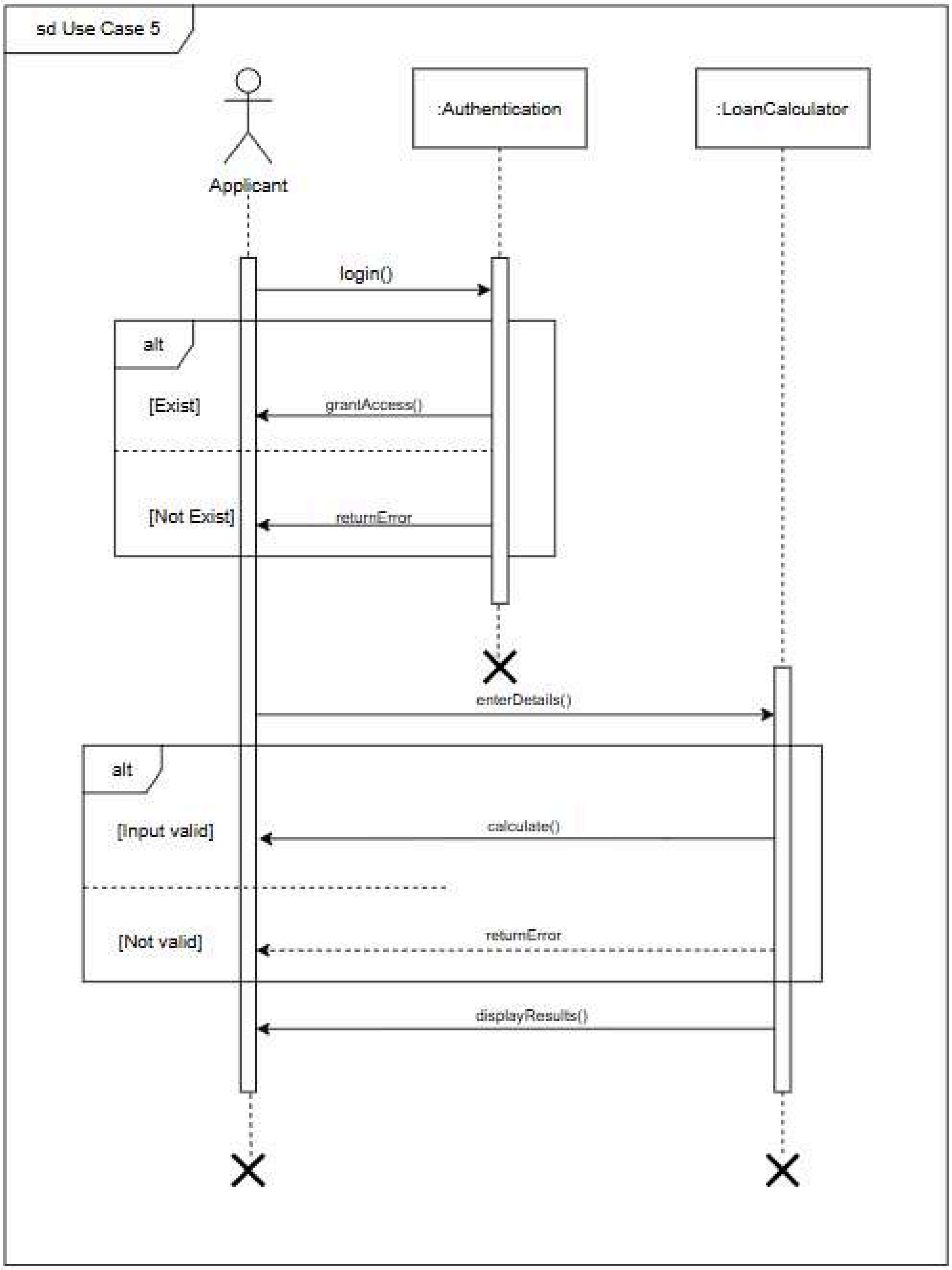


Figure 4.26: SD005 Sequence Diagram for <Calculate Loan> Subsystem

Figure 4.27 shows a sequence diagram for use case approved loan applications. ALK logs into the system, which either grants access or reuters an error. ALK then selects any loan application from the pending list, evaluates and decides whether to approve or not. If necessary, ALK uses the compare button to compare against loan terms before finalizing the decision.

Finally, the system updates the loan status.

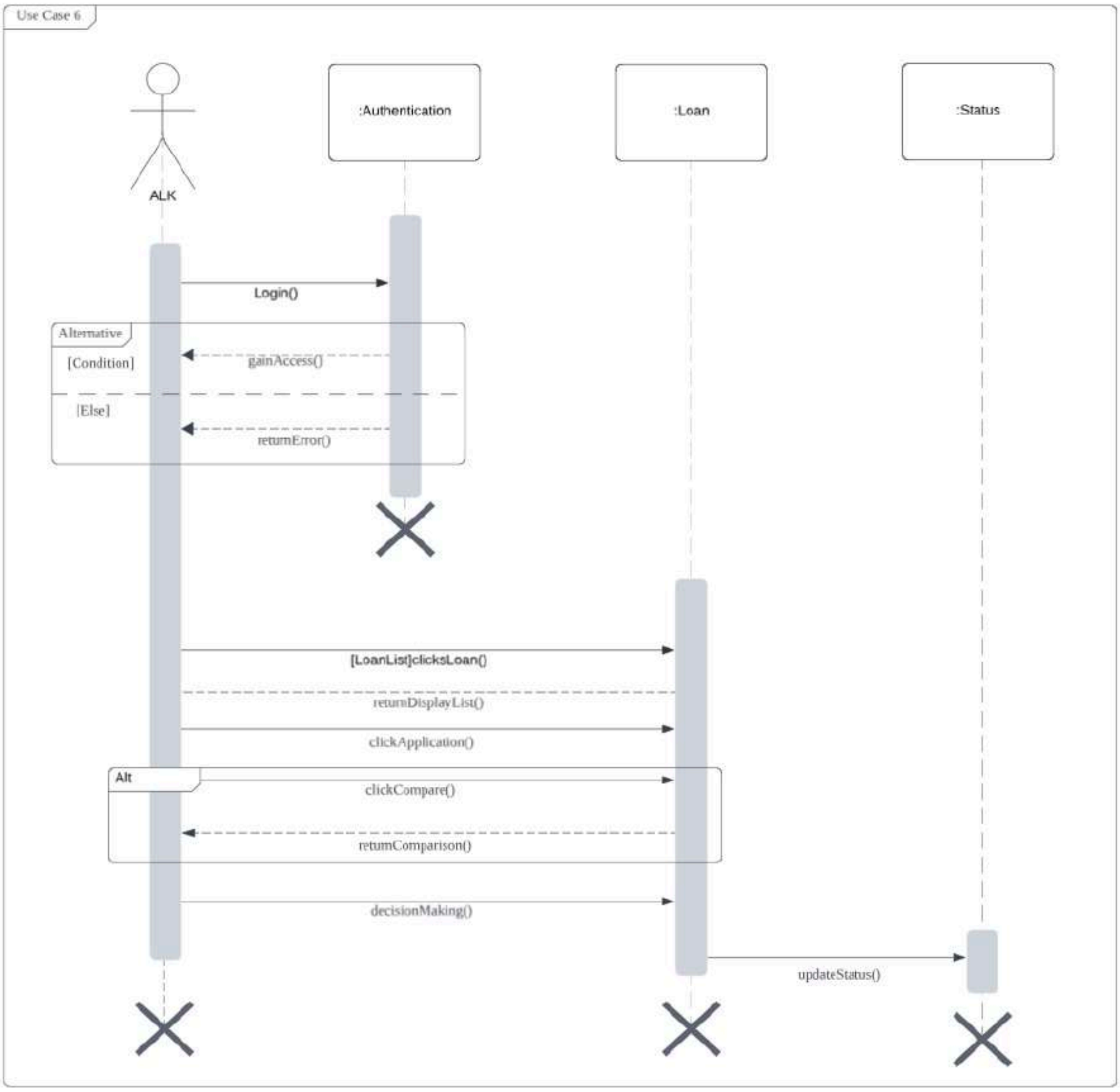


Figure 4.27: SD006 Sequence Diagram for <Approve Loan Application> Subsystem

Figure 4.28 shows the sequence diagram for the process of preparing financial statements and reports by an admin, interacting with the Authentication and Report systems. The admin begins by logging in, where successful authentication grants access to the admin page, while failure triggers an error message. Once authenticated, the admin retrieves member data and generates financial statements for selected members. If statement generation is successful, a confirmation message is displayed; otherwise, an error message appears. Additionally, corporate reports containing membership, loan, and termination statistics are generated, providing monthly and yearly financial summaries.

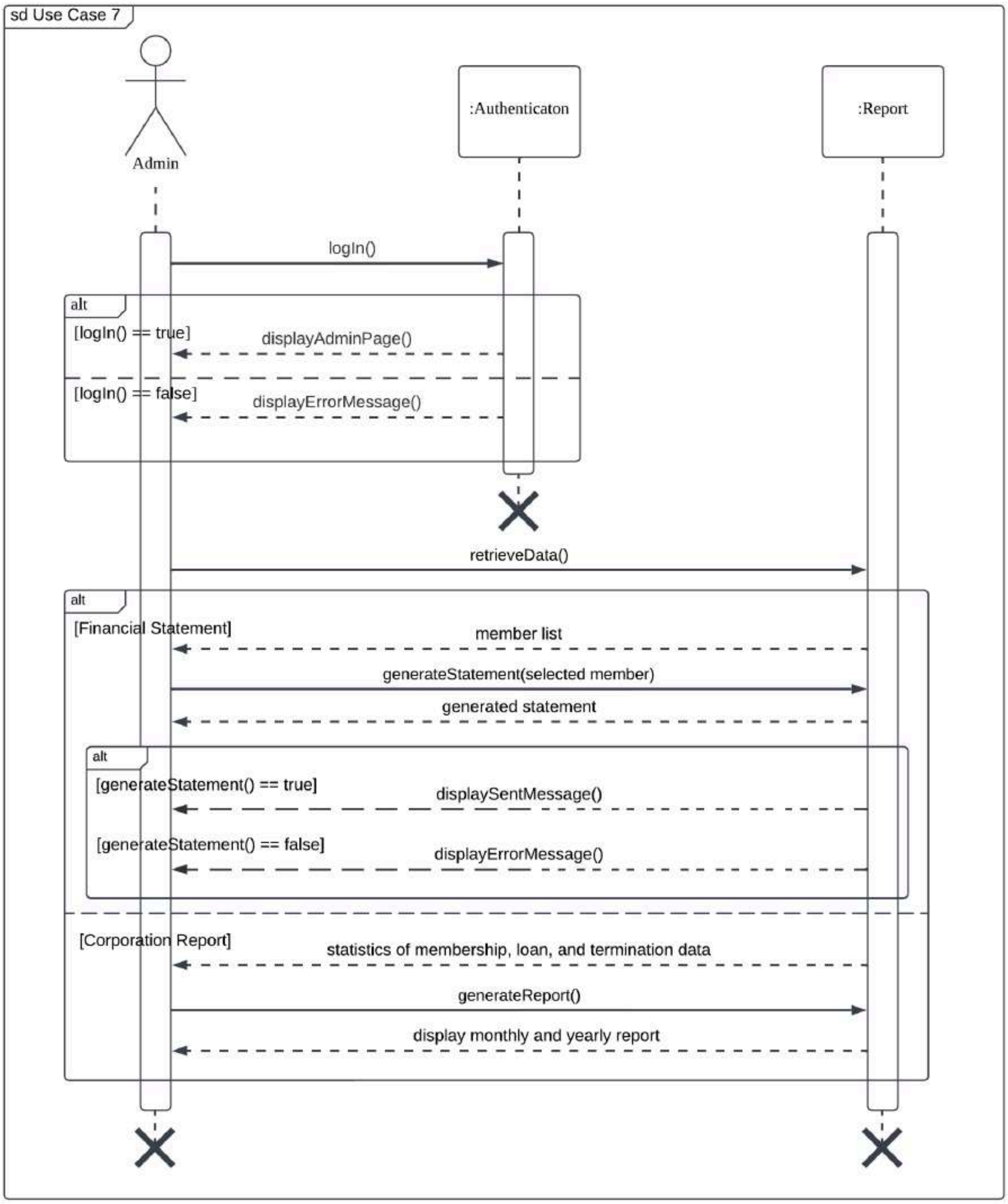


Figure 4.28: SD007 Sequence Diagram for <Prepare Statement and Report> Subsystem

Figure 4.29 illustrates the sequence diagram for the use case Notify via Email, regarding prepared statements and status updates. The Admin initiates the notification process by selecting a notification type, triggering interactions between the Notification and Report components. If a Generated Statement Notification is selected, the system retrieves the prepared statement for the selected member and sends it via email, followed by updating the notification log. Similarly, for Status Notification, an email is sent to the selected applicant, and the send status is updated. The process also supports batch processing, where multiple members or applicants are notified simultaneously. The final step verifies whether the email was successfully sent, displaying either a success or error message accordingly.

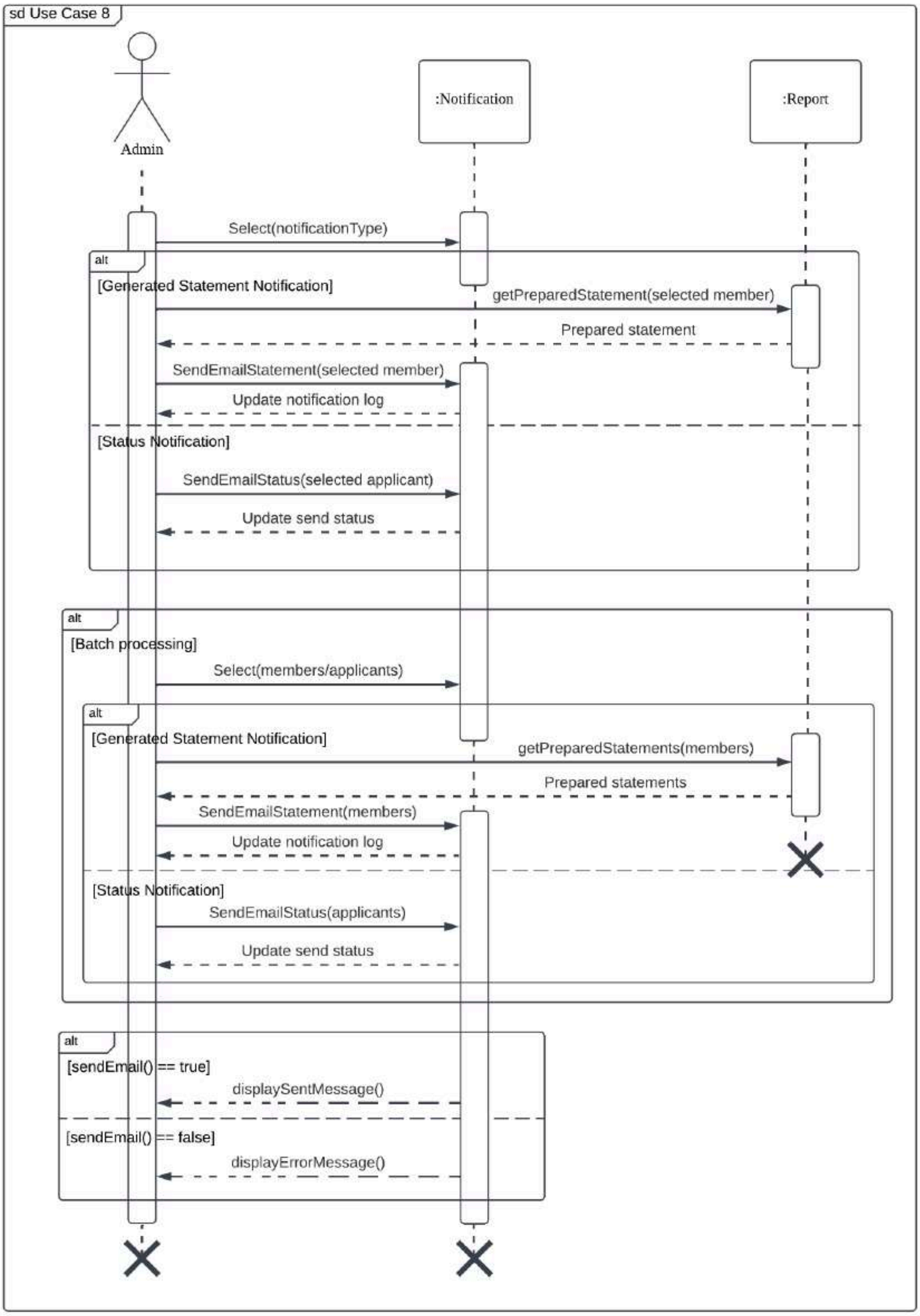


Figure 4.29: SD008 Sequence Diagram for <Notify via Email> Subsystem

Figure 4.30 illustrates the process of managing financial transactions by an admin, interacting with the Transaction and Financial Record systems. The admin selects a transaction type, which can be a Monthly Transaction, an Irregular Transaction, or Batch Processing Monthly. For monthly transactions, the system deducts the salary of a selected member, while irregular transactions involve selecting a member and specifying savings or loan repayment. In batch processing, multiple members are selected for financial updates. Each transaction update is confirmed, and the system displays either a success or error message based on the update status.

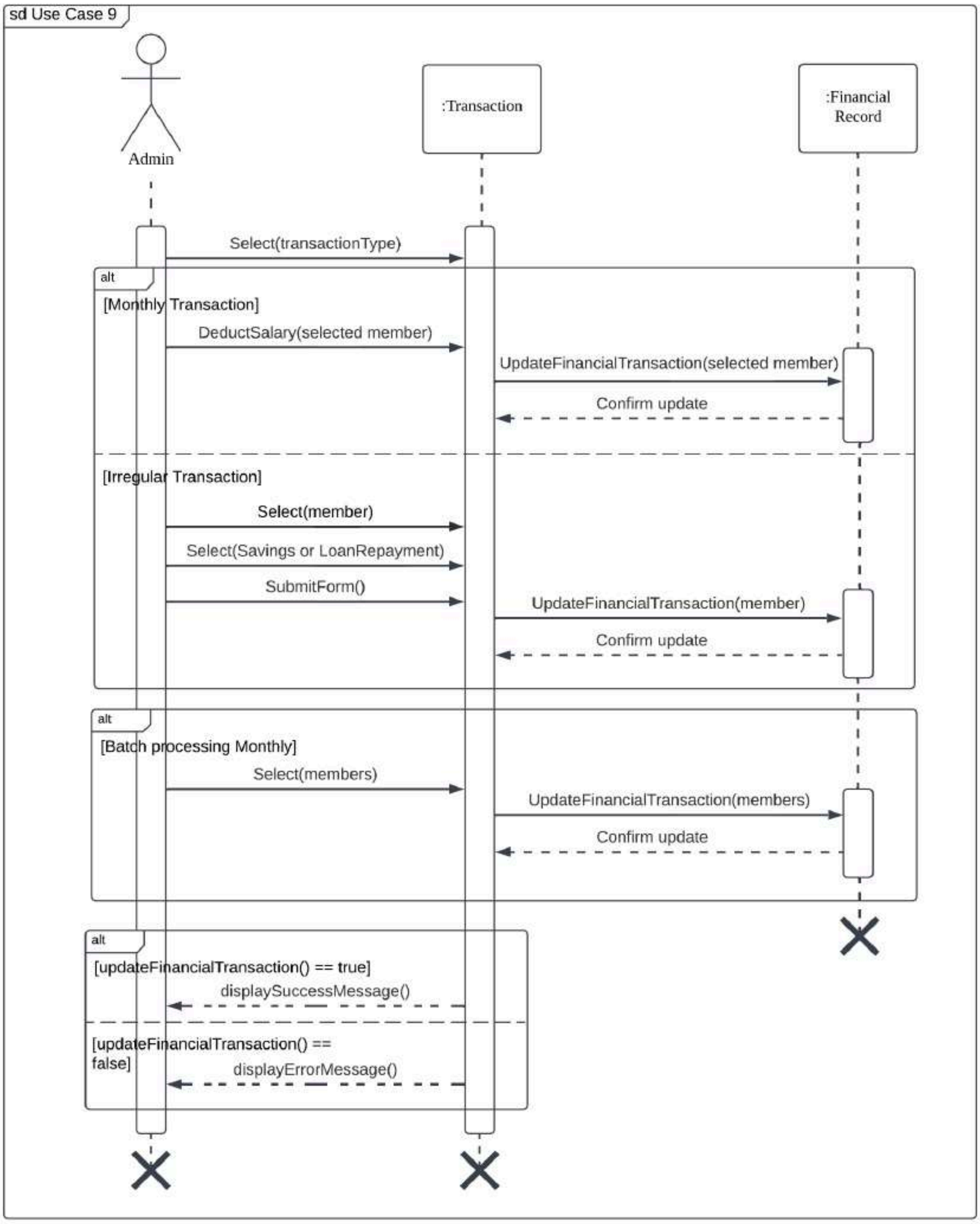


Figure 4.30: SD009 Sequence Diagram for <Manage Transaction> Subsystem

Figure 4.31 is a sequence diagram for Terminate Membership use case that illustrates the interaction between the Applicant, Authentication system and Terminate Membership module. The Applicant logs in and then the system verifies their membership status. If access is granted, the termination form is displayed. The Applicant fills in and submits the form. The system will validate the request, if complete, it updates the status of application. Otherwise, it returns an error to the Applicant.

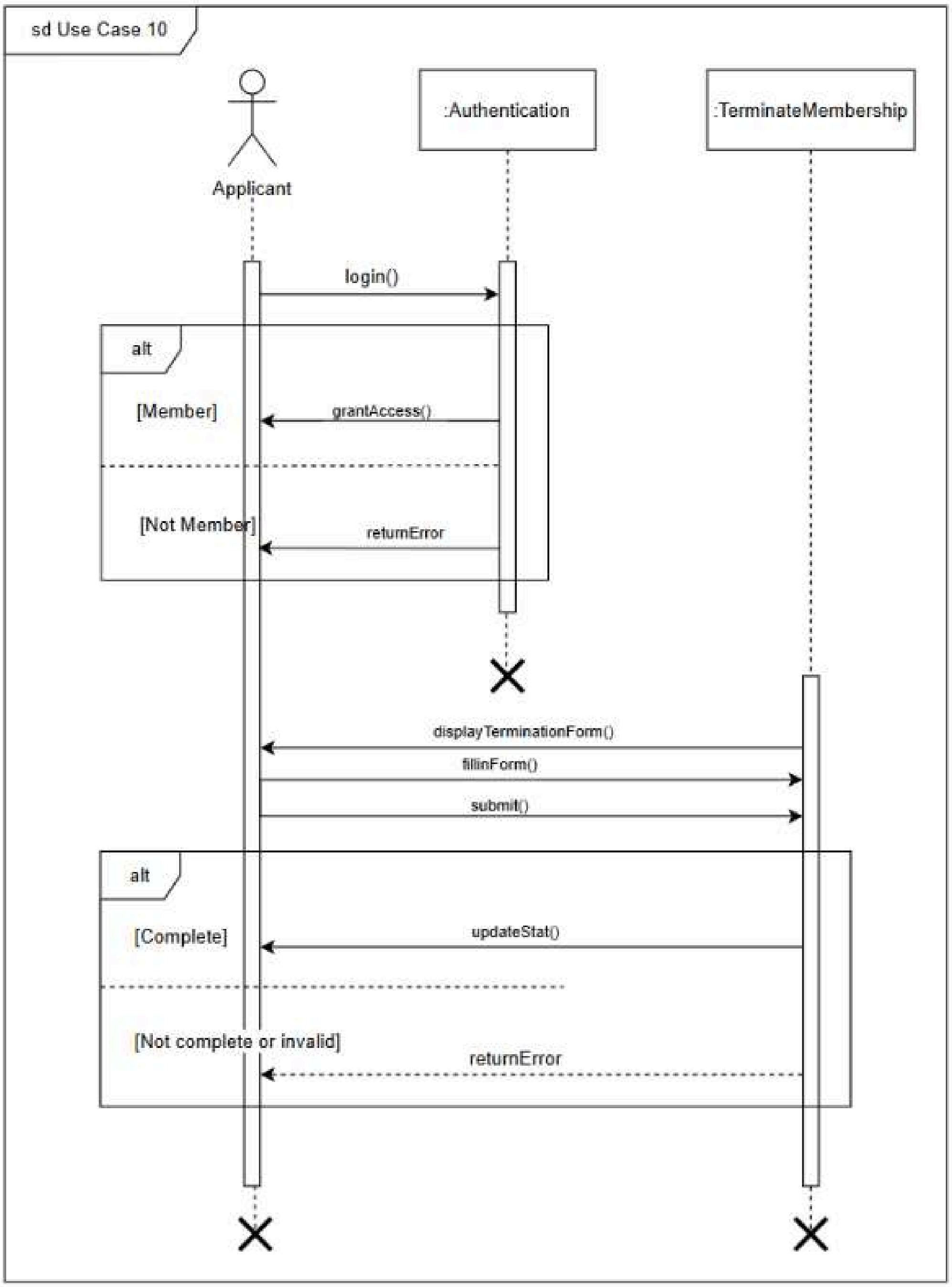


Figure 4.31: SD010 Sequence Diagram for <Terminate Membership> Subsystem

Figure 4.32 is a sequence diagram for the Approve Terminate Application use case that illustrates the interaction between the ALK and Terminate Membership module. The ALK will log in and then the system will display the termination application reviewed by the admin. The ALK then performs decision-making whether to approve or reject the application. If the ALK approved, then the applicant's membership will end, otherwise the membership will continue as usual.

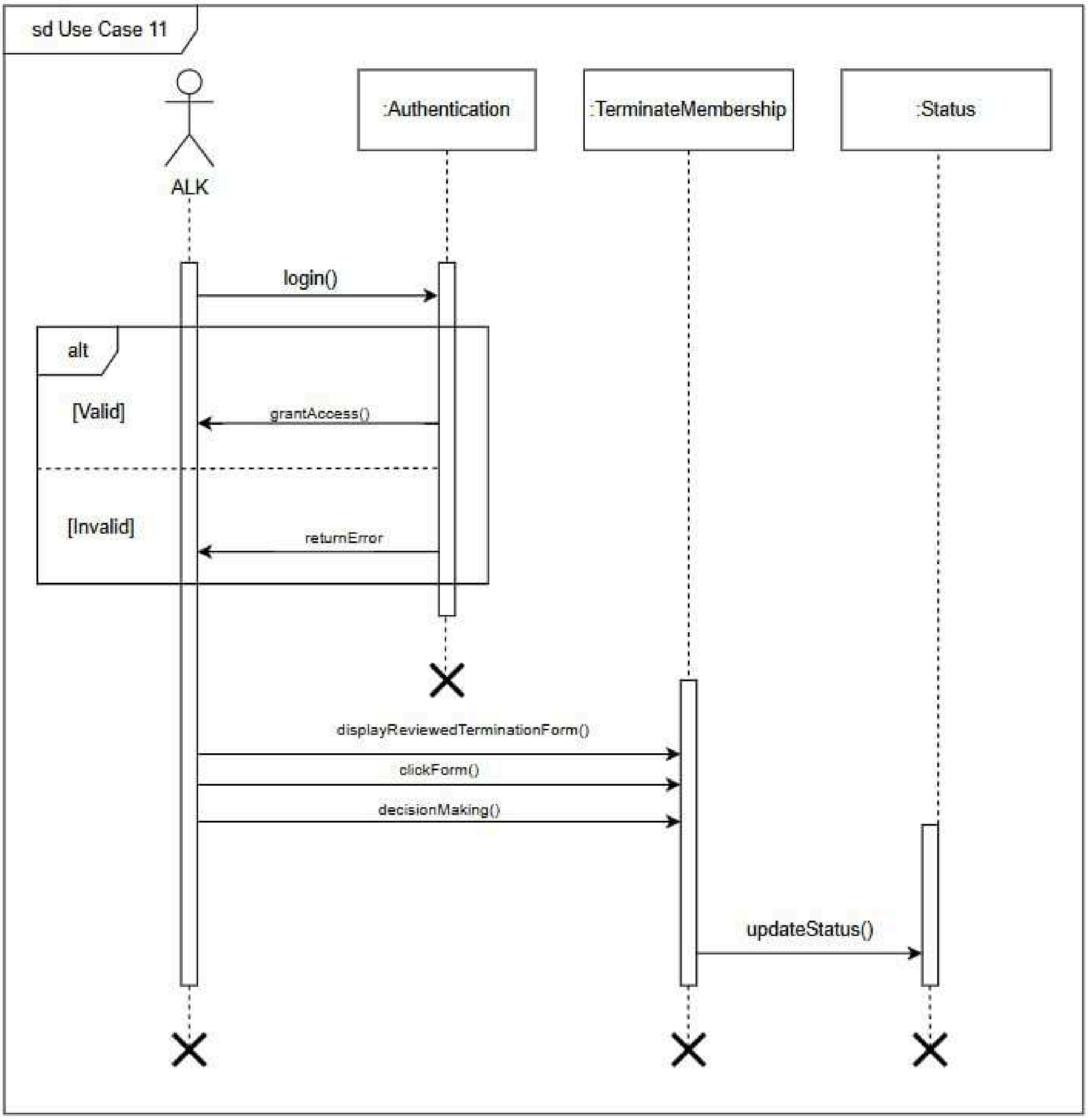


Figure 4.32: SD011 Sequence Diagram for < Approve Terminate Application> Subsystem

4.5 Deployment View - Deployment Diagram

This view focuses on how the system's components are distributed across hardware and software environments. It shows how the software is deployed on servers, devices, or systems, and how they interact with each other. Represented through a deployment diagram, this view illustrates the physical distribution of components across nodes and their communication paths. It helps to understand the system's layout, showing how components are allocated to machines or servers and how they interact over networks. The diagram also highlights the relationships between hardware, software, and external systems, ensuring the system runs efficiently in its environment.

Figure 4.26 illustrates the deployment diagram of the Kada Koperasi Automated System and highlights its key components along with their interactions. The system comprises four main nodes: the Client, Server, Database Server, and External Email Server. The Client node, represented by the artifact <<artifact>> Browser, communicates with the Server node, which includes <<artifact>> Internet Server and <<artifact>> Application Server to handle requests and execute business logic. The Application Server interacts with the Database Server, containing <<artifact>> MySQL DB, to manage and store cooperative data securely. Additionally, the External Email Server, represented by the artifact <<artifact>> Gmail SMTP Service, enables email notifications to send statement reports to members of koperasi KADA via the SMTP protocol.

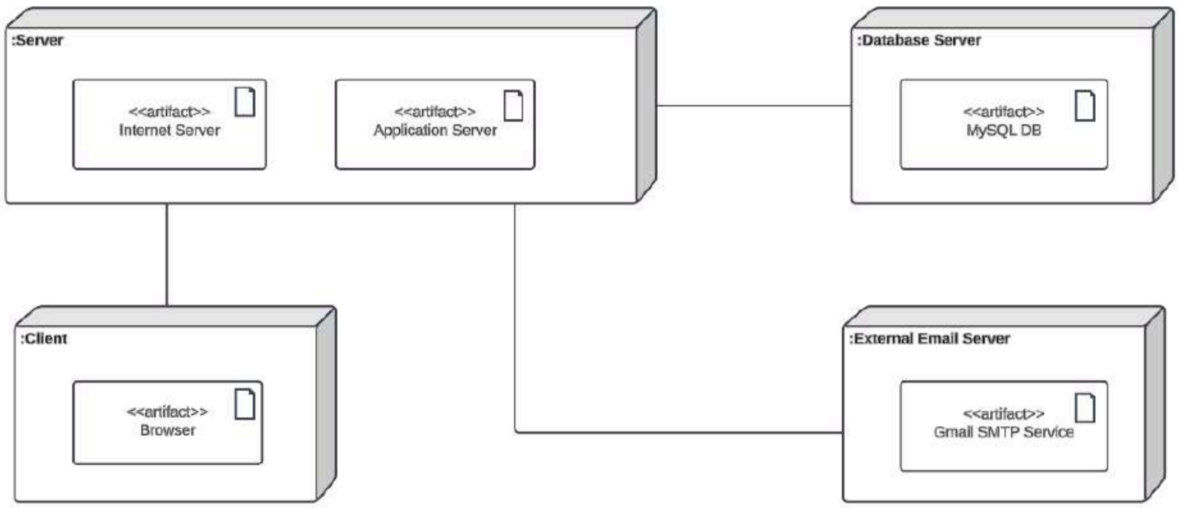


Figure 4.26: Deployment diagram for Koperasi KADA'S Automated System

5. Data Design

This section deals with the system by describing the information domain, using data entities and their relationships to represent the structured data. Various key entities have been listed, including Admin, ALK, Applicant, Loan, Membership, MembershipEnd and Notification. All of these entities are attributed according to their respective properties and behaviours. The major data or systems entities are stored into a relational database. Table 5.1 shows the description of entities in the database.

Table 5.1: Description of Entities in the Database

No.	Entity Name	Description
1	Administrator	General term describing the staff responsible for managing and reviewing membership and loan application
2	AdminApplicationFeedback	General term describing the reason of each application rejection by Admin
3	ALK	General term describing the authorized personnel responsible for approving membership and loan application
4	AlkApplicationFeedback	General term describing the reason of each application rejection by ALK
5	Applicant	General term describing all users logged in as applicants to apply membership.
6	BankInfo	General term describing the bank account information provided by applicants for processing membership applications.
7	FamilyInfo	General term for the details of the family members associated with an applicant.
8	Guarantor	General term describing a person who agrees to take responsibility for repaying a loan if the applicant has not paid for it.
9	Historymodimem	General term describing the past membership applications and the modification allowed
10	Historymodiloan	General term describing the past loan applications and the modification allowed

11	Loan	General term describing the financial agreement where an applicant borrows a specific amount of money to be repaid over a period with interest.
12	LoanRepayment	General term describing the financial activities related to loans, such as disbursements, repayments, and transaction history.
13	LoanType	General term describing the different categories or plans of loans offered
14	Managing	General term describing a notification update related to an approval process
15	Membership	General term describing to the record of applicants who are members of the Koperasi and
16	MembershipEnd	General term describing details of membership termination including reason, date of application, approval and review.
17	Saving	General term describing the funds saved by members, managed through deposits and balance tracking used as collateral for loans.
18	SavingType	General term describing type of savings the company had
19	Status	General term represents transaction records of saving.
20	User	General term describing all the people who used the system.
21	UserType	General term describing type of user who accesses the system

5.1 Data Dictionary

This section covers a data dictionary for the Entity Relation Diagram (ERD) in figure 5.1 which is a structured document that provides detailed information about the data used in a system. It describes the system's entities, attributes and type, and serves as a reference for understanding the data structure. By ordering attribute names alphabetically, table 5.2 until table 5.15 is the data dictionary for developers, designers and stakeholders to communicate with greater consistency, clarity and ease. The data dictionary describes classes and attribute names in the classes to offer a clear overview of the system's data components.

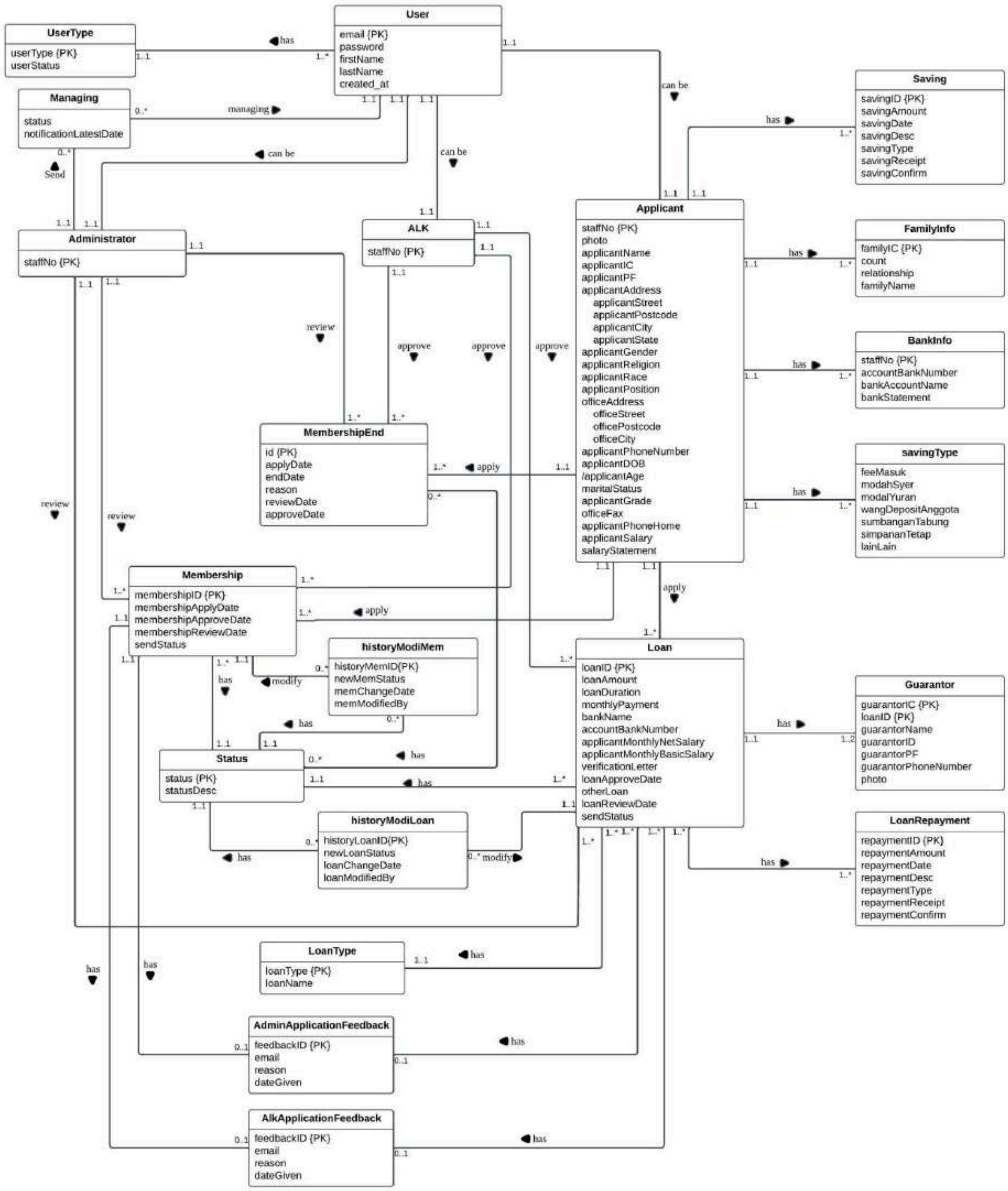


Figure 5.1 Entity Relationship Diagram (ERD)

5.1.1 Entity: Admin

Table 5.2: Data Dictionary for Admin

Attribute Name	Type	Description
staffNo	INT(10)	Admin’s staff number

5.1.2 Entity: AdminApplicationFeedback

Table 5.3: Data Dictionary for AdminApplicationFeedback

Attribute Name	Type	Description
feedbackID	INT (20)	Uniquely identifies feedback comment
email	VARCHAR(30)	Email
reason	LONGTEXT	Feedback messages to send
dateGiven	DATE	Date email was sent

5.1.3 Entity: ALK

Table 5.4 Data Dictionary for ALK

Attribute Name	Type	Description
staffNo	INT(10)	ALK’s staff number

5.1.4 Entity: ALKApplicationFeedback

Table 5.5: Data Dictionary for ALKApplicationFeedback

Attribute Name	Type	Description
feedbackID	INT (20)	Uniquely identifies feedback comment
email	VARCHAR(30)	Email
reason	LONGTEXT	Feedback messages to send
dateGiven	DATE	Date email was sent

5.1.5 Entity: Applicant

Table 5.6: Data Dictionary for Applicant

Attribute Name	Type	Description
applicantAddress	VARCHAR(50)	Home address of the applicant
applicantAge	INT(3)	Current Age of the applicant. This is the derivation of date of birth.
applicantCity	VARCHAR(30)	City of the applicant's address
applicantDOB	DATE	Date of birth of the applicant.
applicantGender	CHAR(1)	Gender of the applicant
applicantGrade	VARCHAR (10)	Professional grade of the applicant's job position
applicantIC	INT(12)	Identification Card number of the applicant
applicantName	VARCHAR(30)	Full name of applicant
applicantPF	VARCHAR(30)	Personal File number of the applicant
applicantPhoneHome	VARCHAR (15)	Home phone number of the applicant
applicantPhoneNumber	VARCHAR(15)	Primary contact number of the applicant
applicantPosition	VARCHAR(30)	Job position of the applicant
applicantPostcode	INT(5)	Postcode of the applicant's address
applicantRace	VARCHAR(25)	Race or ethnicity of the applicant
applicantReligion	VARCHAR(25)	Religion of the applicant
applicantSalary	DECIMAL (10,2)	Monthly Income of the applicant
applicantState	VARCHAR(30)	State of the applicant's address

maritalStatus	VARCHAR (10)	Marital status of the applicant
officeAddress	VARCHAR(50)	Address of the applicant’s workplace
officeCity	VARCHAR(30)	City of the applicant’s workplace
officeFax	VARCHAR (15)	Fax number of the applicant’s workplace
officePostcode	INT(5)	Postcode of the applicant’s workplace
staffNo	INT(5)	Applicant’s staff number

5.1.6 Entity: BankInfo

Table 5.7: Data Dictionary for BankInfo

Attribute Name	Type	Description
accountID	Account Number for transactions	INT(20)
bankAccountType	Name of the bank	VARCHAR(20)

5.1.7 Entity: FamilyInfo

Table 5.8: Data Dictionary for FamilyInfo

Attribute Name	Type	Description
count	INT(20)	Number of family member
familyIC	VARCHAR(15)	Family member Identification Number
familyName	VARCHAR(25)	Name of the family member
relationship	VARCHAR(10)	Relationship of applicant with family member

5.1.8 Entity: Guarantor

Table 5.9: Data Dictionary for Guarantor

Attribute Name	**Type**	**Description**
guarantorIC	VARCHAR(20)	Guarantor identification card
guarantorID	INT(20)	Unique identifier for guarantor
guarantorName	VARCHAR(30)	Guarantor full name
guarantorPF	VARCHAR(20)	Guarantor personal file number
guarantorPhoneNumber	VARCHAR(20)	Guarantor phone number
loanID	INT (5)	Loan identification number
photo	LONGBLOB	Guarantor photo

5.1.9 Entity : historymodimem

Table 5.10: Data Dictionary for historymodimem

Attribute Name	**Type**	**Description**
historyMemID	INT(5)	Unique identifier for each past applications
newMemStatus	INT(3)	New status after modification
memChangeDate	DATE	Date status modified
memModifiedBy	INT(5)	Modifier’s staff number

5.1.10 Entity : historymodiloan

Table 5.11: Data Dictionary for historymodiloan

Attribute Name	**Type**	**Description**
historyLoanID	INT(5)	Unique identifier for each past applications
newLoanStatus	INT(3)	New status after modification
loanChangeDate	DATE	Date status modified
loanModifiedBy	INT(5)	Modifier’s staff number

5.1.11 Entity: Loan

Table 5.12: Data Dictionary for Loan

Attribute Name	**Type**	**Description**
accountBankNumber	INT(20)	Bank account number
applicantMonthlyBasicSalary	DECIMAL(10,2)	Monthly basic salary amount
applicantMonthlyNetSalary	DECIMAL(10,2)	Monthly income amount
bankName	VARCHAR(20)	Bank account name
loanAmount	DECIMAL(10, 2)	Loan requested amount
loanApplyDate	DATE	Loan application apply date
loanApproveDate	DATE	Loan application approved date by ALK
loanDuration	INT(2)	Loan repayment duration
loanID	INT(20)	Unique identifier for each loan application
loanReviewDate	DATE	Loan application reviewed date by admin
loanStatus	INT(10)	Loan application status
loanType	INT(10)	Specified loan types
monthlyPayment	DECIMAL(10,2)	Monthly Payment amount

5.1.12 Entity: LoanRepayment

Table 5.13: Data Dictionary for LoanRepayment

Attribute Name	**Type**	**Description**
RepaymentID	INT(5)	Uniquely identifies saving transaction
RepaymentAmount	DECIMAL(10, 2)	Repayment amount
RepaymentDate	DATE	Repayment date
RepaymentDesc	VARCHAR(200)	Repayment description
RepaymentType	VARCHAR(30)	Repayment type
RepaymentReceipt	VARCHAR(15)	Repayment receipt

RepaymentConfirm	VARCHAR(15)	Repayment confirmation

5.1.13 Entity: LoanType

Table 5.14: Data Dictionary for LoanType

Attribute Name	**Type**	**Description**
loanType	INT(7)	Unique identifier for the loan type
loanName	VARCHAR(25)	Loan name

5.1.14 Entity: Managing

Table 5.15: Data Dictionary for Managing

Attribute Name	**Type**	**Description**
status	VARCHAR(25)	Email sent status
notificationLatestDate	VARCHAR(25)	Date and time of email sent

5.1.15 Entity: Membership

Table 5.16: Data Dictionary for Membership

Attribute Name	**Type**	**Description**
membershipStatus	VARCHAR(20)	Current status of the membership request
membershipApplyDate	DATE	Membership application apply date
membershipApproveDate	DATE	Membership application approved date by ALK
membershipReviewDate	DATE	Membership application reviewed date by ALK

5.1.16 Entity: MembershipEnd

Table 5.17: Data Dictionary for Membership

Attribute Name	**Type**	**Description**
applyDate	DATE	Terminate membership apply date

approveDate	DATE	Approve date
endDate	DATE	Membership end date
id	INT (5)	Uniquely identifies membership end application form
reason	VARCHAR(255)	Reasons message to end membership
reviewDate	DATE	reviewDate

5.1.17 Entity: Saving

Table 5.18: Data Dictionary for Saving

Attribute Name	**Type**	**Description**
savingID	INT(5)	Uniquely identifies saving transaction
savingAmount	DECIMAL(10, 2)	Saving amount
savingDate	DATE	Saving date
savingDesc	VARCHAR(200)	Saving description
savingType	VARCHAR(30)	Saving payment type
savingReceipt	VARCHAR(15)	Saving receipt
savingConfirm	VARCHAR(15)	Saving confirmation

5.1.18 Entity: SavingType

Table 5.19: Data Dictionary for Saving

Attribute Name	**Type**	**Description**
feeCapital	DECIMAL(10, 2)	Fee capital amount
fixedDeposit	DECIMAL(10, 2)	Fixed deposit amount
lainLain	DECIMAL(10, 2)	Others saving amount
memberDeposit	DECIMAL(10, 2)	Member deposit amount
memberSavings	DECIMAL(10, 2)	Member savings amount

shareCapital	DECIMAL(10, 2)	Share capital amount
savingID	INT AUTO_INCREMENT	Unique identifier for the saving record

5.1.19 Entity: Status

Table 5.20: Data Dictionary for Status

Attribute Name	Type	Description
status	INT (3)	Status
statusDesc	VARCHAR (25)	Status description

5.1.20 Entity: User

Table 5.21: Data Dictionary for User

Attribute Name	Type	Description
created_at	DATE	Account created date
email	VARCHAR(25)	Email for communication and receiving any update information.
firstName	VARCHAR(25)	Given name of the user
lastName	VARCHAR(20)	Family Name of the user
password	VARCHAR(15)	Password for user authentication
staffNo	INT(10)	Staff Number of the user

5.1.21 Entity: UserType

Table 5.22: Data Dictionary for userType

Attribute Name	Type	Description
userStatus	VARCHAR(10)	User type in Koperasi Kada
userType	int(5)	Number indicates user type.

6. User Interface Design

The interface includes sections designed for different types of users which are applicants, administrators, and ALK.

For applicants, the interface starts with a login page where users enter their credentials to access the system. Once logged in, they see a menu with options like applying for membership, applying for membership termination, applying for a loan, and using a loan calculator. The membership application involves filling out a form in two steps where users provide personal details and submit their application. For loan applications, users can calculate repayment amounts, fill out forms with personal and guarantor information, sign agreements. Furthermore, the termination of membership application follows a similar process where users fill out a form with their details and state their reason. All applications can check the status of their application after submitting the application. These features ensure users can easily manage their membership and loan applications.

For administrators, the interface focuses on managing the application, including membership, membership termination and loan applications, financial transactions, notifications, and reporting. Admins log in through an admin page to access tools for reviewing applications, updating membership statuses, and verifying applicants' financial and guarantor information before proceeding with approvals. They also manage transactions by recording payments, tracking outstanding balances, and updating payment statuses. Email notifications are sent to inform users about application statuses and financial statements. Additionally, administrators can generate detailed reports on transactions, loan approvals, and membership activities, which are then emailed to relevant parties for record-keeping and decision-making.

The ALK interface is designed for approving or rejecting applications. After logging in, ALK users can choose to review either membership, loan or terminate application. For membership forms, ALK users can approve or reject applications directly. For loan applications, they can check the loan details, compare them with the terms, and review supporting documents like payslips. The system clearly shows whether the application is successful or not, allowing ALK users to make quick decisions.

In summary, the interfaces are designed to help users manage applications, administrators handle reviews, and ALK staff approve or reject the applications efficiently.

6.1 Screen Images

The Koperasi KADA's Automated System's first-time user should see the login page when he/she opens the website like in Figure 6.1. If the user has not registered, he/she should be able to do that on the login page. If the user is already registered, the user can login and will be redirected to which type of user they are either user, admin or Ahli Lembaga Koperasi (ALK). After Logging in, Figure 6.2 allows users to choose from various menu options, including applying for membership, applying for a loan, and calculating loan details as well as applying for a membership termination.

Figure 6.1 <Login Page>

Figure 6.2 <Applicant's Menu>

When users choose to apply for membership, they are directed to Figure 6.3, where they must fill out the membership form. After clicking the "Next" button, they proceed to Figure 6.4 to complete their family information, bank information, and saving information. Once all required information is filled in and they have to agree with an agreement like in Figure 6.7 before submitting the form. Lastly, the application status will be shown as shown in Figure 6.8, indicating whether the membership application is pending approval, is being reviewed, or needs further information.

Figure 6.3 <Membership Form>

Figure 6.4 <Family Information>

Figure 6.5 <Bank Information>

Figure 6.6 <Saving Information>

Figure 6.7 <Applicant's Agreement>

Figure 6.8<Membership Status>

The prerequisite to apply for a loan is, the user must have a Koperasi KADA membership in order to apply for a loan using the website. Before filling out the loan application form as in Figure 6.10 with the loan type, amount, and duration, they can utilize the loan calculator to estimate monthly repayments like in Figure 6.9. After filling out loan information, the system will display member's information as in Figure 6.11. In Figure 6.12, the user must provide the details of a guarantor and have an agreement to the terms like in Figure 6.13 before submitting the form. Lastly, the application status will be shown as shown in Figure 6.14, indicating whether the loan is pending approval, is being reviewed, or needs further information.

Figure 6.9 <Loan Calculator>

Figure 6.10 <Loan Application Form>

Figure 6.11 <Applicant’s Information> Figure 6.12 <Guarantor Information>

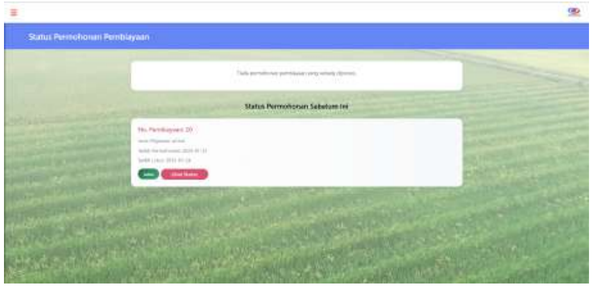


Figure 6.13 <Agreement> Figure 6.14 <Loan Status>

Furthermore, if a member wishes to terminate their membership, they can do so by filling out the application in Figure 6.15 and providing reason as in Figure 6.16. After completing the application, the applicant needs to have an agreement by clicking in the checkbox in Figure 6.17. Then, the application status will also be shown like the other two applications, as in Figure 6.18, indicating whether the loan is pending approval, is being reviewed, or needs further information.

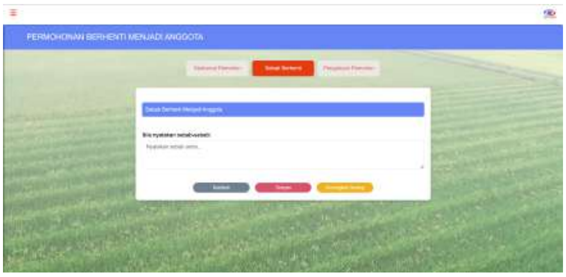


Figure 6.15 <Terminate Application> Figure 6.16 <Reason of Termination>

Figure 6.17 <Agreement> Figure 6.18 <Terminate Application Status>

Admin is prompted to login first by entering Admin ID and Password as shown in Figure 6.1. After successfully logging in as an admin, it will show options to manage various aspects of the organization. These options include reviewing the membership list, reviewing the loan application list, reviewing membership termination application list, and reviewing members approved by ALK as shown in Figure 6.19.

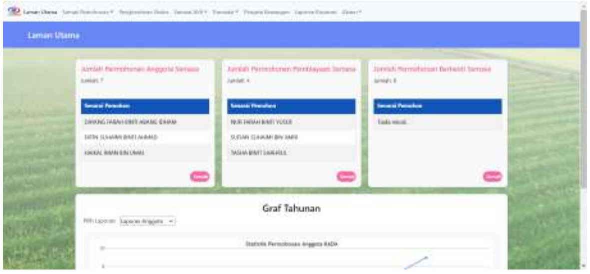


Figure 6.19 <Admin Menu>

After logging in as an admin and selecting the "Membership list", “Loan Application List” or “Membership Termination Application List” option as shown in Figure 6.20, 6.21, and 6.22 admin will see a list of members and their review status (either "Completed" or "Not Completed"). To review a member's form, click the applicant’s name. After completing the review of all details, return to the main review form to update the review status to "Completed" if the application is completed. Otherwise, Admin will update the review status to “Not Completed”. Last but not least, Admin can perform batch processing by clicking any intended names using the checkbox to approve or reject simultaneously.

Figure 6.20 <Membership Applicant’s Name List> **Figure 6.21 <Loan Application Name List>**

Figure 6.22 <Membership Termination Application Name List>

Next, Figure 6.23 illustrates the ALK’s menu which shows the activity handled by ALK. To approve membership, ALK can click the Senarai Permohonan Anggota to view the list of applications reviewed by Admin. Figure 6.24 shows the list of names that are waiting to be approved or rejected. ALK can simply click the button ‘Diterima’ or ‘Ditolak’ to perform decision making for each application.

Figure 6.23 <ALK’s Menu>

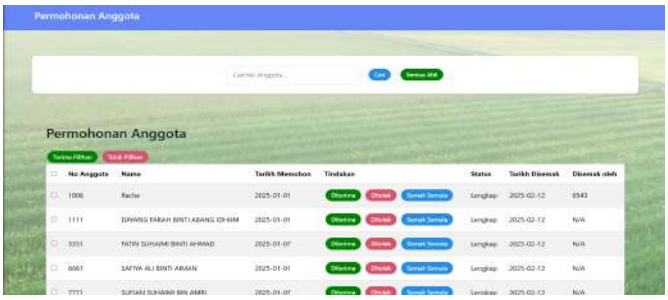


Figure 6.24 <Membership List>

If the decision-maker is unsure and wants to recheck the application, ALK can click the ‘Semak Semula’ button to view the applicant’s application. Figure 6.25 shows the applicant’s information after clicking ‘Semak Semula’. Last but not least for approving membership, ALK can perform batch processing by clicking any intended names using the checkbox to approve or reject simultaneously. Figure 6.26 shows the batch processing function.

Figure 6.25 <Application’s Information>

Figure 6.26 <Batch Processing>

Moving to approving loans, ALK can choose ‘Senarai Permohonan Pembiayaan in ALK’ menu to access the loan applications reviewed by Admin. Same goes for loans, the ALK can directly perform decision-making by clicking the ‘Diterima’ or ‘Ditolak’ button directly from the list. ALK also can view the applications by clicking the ‘Semak Semula’ button. Figure 6.27 and Figure 6.28 show the list of names with buttons and applicant’s loan application respectively. After viewing applicant’s information, ALK can use the ‘Bandin’ button to perform comparison of that particular application with the company's loan policies. Figure 6.29 shows the location of the 'Banding' button and Figure 6.30 illustrates the comparison result.

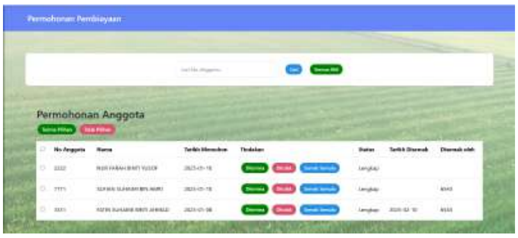


Figure 6.27 <Loan List>

Figure 6.28 <Applicant's Info>

Figure 6.29 <Banding Button>

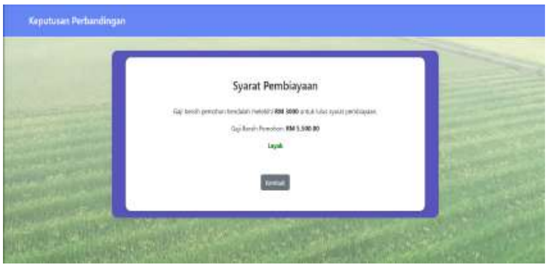


Figure 6.30 <Comparison Result>

ALK is also responsible for approving membership termination applications. Figure 6.31 shows the list of applications reviewed by Admin. ALK can perform decision-making simply by clicking the 'Diterima' or 'Ditolak' button. ALK also can view the applicant's information as shown in Figure 6.32. Lastly, all membership, loan and terminate membership approval supports batch processing using checkbox to perform decision-making to more than one applications simultaneously.

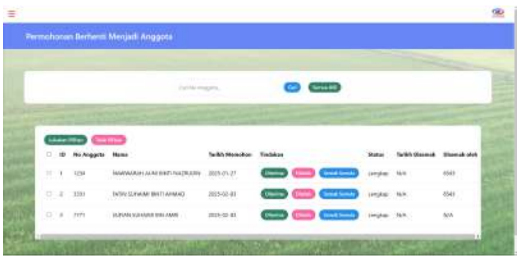


Figure 6.31 <Terminate List>

Figure 6.32 <Application's Info>

Figure 6.33 and Figure 6.34 display the interface for administrators to manage monthly salary deductions and irregular transactions. Administrators can review and update transactions by clicking the "Keluarkan" button or filling the form. Meanwhile, Figure 6.35 and Figure 6.36

show the applicant's interface for viewing savings and loan repayment records. Applicants can check their transaction details to ensure the displayed financial information is clear and easily accessible.

Figure 6.33 <Monthly Salary Deduction>

Figure 6.34 <Irregular Transaction>

Figure 6.35 <Savings Record>

Figure 6.36 <Loan Repayment Record>

Next, Figure 6.37 displays a list of membership, loan, and termination applications that have been reviewed by ALK and require applicants to be notified via email. Figure 6.38 provides an option to generate a statement for the approved loan applicant list. When the administrator clicks the "Hasilkan" button, the system automatically generates the report statement, as shown in Figure 6.39. The administrator can then send the report statement via email for applicant reference. Finally, Figure 6.40 presents the corporation report, which includes both monthly and yearly summaries. This report helps in monitoring the overall financial performance and membership activities within the cooperative.

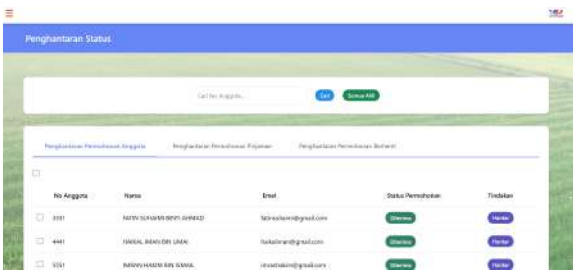


Figure 6.37 <List of Applicant's Status>

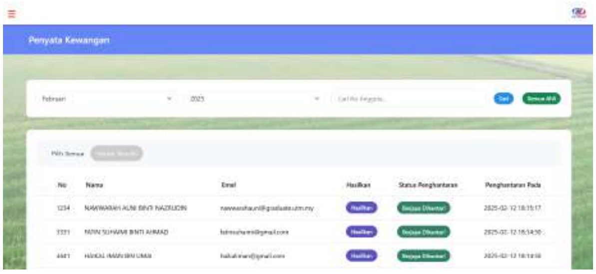
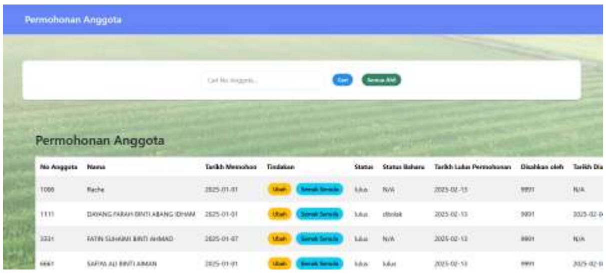


Figure 6.38 <List of Member of Koperasi>

Figure 6.39 <Generated Statement>

Figure 6.40 <Corporation report>

Lastly, ALK can modify the status of each application. This function was used when ALK unintentionally performed a wrong decision-making process. Figure 6.41 and Figure 6.42 illustrate the modification interface of membership and loan application respectively.



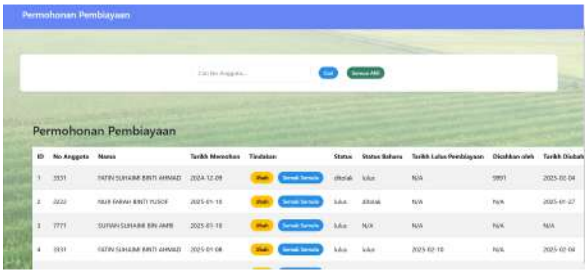


Figure 6.41 <Membership Modification>

Figure 6.42 <Loan Modification>

7. Traceability

This section shows a table that connects user stories to the related subsystems and packages. It helps track which sprint and Agile artifact each story belongs to, making it easier to ensure all parts of the system meet the requirements in the product backlog.

Table 7.1: Traceability

<table>
<tr><th>Sprint #</th><th>Subsystem/Package</th><th>User Story ID</th></tr>
<tr><td>1</td><td>Package Membership Application Module (P001)</td><td>US001</td></tr>
<tr><td>2</td><td>Package Review Module (P002)</td><td>US002</td></tr>
<tr><td>3</td><td>Package Approve Application Module (P004)</td><td>US003</td></tr>
<tr><td>4</td><td rowspan="2">Package Loan Application Module (P003)</td><td rowspan="2">US001</td></tr>
<tr><td>5</td></tr>
<tr><td>6</td><td>Package Approve Application Module (P004)</td><td>US003</td></tr>
<tr><td>7</td><td rowspan="2">Package Statement Report Production Module (P005)</td><td rowspan="2">US002</td></tr>
<tr><td>8</td></tr>
<tr><td>9</td><td>Financial Tracking Module (P007)</td><td>US002</td></tr>
<tr><td>10</td><td>Terminate Module (P006)</td><td>US001</td></tr>
<tr><td>11</td><td>Package Approve Application Module (P004)</td><td>US003</td></tr>
</table>

8. **Test Cases**

This section provides test cases for all the subsystems. It ensures that a subsystem is functional and reliable by describing the test scenarios that include normal, alternative and exceptional cases. Each test case corresponds to a particular package, use case and sequence diagram that provide a systematic way of testing behaviour of the system under some considerations. These test cases should be designed in such a way that they point towards problems while ensuring the system is meeting its requirements.

8.1 TC001: Test <Membership Application Module (P001)> Subsystem: <Login to Apply Membership (UC01)>

This test contains the following test cases:

(a) TC001_01: Test <Login to Apply Membership (SD001)>

8.1.1 **TC001_01: Test <Login to Apply Membership (SD001)>**

This test contains the following scenarios:

(a) TC001_01: Test <normal scenario of Login to Apply Membership (SD001)

DESCRIPTION	VERSION	TEST DATE
Test the Login to Apply Membership Application functionality for applicants.	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
1. Register as an applicant to login. 2. The Membership Application section enables applicants to submit the membership applications online. 3. Input data by applicant insert into the database	SAFIYA	

Test Case ID **(TC001_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to register page	First Name : AMINAH Last Name : HADI Email : aminah@gmail.com Password : 12345678 User Type : Pemohon	Information can be entered and the applicant will be redirected to the login page.	As Expected	PASS	SAFIYA	
2	Navigate to login page	**Email : aminah@gmail.com** **Password : 12345678**	Credential can be entered	As Expected	PASS	SAFIYA	
3	Click Submit	-	Customer is logged in	As Expected	PASS	SAFIYA	
4	Navigate to membership application page	-	The membership application page should be open	As Expected	PASS	SAFIYA	
5	Enter applicant's information	applicantName : AMINAH applicantIC : 547656 applicantDOB : 2025-01-01	All information required completed	As Expected	PASS	SAFIYA	

		applicantAge : 1 maritalStatus : Bujang applicantGender : PEREMPUAN applicantReligion : Islam applicantRace : Melayu applicantStreet : No. 11A Jalan Kantansari 3 applicantCity : Rawang applicantPostcode : 48010 applicantState : Selangor staffNo : 10001 applicantPF : 10001 applicantPosition : MANAGER applicantGrade : S21 officeStreet : No. 11A Jalan Kantansari 3 officeCity : Rawang officePostcode : 48010 officeState : Selangor officeFax: 0360223345 applicantPhoneNumber : 0175969209 applicantPhoneHome : - applicantSalary : 1234.00					
6	Click “SIMPAN”	-	Applicants will be redirected to the family information page.	As Expected	Pass	SAFIYA	
7	Enter family information	relationship : SUAMI familyName : ABID familyIC : 8977655	All information required completed	As Expected	PASS	SAFIYA	
8	Click “SIMPAN”	-	Applicants will be redirected to the bank information page.	As Expected	Pass	SAFIYA	
9	Enter bank information	bankAccountName : OCBC Bank Malaysia accountBankNumber : 786554564	All information required completed	As Expected	PASS	SAFIYA	
10	Click “SIMPAN”	-	Applicants will be redirected to the saving information page.	As Expected	Pass	SAFIYA	
11	Enter saving information	feeMasuk : 1 modahSyer : 1 modalYuran : 11.00 wangDepositAnggota : 0.00 sumbanganTabung : 0.00 simpananTetap : 0.00 lainLain : 1.00	All information required completed	As Expected	PASS	SAFIYA	
12	Click “SIMPAN”	-	Applicants will be redirected to the applicant's statement page.	As Expected	PASS	SAFIYA	
13	Tick the box if the applicant’s agree with the statement page.		Applicants need to tick off the box.	As Expected	PASS	SAFIYA	
14	Click “HANTAR”	-	Status application is displayed	As Expected	PASS	SAFIYA	

8.2 TC002: Test <Review Module (P002)> Subsystem: <Review Applicant's Form (UC02)>

This test contains the following test cases:

(a) TC002_01: Test <Review Applicant's Form (SD002)>

8.2.1 TC002_01: Test <Review Applicant Form (SD002)>

This test contains the following scenarios:

(a) TC002_01: Test <normal scenario of Review Applicant's Form (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the Review Membership Application Form and Loan Application Form Functionality for Admin	1.0	14-Jan-2025
TEST OBJECTIVE	**AUTHOR**	**REVIEWER**
1. Review membership application form and loan application form. 2. Can update the status of the forms either "Lengkap" or "Tidak Lengkap"	SAFIYA	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Enter User email & Password	email: aliaabu@gmail.com password: 12345678	Credential can be entered and Admin dashboard page is displayed	As Expected	PASS	SAFIYA	
	Navigate to Admin applicant page	-	The page should display three options which is Senarai Permohonan Anggota, Senarai Permohonan Berhenti and Senarai Permohonan Pembiayaan	As Expected	PASS	SAFIYA	
	Click Senarai Permohonan Anggota	-	The page display the list of membership applications	As Expected	PASS	SAFIYA	
	Click on one of the applicant's name	-	The page will display all the information entered by the applicant.	As Expected	PASS	SAFIYA	
	Choose one applicant or more than one		If choose more than one, the checkbox are highlighted	As Expected	Pass	SAFIYA	
	Click "Lengkap"	adminStaffNo = 6543	The status column changed to Lengkap	As expected	PASS	SAFIYA	
	Click "Tidak Lengkap"	adminStaffNo = 6543 reason = Slip Gaji tidak sah	The status column changed to Tidak Lengkap	As Expected	PASS	SAFIYA	
	Click Senarai Permohonan Pembiayaan	-	The page display the list of loan applications	As Expected	PASS	SAFIYA	
	Click on one of the applicant's name	-	The page will display all the information entered by the applicant.	As Expected	PASS	SAFIYA	

	Choose one applicant or more than one		If choose more than one, the checkbox are highlighted	As Expected	Pass	SAFIYA	
	Click “Lengkap”	adminStaffNo = 6543	The status column changed to Lengkap	As Expected	PASS	SAFIYA	
	Click “Tidak Lengkap”	adminStaffNo = 6543 reason = Slip Gaji tidak sah	The status column changed to Tidak Lengkap	As Expected	PASS	SAFIYA	
	Click Senarai Permohonan Berhenti	-	The page display the list of membership termination applications	As Expected	PASS	SAFIYA	
	Click on one of the applicant’s name	-	The page will display all the information entered by the applicant.	As Expected	PASS	SAFIYA	
	Choose one applicant or more than one		If choose more than one, the checkbox are highlighted	As Expected	Pass	SAFIYA	
	Click “Lengkap”	adminStaffNo = 6543	The status column changed to Lengkap	As Expected	PASS	SAFIYA	
	Click “Tidak Lengkap”	adminStaffNo = 6543 reason = Alasan tidak munasabah	The status column changed to Tidak Lengkap Confirmation message “Status berjaya dikemaskini!” and the application are removed from the list	As Expected	PASS	SAFIYA	

8.3 **TC003: Test <Loan Application (P003)> Subsystem: <Apply Loan (UC04)>**

This test contains the following test cases:

(a) TC003_01: Test <Apply Loan (SD004)>

8.3.1 TC003_01: Test <Apply Loan (SD004)>

This test contains the following scenarios:

(a) TC003_01: Test <normal scenario of Review Applicant’s Form (SD001)>

DESCRIPTION	VERSION	TEST DATE
Test the Loan Application section functionality for applicant	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
1. The Loan Application section enables members to submit the loan applications online. 2. Input data by member insert into the database	FARRA	

Test Case ID **(TC001_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to loan applicant page	-	The loan application page should be open	As Expected	Pass	Farra	

2	Enter loan information required	**loanType**= Al-Bai **loanAmount**= 50000 **loanDuration**=100 **bankName**=RHB **accountBankNumber** =1501105123466544 **netSalary**=3150 **basicSalary**=3500 **gName1**=BUNGA BINTI RASHID **gIC1**=9008176790 **gPF1**=9076 **gID1**=6790 **gPhoneNumber1**=01 907667098656 **gName2**=CAMELIA HONG **gIC2**=911111112234 **gPF2**=2234 **gID2**=4322 **gPhoneNumber2**=01 223443221	All information required completed	As Expected	Pass	Farra	
3	Click Submit		Status application is displayed	As Expected	Pass	Farra	

8.4 TC004: Test <Loan Application (P003)> Subsystem: <Calculate Loan (UC05)>

This test contains the following test cases:

(a) TC004_01: Test <Calculate Loan (SD005)>

8.4.1 TC004_01: Test <Calculate Loan (SD005)>

This test contains the following scenarios:

(b) TC004_01: Test <normal scenario of Calculate Loan (SD005)>

DESCRIPTION	VERSION	TEST DATE
Test the functionality of calculator in the system	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
1. Function can correctly calculate results.	FARRA	

Test Case ID **(TC001_01 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to loan calculator page		The loan calculator page should be open	As Expected	Pass	Farra	
2	Enter Loan Amount& Loan Duration (Years)	**loanAmount**= 1500 **loanDuration**=-3	Loan monthly repayment calculated and displayed	As Expected	Pass	Farra	
3	Click Submit		Monthly repayment is calculated	As Expected	Pass	Farra	

8.5 TC005: Test <Approval Module (P004)> Subsystem: <Approve Membership (UC03)>

This test contains the following test cases:

(a) TC005_01: Test <Approve Membership (SD003)>

8.5.1 TC005_01: Test <Approve Membership (UC03)>

This test contains the following scenarios:

(a) TC004_01: Test <normal scenario of Approve Membership (SD003)>

DESCRIPTION	VERSION	TEST DATE
Test the Login Functionality in Banking	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
1. The system enable ALK to approve membership applications	Dayang	

Test Case ID **(TC004_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to ALK page		The page should display ALK's menu	As Expected	Pass	Dayang	
2	Click Senarai Permohonan Anggota		The page display the list of membership applcations	As Expected	Pass	Dayang	
3	Choose one applicant or more than one		If choose more than one, the checkbox are highlighted	As Expected	Pass	Dayang	
4	Click Diterima	alkStaffNo = 9991	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	As Expected	Pass	Dayang	
5	Click Ditolak	alkStaffNo = 9991 reason = Slip Gaji tidak sah	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	As Expected	Pass	Dayang	
6	Click Semak Semula		ALK redirected to applicant information page	As Expected	Pass	Dayang	

8.6 TC006: Test <Approval Module (P005)> Subsystem: <Approve Loan Application (UC06)>

This test contains the following test cases:

(a) TC006_01: Test <Approve Loan Application(SD006)>

8.6.1 TC006_01: Test <Approve Loan Application (UC06)>

This test contains the following scenarios:

(a) TC006_01: Test <normal scenario of Approve Loan Application (SD006)>

DESCRIPTION	VERSION	TEST DATE
Test the Login Functionality in Banking	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
1. The system enable ALK to approve loan applications	Dayang	

Test Case ID (TC001_01_01)	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to ALK page		The page should display ALK's menu	As Expected	Pass	Dayang	
2	Choose Senarai Permohonan Pembiayaan		The page display the list of loan applcations	As Expected	Pass	Dayang	
3	Click Ditolak	alkStaffNo = 9991 reason = Slip Gaji tidak sah	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	As Expected	Pass	Dayang	
4	Click Diterima	alkStaffNo = 9991	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	As Expected	Pass	Dayang	
5	Click Semak Semula		The page display applicant information	As Expected	Pass	Dayang	
6	Go through the applicant details by clicking every section		The page display applicant information	As Expected	Pass	Dayang	
7	Click Banding button		Comparing applicant's information with loan term	As Expected	Pass	Dayang	

8.7 TC007: Test <Approval Module (P004)> Subsystem: <Approve Terminate Application (UC11)>

This test contains the following test cases:

(b) TC007_01: Test <Approve Terminate Application(SD011)>

8.7.1 TC007_01: Test <Approve Terminate Application(UC011)>

This test contains the following scenarios:

(b) TC007_01: Test <normal scenario of Approve Terminate Application (SD011)>

DESCRIPTION	VERSION	TEST DATE
Test the Login Functionality in Banking	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
2. The system enable ALK to approve membership termination application	Dayang	

Test Case ID **(TC004_01 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to ALK page		The page should display ALK's menu	As Expected	Pass	Dayang	
2	Click Senarai Permohonan Berhenti		The page display the list of membership termination applcations	As Expected	Pass	Dayang	
3	Choose one applicant or more than one		If choose more than one, the checkbox are highlighted	As Expected	Pass	Dayang	
4	Click Diterima	alkStaffNo = 9991	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	As Expected	Pass	Dayang	
5	Click Ditolak	alkStaffNo = 9991 reason = Belum boleh nyahaktif	Javascript display "Status berjaya dikemaskini!" and the application are removed from the list	As Expected	Pass	Dayang	
6	Click Semak Semula		ALK redirected to applicant information page	As Expected	Pass	Dayang	

8.8 TC008: Test <Statement Report Production Module (P005)> Subsystem: <Prepare Statement and Report (UC07)>

This test contains the following test cases:

(a) TC008_01: Test <Prepare Statement and Report (SD007)>

8.8.1 TC008_01: Test <Prepare Statement and Report (UC07)>

This test contains the following scenarios:

(a) TC008_01: Test <normal scenario of Prepare Statement and Report (SD007)>

DESCRIPTION	VERSION	TEST DATE
Test the functionality of Prepare Statement and Report	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
Validate and verify the accuracy for the statements and reports	Auni	

Test Case ID **(TC001_01 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Enter User email & Password	email: aliaabu@gmail.com password: 12345678	Admin dashboard page is displayed	As Expected	Pass	Auni	
2	Click Penyata Kewangan on navigation bar		List of koperasi members displayed	As Expected	Pass	Auni	

3	Click the Hasilkan button in the same row as the desired members		System processes and prepares the statement	As Expected	Pass	Auni	
4	View the generated statement		Financial statement of selected members displayed	As Expected	Pass	Auni	
5	Verify the statement data accuracy		Statement contains correct applicant financial information	As Expected	Pass	Auni	
6	Click Laporan Koperasi in navigation bar		Corporation report displayed	As Expected	Pass	Auni	
7	View the corporation report for monthly and yearly		Report contains correct statistic of membership and loan applications	As Expected	Pass	Auni	

8.9 TC009: Test <Statement Report Production Module (P005)> Subsystem: <Notify via Email (UC08)>

This test contains the following test cases:

(a) TC009_01: Test <Notify via Email (SD008)>

8.9.1 TC009_01: Test <Notify via Email (UC08)>

This test contains the following scenarios:

(a) TC009_01_01: Test <normal scenario of Notify via Email (SD008)>

DESCRIPTION	VERSION	TEST DATE
Test the Email functionality in the system	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the email functionality in the system is working correctly	Auni	

Test Case ID **(TC001_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Click Penyata Kewangan on navigation bar		List of koperasi members displayed	As Expected	Pass	Auni	
2	Locate prepared statements		Prepared statements is visible	As Expected	Pass	Auni	
3	Click send button		Notification of email berjaya dihantar!	As Expected	Pass	Auni	
4	Click Ok		Status penghantaran and penghantaran pada column is updated	As Expected	Pass	Auni	
5	Click penghantaran Status on navigation bar		List of membership and loan applicants displayed	As Expected	Pass	Auni	

6	Click send button for desired applicants		Notification of successful email notification	As Expected	Pass	Auni	

8.10 TC010: Test <Terminate Module (P006)> Subsystem: <Terminate Membership (UC010)>

This test contains the following test cases:

(a) TC010_01: Test <Terminate Membership (SD010)>

8.10.1 TC010_01: Test <Terminate Membership (UC010)>

This test contains the following scenarios:

(b) TC010_01_01: Test <normal scenario of Terminate Membership (SD010)>

DESCRIPTION	VERSION	TEST DATE
Test the Terminate Application section functionality for applicant	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
1. The Terminate Application section enables members to submit the terminate applications online. 2. Input data by member insert into the database	Farra	

Test Case ID **(TC001_01 _01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Navigate to terminate applicant page		The terminate application page should be open	As Expected	Pass	Farra	
2	Enter terminate information required	Reason: Pencen	All information required completed	As Expected	Pass	Farra	
3	Click Submit		Status application is displayed	As Expected	Pass	Farra	

8.11 TC011: Test <Financial Tracking Module (P007)> Subsystem: <Manage Transactions (UC009)>

This test contains the following test cases:

(b) TC011_01: Test <Manage Transactions (SD009)>

8.11.1 TC011_01: Test <Financial Tracking Module (P007)>

This test contains the following scenarios:

(b) TC009_01: Test <normal scenario of Manage Transactions (SD008)>

DESCRIPTION	VERSION	TEST DATE

Test the transactions functionality in the system	1.0	14-Jan-2025
TEST OBJECTIVE	AUTHOR	REVIEWER
To verify that the financial transactions in the system is working correctly	Auni	

Test Case ID **(TC001_01_01)**	Test Steps	Input Data	Expected Results	Actual Results	PASS/ FAIL	Tested By	Tester Comments
1	Click Transaksi Bulanan on navigation bar		List of koperasi members savings and loans transactions before and after deductions displayed	As Expected	Pass	Auni	
2	Click Keluarkan button at any applicants		Notification of successful transaction	As Expected	Pass	Auni	
3	Click Transaksi Tidak Tetap on navigation bar		Transaction form displayed	As Expected	Pass	Auni	
4	Enter Transaction Form	Pilih Staf: 3331 - FATIN SUHAIMI BINTI AHMAD Jenis Transaksi: Simpanan Jenis Simpanan: Simbangan Tabung Jumlah: 30.00 Keterangan: Bayran Lebih Jenis Bayaran: Tunai	Completed transaction form displayed	As Expected	Pass	Auni	
5	Click Hantar Transaksi		Notification of successful transaction	As Expected	Pass	Auni	
6	Click OK		Notification of successful transaction disappears	As Expected	Pass	Auni	

9. Traceability Matrix

A Traceability Matrix (RTM) is used to show the connections between different parts of a project, like user and system requirements, design documents, test cases, and issues. It helps koperasi KADA track the progress of requirements from their start to the final implementation and testing. The RTM creates links between package items, related User Stories, Sequence Diagrams, and their test cases. It connects the IDs of User Stories and Sequence Diagrams, making the relationships clear, and includes the Test Case IDs for each User Story and Sequence Diagram.

Table 9.1: Example of Traceability Matrix for <koperasi KADA's automated system>

Sprint No	Package Item	User Story ID	User Story Description	Sequence Diagram ID	Sequence Diagram Description	Test Case ID
1	Package 001	US001	As an applicant of Koperasi KADA, I want to apply for a membership through the Koperasi KADA Automated System, so that I can apply for a loan.	SD001	The membership application process required applicants to fill out the application. The system will save the submissions to the database	TC001
2	Package 002	US002	As an admin of Koperasi KADA, I want to review the membership application form, membership termination application form and loan application form whether it is completed or not.	SD002	The Admin decides whether to mark the status of membership, membership termination and loan application forms as either 'Lengkap' or 'Tidak Lengkap'.	TC002
3	Package 004	US03	As an ALK, I want to approve applicants' membership so that I can ensure that only eligible individuals can	SD003	The ALK makes decision to either approve or disapprove membership	TC005

			become membership and earn membership benefits		applications reviewed by Admin	
4	Package 003	US001	As a member of Koperasi KADA, I want to apply for a loan through the Koperasi KADA Automated System, so that I can request financial support for my needs.	SD004	The loan application process required members to fill out the application. The system saves the submissions to the database.	TC003
5	Package 003	US001	As a member of Koperasi KADA, I want to calculate my loan repayment and dividend returns, so that I know about my financial commitments and also benefits.	SD005	The loan calculator processes input to generate a monthly repayment amount.	TC004
6	Package 004	US003	As an ALK, I want to approve loan application so that I can let only eligible applicants to receive loan in line with company's policies	SD006	The ALK makes a decision to either approve or disapprove loan applications reviewed by Admin. ALK also can utilize button 'Banding' to automatically compare the loan applications with loan terms before decision-making.	TC006
7	Package 005	US002	As an admin, I want to generate online statements	SD001	First, the Admin logs into the system. After	TC008

			and reports so that I can provide correct reports' records to the stakeholders and members of koperasi KADA.		successfully logging in, the interface displays a list of Koperasi KADA members. Finally, the Admin generates a statement for each member. Admin also can choose to review the corporation report monthly or yearly.	
8	Package 005	US002	As an admin, I want to send automated email notifications to applicants so that they are updated about their financial statement and application status.	SD002	The Admin begins by retrieving the prepared report or status aplication. Next, they send an email notification to the intended recipient. Finally, they receive feedback confirming whether the email was successfully delivered.	TC009
9	Package 006	US001	As a member of Koperasi KADA, I want to terminate my membership application so that I can discontinue my association with Koperasi KADA.	SD010	The terminate application process required members to fill out the application. The system saves the submissions to the database.	TC010
10	Package 007	US002	As an admin, I want to manage monthly and irregular	SD002	The admin first selects whether the transaction is monthly or	TC011

			transactions for saving and loan repayments to ensure financial records are accurate and up-to-date.		irregular. If it is monthly, a button is pressed to process salary deductions. For irregular transactions, the admin chooses between loan or savings, fills out the form, and submits it. Finally, the transaction is recorded in the system.	
11	Package 004	US011	As an ALK, I want to approve membership termination application so that I can deactivate members' membership	SD011	The ALK makes decision to either approve or disapprove membership termination applications reviewed by Admin	TC007